

The PLANET MAPPERS

*An Adventure Story of the Space World
of Tomorrow*



by
E. EVERETT EVANS

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*** START OF THE PROJECT GUTENBERG EBOOK THE PLANET
MAPPERS ***

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E. EVERETT EVANS

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Books by E. Everett Evans
MAN OF MANY MINDS
ALIEN MINDS
THE PLANET MAPPERS

The characters and situations in this book are wholly fictional and imaginative: they do not portray and are not intended to portray any actual persons or parties.

*To my boys—
Carl, Dave,
Tommy, Billy,
Edward, Freddy*

THE PLANET MAPPERS

1

As he heard that dread yet telltale *spang* against the hull of their spaceboat, young Jon Carver dropped his reelbook and sprang to his feet. His eyes looked swiftly to help his ears trace the sudden *hiss* he knew was their precious air escaping.

In the back of his mind he heard the sudden grunt his father made, the sound of a falling body, his mother's frightened scream, and his brother's "What's wrong?" But he did not stop his own lanky, gangling body in its leap toward the outer bulkhead. And as he jumped, he pulled his handkerchief from his hip pocket.

"Leaping tuna! If that isn't fixed quick, we'll lose our air," was his near-panicked thought. "We won't be able to get where we're going. Be lucky if we come out of it alive!"

So, guided by the whistling, escaping air, Jon found the hole, nearly half an inch in diameter. Into it he wadded the corner of the cloth as best he could. The outward loss of their precious air slackened, although there was still some leakage he could not stop this way. He jumped to the nearest of the many emergency repair kits scattered about the ship. From it he grabbed a metal patch and an electric torch.

Swiftly he plugged the latter into a wall socket. With it he quickly welded the patch into place, after pulling—with considerable difficulty—his handkerchief from the hole. "It'll do for now," he decided, after carefully examining his work and listening closely to make sure there was no more whistling-out of air. "But we'll have to go outside and really fill in and weld-plug that hole in the hull, but quick."

He re-stowed the torch, then opened a flagon of emergency oxygen-helium mixture in front of the electric blowers that kept their air circulating—to replenish what had been lost. Only then—although it had been less than two minutes, really—did he turn back to the rest of the family. He had been somewhat surprised that his father had not come to help him; he had not been at all surprised that his brother had not. Jak was a grand guy—Jon

thought the world of him—but he just wasn't worth a dead salmon in an emergency like this; he did not have a mechanical type of mind.

Now, as he turned, Jon saw his mother and brother kneeling beside the prone body of his father, and noted with astonishment that she was crying. There was something stiff and unnatural about the man's body, too, lying there on the deck beside his recline seat.

A sudden fear sent the boy leaping across the room. "What ... what happened? Pop isn't dead, is he?"

"No. Something made him fall, and he hit his head on the deck and knocked himself out," Jak said without looking up. "His foot caught in the footrest, and as he fell over the seat arm his leg broke."

Jon dropped to his knees beside his weeping mother and threw an arm about her. His eyes were wide and damp with swift tears, for, in spite of the rapid growth his body had undergone in the past few years, he was still only sixteen—and he loved this splendid father of his with genuine devotion.

It just couldn't be that Pop wouldn't live, he thought in panic. He couldn't make himself believe that he might no longer have the wonderful companionship and guidance and counsel of this grand man who had been his world.

His mother, seeming to realize what the boy was undergoing, forced back her own grief to turn and gather this younger son into her arms, comforting him as only mothers can.

They watched the elder brother's swift, competent hands as he bathed with soft cotton, soaked in some kind of medicine taken from the open first-aid kit beside him, the bruised place on the back of the father's head. Jak had already shaved away the hair about this bruise. Now he took an atomizer and sprayed on a clear, plastic bandage.

Mrs. Carver turned anxiously to her younger son. "Jon, you know how to run the ship. Turn it around and get us back to the nearest hospital as fast as it will go."

Jon looked at her in astonishment, for it had never before occurred to him that she did not know at least something about inter-stellar astrogation. "We can't, Mom. You don't run a ship in space like you do a ground car. We're on

negative acceleration now, but it'll be close to two days before we've slowed enough for any kind of maneuvering."

"That's right, Mother," Jak came unexpectedly to his brother's aid. "You can't stop or turn a spaceship at will. But I don't think we need worry too much. Father's head wound is not serious, although there's a slight concussion. And we can set his leg so it will heal straight—it's a clean break."

"Besides, it would take at least a month to get back to the nearest colonized planet," Jon took up the explanation. "You know we're almost six weeks out of Terra."

Mrs. Carver still looked doubtful, but responded, as did Jon, when Jak began issuing instructions to them to help him in setting the broken leg. He had cut away the trousers and removed the boot and sock. Now he asked his mother to grasp his father's shoulders and hold tightly. He then showed Jon how to hold the toes and heel of the injured leg, and pull steadily downward while he manipulated the bone ends into place.

When the break had been adjusted, Jak dissolved certain plastics into a heavy, viscous liquid which he sprayed onto the leg. This mixture hardened almost instantly, forming a cast that was far stiffer and yet less weighty than either the ancient plaster casts or cumbersome splints.

When it was finished, they all rose, and while their mother hurried ahead to prepare the bunk, the boys stooped and lifted their father's inert body. Staggering a bit under the load, yet handling him tenderly, they carried him to his wall bunk and lowered him onto the sheeted mattress. After their mother had tucked in the top sheet and blankets, the boys buckled the acceleration straps about the bunk, and Jak made an extra binder with a folded blanket about the broken leg. Now, if their father regained consciousness, or moved about restlessly in partial awakening, he could not fall out and perhaps hurt himself more.

When all had been done to make the wounded man as comfortable as possible, Mrs. Carver turned to Jon questioningly.

"What happened, Son? Do you know?"

"Meteoroid broached the hull, then must have gone on and almost hit Pop. If it was a close miss, the force of its passage must have made him duck and

fall."

"But I don't feel any air escaping."

"There isn't now. I patched the hole, inside. Temporary job, though. Pop'll —" He stopped in sudden realization, then straightened resolutely and his voice was calmer, more sure, as he went on. "I mean, I'll have to go outside and make a permanent weld. Might as well do it now."

His mother's face showed the pride she felt in this young son who could plan and do the things that had to be done, even while she knew he was upset by his father's accident.

"Yes, it should be done at once." But she gripped his arm convulsively. "Be sure your lifeline is fastened securely, Jon."

He patted her hand awkwardly. "I will, Mom. I've been outside a lot, you know, and understand just what to do."

He broke away and ran toward the airlock. From the closet just inside the inner lockdoor he took his spacesuit, and put it on as quickly as he could. He was still working on the zippered seam down the front, smearing on the quickly-drying plastic that made it doubly airtight, when his brother came in.

"Can I help, Chubby?"

"Sure, give me a hand with my helmet. Say, Owl, will Pop really be all OK?"

"I ... I think so. He got a bad smack when he fell. But his heart seems to be beating strongly, and I think the concussion'll wear off soon. The leg'll heal, but he'll be out of commission about six weeks."

He picked up the quartzite "fishbowl" and slipped it over Jon's head. They settled it firmly in place on the suit-ring, and screwed tight the lugs that held it in place. As Jon turned on his oxygen he motioned to the plastic, and Jak smeared it carefully all around the seam.

When he had finished, Jon increased the oxygen flow until the suit bulged, while Jak minutely inspected every point for any possible leakage. Finding none, he made the OK sign with thumb and bent forefinger, and Jon

reduced his air-flow and opened the escape valve until the suit deflated enough so he could move about easily.

From a chest of repair supplies the younger boy took a can of metal-seal and a self-contained acetylene torch. These he fastened to his belt while Jak was getting, from a wall hook, a coil of thin but terrifically strong, light, plastic rope that would neither freeze nor lose its pliability in the utter cold of space. While spacesuits had magnetic shoe soles to keep their wearers in contact with the hull, a lifeline was a safety factor in case they happened to break that contact and drift away from the ship.

Jon checked his suit and equipment again, making sure he had all the tools he might need, and that they were firmly in place. He snapped one end of his lifeline into a ring at his belt, tugging strongly on it several times.

Then he turned and grinned through the helmet at his elder brother. He waved him away from the inner lockdoor, then pressed a button. The inner door swung open and air rushed in to fill the vacuum between the inner and outer lockdoors.

Jon stepped into the narrow space, skirted the handling mechanism there, then pressed another button to actuate the motor that closed and locked the inner door. When the red signal light told him it was airtight, he switched on the pump that returned the air to the body of the ship. The lock empty, he twisted the knob that opened the outer lockdoor, then snapped the other end of his lifeline to a ring just beside the opening doorway. He switched on his suit-heater as he felt the chill of space.

Slowly, ponderously, the mechanism swung the great eighteen-inch-thick outer door partially open, and Jon was facing deep space. Although he had spent nearly a third of his life out here, it was a sight that never tired the boy's active, imaginative mind, and even now he stood for a long minute, eagerly looking outward.

The awesome blackness of the void seemed alive with millions upon countless millions of tiny, distant, pinpointed lights he knew were giant suns. On and on they stretched, as far as the eye could see—and beyond. In the far, far distance were blotches of light Jon knew were the incredibly distant nebulae—other uncounted billions of suns that made up the far-off galaxies and universes.

He looked overhead, picking out against the backdrop of the nearer suns of our own galaxy—the Milky Way—some of the larger giant suns ... Canopus, Rigel, Deneb, Betelgeuse, Antares and others he knew by sight. The patterns familiar on Terra were somewhat distorted here because of the difference in distance and his line of sight, but those suns could not be mistaken.

He only stood there for a moment, then he reached out carefully and grasped the rung of the metal ladder welded onto the hull, and which ran completely around the ship. He pulled himself onto this, and held there while he estimated where that hole should be.

"About twenty-four feet to the left, and one or two lower than the doortop, I think," he muttered to himself. He climbed several rungs, then half-straightened and set first one foot and then the other firmly and flatly onto the hull beside the ladder rungs. He tried each of his shoes, making sure their magnetic soles were gripping tightly against the hull surface. Then he let loose the ladder and stood upright. Compared to the decks inside, he was at right angles, but there is no up, down or sideways in space—except that your feet always seem "down."

Assured that his shoes were holding firmly, he slid first one foot and then the other along the hull. In this way he walked ahead, always in full contact, yet able to progress almost at a normal pace. He counted his steps, and when he felt he was near the hole for which he was looking, stooped and began searching about the surface more minutely.

His estimate had been close, and it took him only a moment to find the place where the meteoroid had struck. He drew his lifeline taut and tied the loop to his belt, leaving the end of the line still snapped in place. Now, even though his knot might come loose, he was still fastened to the ship.

He took the can of metal-seal from his belt pouch, fumbling a bit because it was difficult working with such heavy gloves as those attached to his spacesuit. There was plenty of light from the billions of stars, nor did it matter what hour the ship's chronoms might indicate inside, it was always the same out here.

He squatted down, still keeping both feet flat against the outer skin of the ship. Carefully he poured some of the sluggish, viscous liquid metal into the

funnel-shaped hole, which was over an inch wide at the hull surface. Then he unslung his torch. He snapped the lighter and adjusted the flame to a narrow, pencil beam.

With the beam he melted the metal-seal he had poured into the hole. In the cold depths of space, where the temperature was about absolute zero, the metal cooled almost instantly as he turned his torch away. He then added more seal, melted that, then more seal, and so on, a bit at a time, until the hole was completely filled, and the hull surface once again smooth and even.

Satisfied at last that the damage to the ship was completely repaired, he hooked his torch to his belt once more, recapped the can of remaining metal-seal and stored it in his belt pouch. He rose and stood again for a few short moments, looking at the glory of the universe as it can only be seen from a spaceship. Then he made his way back to the lock and entered the ship.

He touched the stud and the motor slowly closed the great outer door. When the red signal light showed it was airtight, he punched the other button, air filled the entry, and then the inner door opened. He went through into the ship, closed the inner door, and when that was tight, started the motor that pumped the precious air from the lock back into the ship.

His brother had not stayed around to help him, so Jon had to strip the plastic from his zipper and around the base of his helmet by himself. It was an awkward job, as was trying to unscrew the lugs at the back of his shoulders, and he growled a bit beneath his breath because Jak had not waited, nor come back to help him.

But his irritation quickly passed and he grinned to himself. He knew his brother so well—Jak simply had not thought to stay and help, or he would willingly have done so.

Jak's tastes and desires ran more to other things, Jon knew. To medicine, and to all growing things, whether plant, animal or human. Jak had always been far more interested in what made *life* grow and perform its miracles, than he had in how and why *machines* operated.

And, Jon acknowledged honestly, it was a good thing for them all in this present emergency. If good old Jak wasn't half a doctor already, Pop would

really be in a bad way ... and so would all of them, if they lost that steady and competent prop on whom they all leaned so confidently.

"I sure wouldn't have known what to do," Jon admitted to himself, as the thought of his father made him hurry the removing of his suit. "I probably would have run for my tool kit, not the first-aid one."

He finally got the suit off and hung it back in the closet. He gathered up the scraps of used plastic and stuffed them into the near-by trash disposal chute. Then he ran into the living room and on to the side of his father's bunk, where his mother and brother were standing, watching.

"How is he?"

"Just the same."

"You're sure he ... he isn't...?"

"No, he's still alive, and I'm sure he'll pull out in time. Only question is, how long it'll take?"

Jon's mind began churning with problems. What would they do while Pop was "out"? Who was to run the ship; make the calculations on orbits and trajectories? Who's to handle the controls of landing when we reach our destination, which won't be very long now? Who'll do the thousand and one things Pop has always done? Who'll make the decisions?

Again the sense and knowledge of his personal loss came home—and young Jon Carver sank onto the deck of the bunkroom. Again he was just a boy who had lost his dearest pal, his ideal. Pop just couldn't die! Who'd help him with his problems; teach him the many things he was always wanting to know?

It just couldn't be that there would be no more of those tussles of friendly play; those boxing matches or wrestling bouts by which his growing body adjusted to swift action and hard knocks. He could not make himself believe that there would be no more of those hours of practical instruction, or the long, pleasant evenings when the big man would talk of the places where he had been, the things he had seen and done in his travels about the galaxy.

For Tad Carver was one of the real pioneers of deep space. He had been an officer of the first ship to reach the stars—the planets of Sirius.

Deep-space travel was not yet a commonplace thing, although it was becoming so more swiftly with each passing year. Jon knew that there were now regular trips to the planets and some of the moons of his home solar system. One could have a two weeks' vacation trip from Terra to Luna for a thousand credits, or a month's cruise to Mars or Venus for forty hundred.

Merchant ships made fairly regular voyages to the planets of Sirius and Vega and, less often, to one or two other even more distant worlds which had been found to contain friendly and civilized beings—not all of them humanoid—who were glad to engage in inter-stellar commerce. Other spaceships plied between Terra and the many newly discovered worlds that were being colonized by Earth people.

But it had been men like Tad Carver who, co-operatively, had bought ships and surveyed the spaceways. It was they who had opened up those parts of the galaxy so far charted and who, incidentally, had made fortunes for themselves from the metals, strange jewels and other rare objects they had discovered and brought back, and for which the rich of Terra had paid so willingly and so handsomely.

That was why, after a number of years and many such trips, Carver had been able to buy his own small ship, outfit it for deep space travel, and take his family with him on his further voyages of exploration and survey. They were now en route to a new portion of the galaxy, one never—so far as they knew—visited by human beings.

"But what'll we do without Pop?" Jon's mind went back to his problem. "Who would be in command of their ship now? Mom didn't know a thing about the navigation of space. Look how she'd demanded he turn around 'right now'!" She was wonderful, and Jon loved her dearly. But he also knew she would be absolutely out of place trying to make their decisions about where to go, how to get there, how to run the ship, and so on. She had always seemed content to "keep house" on the ship, just as she had on Terra, and paid but little attention to what else was going on.

And Jak was just about as bad. The older boy was quick-and-logical thinking, and knew a lot—but not about such things. Jon had been the one who was always tagging their father around, forever asking questions about how to do this, why was that done, what did this machine do and what was the theory behind it, and so on? He had always been working with

machines, almost since he could toddle. He took them apart, not destructively but questioningly, and was very soon able to put back together again correctly an endless succession of ever-more-complicated mechanisms.

Recently he had begun the study of astrogation—he had also long been a "math shark"—and now knew enough to realize how little he really did know about this complicated subject—although actually it was a great deal.

Sobered, and suddenly aware of a growing maturity brought on by the terrific problems they faced, Jon sat up. He rose and went over to his mother's side. He touched her softly on the shoulder, and she looked up at him. At sight of his anxious face she threw her arms about him.

"Jon, boy, what will we do now? How will we ever manage without Mr. C?"

At this echo of his own questionings and doubts, the boy straightened. "We'll make out all right, Mom," he said with a bravado he certainly did not feel, but which he hoped she would think was genuine. "We'll have to make up our minds what we're going to do, then do it. We'll keep on with Pop's plans, of course." This was a statement rather than a question.

"Why ... why...." She seemed startled by the realization that she had to make a decision. "I hadn't thought about that yet." She was silent a moment, then turned to her elder son, who had also risen and was listening intently. "What do you think, Jak? You're older, so you'll have to take charge now and be the man of the family."

The slender, studious eighteen-year-old looked startled. "I ... I don't know," he stammered, his eyes suddenly filled with strange fears. "I ... I suppose we might as well go home. We don't know where we're going, or what we were to do when we got there...." He suddenly looked like a little boy who has lost everything and everyone in whom he had looked for and found comfort and security. "Don't ask me, Mother. I don't know what we're going to do. We're apt to die, without Father to keep us going safely!"

Jon stared at him, this brother he had always loved and to whom he had looked up as a strong, elder companion ... in spite of their almost continuous, although friendly, bickerings, which never disturbed the warm affection underneath.

Now he just couldn't believe his eyes and ears. This couldn't be Jak—the strong, reliable Jak!

Suddenly he felt a surge of anger and distrust. Yet immediately he was ashamed of himself for such feelings. This wasn't any minnow of a predicament they were in—it was a very whale of a mess. He was scared, himself, and could understand just how Jak must feel. But, by the great horned catfish, he wasn't going to let himself cry about it any more—especially in front of Mom! Something had to be done, and it would be done!

A thought flashed through his mind, and he straightened with resolve. "*Shut up!*" he yelled at his brother ... and when Jak and their mother stared at him in amazement Jon grinned calmly and said, half apologetically, "Just trying to snap you out of the dumps. I say we've got to think this out carefully, and not make any snap decisions—or give up like this. The ship's on automatic drive and decelerating, so we don't have to worry about running it for some time. But Pop wouldn't like it if we didn't keep on. You know how important this trip is to him. Besides, he'll be waking up soon, and even if he has to stay in his bunk, he can tell us what to do."

"Do you know where we're going, and why?" Jak was still upset.

"Sure. Pop talked with me a lot about it."

Their mother looked from one to the other doubtfully, then smiled in a constrained manner. "You ... you're probably right, Jon. Mr. C. did say this would make or break us. I leave it up to my two big boys to discuss and suggest plans until your father is able to take charge again."

With an effort she pulled herself together, and now her smile was firmer, brighter. "Meanwhile, I think we'd better have something to eat. We have to keep up our strength for whatever is coming, you know."

2

When the boys woke up the next morning, their mother reported that their father had apparently had a restful night, coming out of his coma briefly a couple of times.

After breakfast the two boys went into the control room and began examining the various instruments and recorders on the panel, to see if they could figure out how much longer it would take them to reach the system their father was seeking. Through the visiplates they could now see not only the sun toward which Jon said they were heading, but even its nearer planets were beginning to show appreciable discs.

As they were studying these, Jak suddenly asked, "How do you suppose we happened to run into a meteor way out here in space like that?"

Jon shook his head helplessly. "Darned if I can figure it out. I always supposed such stuff was only found inside a planetary system. Must be there's some in deep space, though, since we sure as perch got hit by one." He reached in his pocket and pulled out a small marble-sized stone. "Here it is. I hunted around and found it last night. It dented the farther bulkhead, but must have lost so much momentum it couldn't penetrate."

"Just one of those billions-to-one chances, eh?" Jak looked up from his examination of the stone.

"Yes, there's still so much about space nobody knows yet."

Jak thought silently for a moment, then asked, "Well, what do you think we should do next?"

"Keep going, natch." Jon's voice was earnest. "We can't be more than a couple of days away from the nearest planets—and we're over six weeks out of Terra. Pop said this system we're heading for has four or five planets, at least, and that probably Two and Three, and maybe Four, would be fairly Earthlike and habitable. So long as we're so close, it would be wrong if we didn't at least take a close-up looksee at them."

"Yes," slowly, "Father'd want us to do that."

"You know darned well he would. He's sunk almost everything he's got into this ship and this trip, and if we miss now, the government probably wouldn't give us another exclusive crack at it, even if we could scrape up the credits to come out here again."

"Didn't Father say something the other day about his spectro-analyzer—you know, 'Annie'—showing there was...?"

"Yes, 'Annie' popped up strong on that, and that's another reason we've got to keep going—especially since you think Pop'll snap out of it in a few days. You're sure of that, aren't you?" He peered intently into his brother's eyes.

"Yes, as far as I can tell. There's a concussion where his head hit the floor, but I don't think it's too bad, and it should wear off soon."

Jon sighed with relief. "If ... if he was dead, or dying, it would be different, and I'd say go home. But there's another thing. Before we left Terra we heard a rumor Slik Bogin was chasing around out in this sector, and we don't want to let him beat us to this system."

"Bogin? That's the notorious pirate, isn't it? No, if he's out here, we don't want to let him beat us—though what we could do if he did try, I don't know."

"We'll figure that out if he tries to hijack us."

"You hope!" There was a long silence while the boys studied their instruments again. Then, "What about landing, Chubby? Can you do it?"

"I've been studying up on it—put on the sleep-instructor last night." Jon was suddenly half-frightened with the prospect, but determined to keep his voice level. "I've helped Pop land the crate several times—even handled the controls under his instructions—so I think I can do it, with you reading off the manual to me. Anyway, if—if Pop gets worse, we've got to land some time, so we might as well try it here as any time or anywhere."

Jak stood silent a long moment, rubbing his hand through his hair as he did when concentrating. Then he looked up with determination. "Jon, you and I have got ourselves a job to do." And now his voice was steady and earnest. "It's up to us to take care of Mother and keep her from worrying. So,

whenever we're where she can hear us, we've got to act brave and sure of ourselves, no matter how we feel inside."

"Yes, she's all broken up about Pop. We ought to do most of the work, too, so she...."

"No," Jak shook his head, "that'd be the worst thing we could do. She isn't sick, physically, and if she keeps busy, she won't have time to worry so much. So we must keep her from having too much idle time."

"Oh ... maybe you're right, Owl ... yes, guess you are, at that—that's more your dish. But we can act like everything's going to jet fair. It's a deal." He held out his hand, and the two brothers clasped in agreement.

They went into the living quarters. "Hi, Mom, lunch ready yet? I'm starved."

"As usual," Jak bantered.

Mrs. Carver looked up apathetically from the recline seat where she had been sitting, worrying, during the several hours the boys had been in the control room. She looked as though she were almost shocked at their seemingly heartless question, forgetting that she, herself, had used the same excuse the night before.

But in a moment she smiled tremulously. "I guess I let myself forget my job, and that we have to go through the motions of living." She rose slowly, and the boys came and put their arms about her. "Mr. C. wouldn't want me to break down like this. I'll try to do better."

She gave her sons a quick hug and went into the little galley, where they heard her moving about from the deep-freeze to cupboards to induction-cooker. Soon the smells of appetizing food spread throughout the ship.

Jon had gone back into the control room and picked up the reelbook on astrogation, opening it to the chart of the pilot panel. He was still studying this and tracing, from the diagrams in the book, the controls, switches and recorders on the panel itself. He memorized each one as he went along, and made sure he knew its functions.

When Jak called him to lunch, Jon carried the reel with him and continued studying it as he absentmindedly ate. His preoccupation with it raised his mother's fears again. "Can you make anything out of it, Son?"

"Huh?" He roused himself then, and grinned at her. "Sure, Mom, it's easy. Pop taught me most of it already, and I'm just refreshing my mind. I'll set us down in one piece, don't you fear."

"How soon will we arrive?"

"About tomorrow noon, I think, by our clocks. No telling what time it'll be there. I'll take measurements again and make sure, right after I'm through eating. We must be about ready to step up our deceleration."

He looked at his mother more intently, and his voice was so earnest it broke from baritone to a childish treble in places. "Mom, I'm not questioning your authority or anything, but you said yesterday that Jak was to be in charge until Pop wakes up. Now, Jak doesn't know anything at all about astrogation, and while I don't know it all, I do know more than he does, and I'll have to handle it. So what about me being in charge of the ship when we're in flight or on landings and take-offs, and Jak in charge other times? Though whatever you say goes, of course," he added hastily.

Somewhat to his surprise, his brother sided with him. The elder seemed to realize this was no time for one of their friendly squabbles about which was to be "top man"; that their very safety depended on the fact that whichever knew the most about any one thing should be the one to have the say about it.

Their mother looked from one to the other helplessly. "I ... I guess that will be all right. You two figure out things between you. You're all the men I have now until your father..." She almost broke into tears then, but pulled herself together. "Yes, you do whatever you think is best about such things."

"We'll handle it," Jak assured her. "But you'll still be boss in chief."

"You say 'when' and 'what,' and Jak and I'll figure out 'how.'" Jon grinned.

She stretched out her arms and grasped each by a hand. "My big boys! I'm sure we'll come through safely. You're getting to be real men." Then she changed her tone and asked, "You're going to land on one of those planets, then, as Mr. C. planned?"

"Being so close, it seems best," Jak answered. "How long we stay will depend on what we do or don't find there."

"Yes, we need a few days' rest on firm ground before we start back to Terra, at least. We want to freshen our air, if we can, and maybe get some fresh food. Besides, we ought to try to get all the necessary data to prove Pop's discovery, if the planets are uninhabited but worth colonizing."

"I agree," seconded Jak, "even if we have to land in some secluded spot and just rest."

"I'll leave it up to you, then." Their mother appeared more like her usual happy self than the boys had seen her since the accident. "I'll keep house like I always have, and you boys do whatever else you think best."

Jak laughed. "We'll be like those Musketeers in that old book I read some time ago. 'All for one and one for all.'" He held out his hand dramatically. "Put your hands on mine, and we'll all swear to it."

Laughing, they did as he suggested, although their mother pretended severity. "You know I don't like swearing, Boys."

Jak grinned. "But I meant this in the sense of 'taking an oath,' not of 'cussing.'"

"Oh," she wrinkled her nose at him, grinning with her old-time impishness, "that's different."

Jon rose from the table. "I'll get back to my studying."

"You listen to your mother, and don't study too hard," she warned, knowing how he was apt to "lose himself" in his books. "You need plenty of rest for tomorrow."

"All right, Mom."

But when she went into the control room long after dinner, he was still deep in his reelbook. She took it away from him. "Get to bed, Jon. You promised."

"I'm sorry, Mom. Just got so interested I forgot time." He kissed her. "Night, Mom. And don't worry. We'll make out swell."

"I'm sure of it." Her words were brave but he could see the tears were perilously close.

"You'd better ask Jak for some barbit, or you won't sleep any better than you did last night," he counseled. "Remember, he and I are going to take turns watching Pop."

"Thank you, Son. Good night."

He touched a switch and the glolights dimmed and went dark as he followed her out.

All the next morning Mrs. Carver and her two sons were in the control room—except for their frequent trips to Mr. Carver's bunk, to see how their patient was getting along. They were studying through the telescopic visiplates the solar system they were rapidly approaching. Jon had figured the sun was a Type G Dwarf, much like Sol, but a little larger. It had, they now knew, only five planets. Three of these—Two, Three and Four—had seven satellites among them. From their distances from the sun, the boys figured that probably Two and Three would have climates that human colonists, with some adaptation, could stand.

Now they were peering even more closely into their plates, as their ship circled the globe beneath them. Jon had maneuvered it into a spiral course about Planet Two, in such a manner that, from a height of about a hundred miles, they could get a good view of the world beneath them, in their telescopic plates.

"Lots of plant life, but I haven't seen anything that looks like cities," Jak said at last.

"Nor I," from their mother and, "Me neither," Jon added.

Their first measurements of this new planet had shown it to be almost the size of Terra, and they had been delighted to see that there was a moon of considerable size, although not as large as Luna. It was about one hundred and fifty thousand miles out.

"There's a number of large seas or oceans," Jak commented without taking his eyes from his visiplate. "Look at that plant life, though—it evidently coats the whole planet. From here it looks like jungle."

"Lots of lakes and rivers on it, and in those plains we saw." Jon was excited. "It sure looks like a wonderful world where men can live."

As they crisscrossed the planet from pole to pole, they saw small ice fields about each.

"That means there'll be varied seasons here," Jon stated.

"Not necessarily," Jak argued. "In fact, while possible, it's not even probable."

"Says you," Jon sniffed. Then later, "I figure the year here at about three hundred days. Just an approximation, of course, but probably within five per cent. I'm not too good at such things."

"You're probably wrong," Jak snorted, and their mother interrupted what she thought was the beginning of another of their interminable arguments.

"Are you going to land here, or go on to another planet first?" she asked Jon.

"I'm going low enough to test atmosphere and temperature before I decide," he told her.

"Well," resignedly, "do as you boys think best."

Jon manipulated his controls and as the ship tilted slightly, they could see in their plates the ground coming closer. Slowly, under the increased reaction of the powerful bow tubes, the ship slowed until it was cruising at about one thousand miles an hour and about a mile above the surface—or the tops of the vegetation, at least. Then Jon leveled it off.

"You know how to test atmosphere, Jak?" he asked. "The temp now is about 99.4 degrees Fahrenheit, so it probably isn't over 110 at ground level."

"Yes, Father taught me that." Jak moved over to the hull wall where there was an atmosphere-trap and the mechanism that tested and recorded the contents of any air they might encounter on a new planet. He worked this and studied the results.

This latest invention of Terran aeroscopic technies was simple to operate. A chart, already prepared to show the constituents of Earth's atmospheric limits compatible to human needs, was placed beneath a stylus. The latter drew a curve showing the components of the new air, and if the lines did not go above or below the red one on the prepared chart, the atmosphere was safe for human consumption.

"Carbon dioxide a little higher, and when I tested density with a spring balance the ten-pound weight showed nine and a half," Jak reported. "That means we'll feel a trifle lighter, and won't find walking and lifting as hard."

Their mother had been hovering nervously in the background. Now she stepped up and asked, "Are you sure it is safe here?"

"We will be before we go outside, Mother," Jak assured her, then turned to Jon. "Where are you going to land?"

"Soon as I find a good spot. Keep your eyes peeled for a large clearing."

But they had gone only a few more miles when Jak yelled, "There, Jon! Off to the left a mile or so."

At his first words Jon had increased the negative acceleration. His darting eyes spotted the clearing, and he put the ship into a circle and elevated the nose so they climbed to a height of some twenty miles.

"Grab that astrogation book and get ready to read me the checks, Owl. Mom, you strap in. Is Pop all right?"

Mrs. Carver assured him that on her recent trip to her husband's bunk she had seen to it that he was safely fastened down, in anticipation of their landing.

Jak picked up the book and opened it to the book-marked page. He sank into the co-pilot's seat, and fastened the safety belt. "Ready when you are."

Their mother now reported, "All fast, Jon."

A moment while the younger boy glanced quickly at his various dials, then he said tensely, "Shoot."

"Check decelerometer."

"On the hairline."

"Check outside air pressure."

"Seven four two."

"Terrain indicator."

"Level."

"Altimeter."

"Four thousand three hundred. Going down a hundred per second."

"Let her down."

Anxious seconds of jockeying, Jon's eyes flashing from indicator to gauge to telltale to screen, his hands and feet moving here and there on the controls.

The two others gasped as they saw the ground rushing toward them so swiftly. The ship landed—but with a jar that shook them all.

"Off bow retarders," Jak yelled.

The roar of the tubes ceased and they were almost stunned by the sudden silence.

"Down landing props."

The grind of a motor, then a gentle jar and the ship seemed to straighten a bit.

"Props down."

"Close fuel petcocks."

"Closed."

"Shut off fuel pump."

"Shut."

"All controls in neutral."

Jon's hands flashed over several levers, knobs and switches.

"Everything neutral." He turned in his seat then, and his face wore a wide grin of triumph. "We did it. We're down."

He noticed his mother's white, strained face, and called to her, "Relax, Mom. I set you down in one piece, just as I said I would."

Jak broke in with a scoffing comment—although his eyes showed the secret pride he felt in his younger brother's ability—"Lousy landing. What's the big idea, jolting us all like that? Want to bust up the ship?"

"Now, Boys," their mother hastened to break up this incipient quarrel before it had the chance to get started—which was exactly what Jak intended—"I

think Jon did exceptionally well, considering it was his first solo landing. I'm not hurt at all, and I'm sure the ship isn't, either."

Jak pretended to look ashamed, although neither of the boys could completely hide their grins, and had to face away from her. "Yes, I was just steaming off. It was really a swell job, Chubby."

But Jon had already pushed out of his seat and was at one of the window-ports, peering eagerly outside. However, he did fling back over his shoulder, "You helped a lot, Owl. Couldn't have done it without you."

The other two came up quickly to stand beside him, staring at this strange, new world. The clearing in which the ship rested, they could see now, was about a hundred acres in extent. Near the ship the strange grass with which the clearing was carpeted was seared and black from their landing blasts, and burning in places. But toward the huge trees that walled the clearing, the grass was in its natural green state, covered with tiny, whitish blossoms.

The trees visible from the ship were mostly very tall, averaging well over three hundred feet, the Carvers estimated. They looked somewhat like Douglas firs, but with a difference the Terrans could not at the moment figure out.

The three could see no animal or bird life, but guessed this did not mean there was none. The jungle might be teeming with life, but it would probably have been frightened away for the time being by this strange, fire-breathing monster that had descended from the skies to land on their world.

"Think it's safe to go out?" Jak asked.

"Now you listen to your mother, and don't take any chances."

"We won't," Jon told her, then answered his brother. "We'll wait an hour and see what we can see from here, then decide."

"I sure want a closer look at that plant Life." Jak's eyes glistened, and he ran to get his binoculars to see better.

"I ought to examine the hull and tubes, too, to make sure they aren't fouled or corroded," Jon told his mother.

"It's nearly time for lunch." She turned away. "At least you must stay in until after that." It was plain she was still worried, and the boys tried to

reassure her and quiet her fears.

When she called they reluctantly left their vantage points at the ports and went in to eat the lunch she had prepared. Several times she had to caution them against bolting their food, as they talked eagerly of what they might find here.

Finally finished, Jon rose. "Come on, Owl," he urged, "let's go outside and give it the once-over lightly."

"Better break out our rifles first," the elder advised. "No telling what we'll run into."

"If it's dangerous enough for guns, I wish you wouldn't go." Their mother was worried again.

"They're just a precaution, same as Father would take if he was in charge," Jak soothed. "We won't go out of this clearing this first time."

"You'd better give Mr. C. another feeding first, hadn't you?"

Jak consulted his wrist-chronom. "Yes, it's nearly time, and we might not be back by the regular hour."

The problem of keeping their father fed and in good health, apart from his head and leg injuries, had not proven too hard when they became convinced that he was not going to wake up often enough to eat normally.

Jak, while working as orderly in the Centropolitan Hospital the previous summer, had assisted the interns and nurses in giving intravenous feedings to unconscious patients. So he knew the general procedure, as well as the composition and quantity of the nutrient liquid to be administered.

"Will you come help me, Mother," he had asked when he was sure he was ready for that first feeding. "We've got to find certain things in our food stores."

"You're sure you know how to do it?"

"Yes, it's not hard. We need liquid proteins, salt, sugar and glucose." With his mother helping, he had gathered these from their stores, and taken them into the galley. There he had carefully measured out and mixed these ingredients in the proportions his books stated.

Then he and Jon had gone into the workshop and there the younger, under his brother's supervision, and with pictures of the apparatus as a guide, had rigged up a drip-regulator to go into the mouth of a large bottle. To this they had attached a long, slender, plastic tube, and to the far end of that a large, hollow feeding needle.

As the others watched anxiously, Jak had inserted the needle into the large vein on the inside of his father's left elbow. With his thumb Jak had softly rubbed the vein just above the needle's point, to assist the flow of the nutrient. Soon it was done. Mr. Carver had stirred and his eyelids had fluttered when the needle was inserted but he had not fully regained consciousness.

That first feeding so successfully accomplished, Mrs. Carver did not seem to worry quite so much about her husband, although she was careful to keep track of the feeding times, and to remind her sometimes forgetful son of his duty.

The feeding given this day, the boys consulted together.

"Shall we wear our spacesuits?" Jon asked.

"I don't see why. It's hot outside, but bearable, and the air's all right," Jak answered positively. "I not only tested it, but I breathed the sample I took in through the trap. It smells good, and hasn't hurt me any. We'll take our guns, and I want my magnifying glass and knapsack for specimens."

"And I'll put some multiform tools in my belt. Then, in case there's anything that needs doing on the tubes or hull, I can do it quickly."

The two brothers assembled their gear and Jon was just reaching for the button to open the inner door when they stopped short and shrank back.

For a terrific roar came from outside ... such a tremendous sound it penetrated even the hull of their ship!

3

At that horrid noise, the two boys stood frozen a moment, then with one accord raced to the control room, where they peered out of the quartzite ports.

"Great whales, look at that thing!" Jon shouted as they caught their first glimpse outside.

"Yeow!" Jak yelled in amazement. "What do you suppose it is?"

"Never saw anything like it before."

They stared in awe at the tremendous creature standing in the little clearing, looking belligerently toward their ship. It was so unlike any Earth beast it was no wonder the boys were startled. The huge body was covered with heavily matted fur. It must have been at least a dozen feet long, and stood about eight feet tall. But the striking thing was that the body was triangular, and the beast was three-legged—two at the back and one in front.

There was no tail, and the blocky legs—one at each corner of the weirdly triangular body—seemed to end in clawed feet. The head was shaped something like that of a horse, but the huge mouth, now partly open, was seen to contain great fangs, larger than those of any beast the boys had ever seen in Terran zoos or on any planet they had visited. Two of the tusks were almost like the ones they had seen in pictures of ancient saber-toothed tigers.

The whole getup gave such an effect of fierceness that both boys felt a shiver run down their spines. Jak's voice was tremulous as he spoke. "Yipe! I'm sure glad I'm not out there with that."

Jon was slow in answering. "Yet, if we're going out at all...." He hesitated, then continued, "We'll either have to chase it away, or kill it."

"If we can," his brother retorted.

"I think our guns'll handle it," Jon said. "The question is, how are we going to do it without exposing ourselves?"

Jak thought swiftly. "Maybe we could open the outer lock door a crack, just enough to see through and aim our guns."

"Yes, I guess that's it."

"Don't say anything to Mother," Jak cautioned.

"Of course not, silly. Come on, let's see if we can kill it."

The two ran to the airlock and opened the inner door. Leaving it open, they examined their guns to make sure they were fully loaded. Then Jon punched the button to open the outer lockdoor. It was possible to do this while the inner one was still open, since there was now air outside to equalize the pressure. When the door had swung open a couple of feet, Jon stopped the motor, and joined his brother, who was peering through the opening.

The huge creature was still facing them, about forty yards from the ship. One of its rear feet was now pawing at the ground, tearing up great chunks of sod, while it roared its mighty challenge time and again.

"I'll kneel and aim for the left eye. You stand over me and try to hit the other." Jak took swift command. "Then try for the brain or heart, and keep pumping while our ammo lasts or until we kill it."

"Right." Jon took his place and aimed his gun. "Count three and we'll fire together."

Jak knelt and steadied his rifle with one elbow on his extended knee. "One ... two ... three ... fire!"

As the two shots crashed out the creature sprang into the air a couple of feet. A great scream of pain and rage shook the very ground and made the air tremble. It hesitated only a moment, then charged toward the ship at terrific speed.

The boys pumped shots as fast as they could. Both had hit the head, but neither had put out an eye as they had hoped. They kept firing as fast as they could work their guns. Blood spouted from numerous wounds on the beast. But still it came on madly, with swift though lumbering bounds.

"Back quick, and shut the door," Jak yelled as his hammer clicked on an empty gun. He pushed backward and scrambled to his feet as Jon leaped to

the door controls. The heavy door swung shut ... and the boys breathed a sigh of relief.

But almost at once their eyes filled with fear. They cringed back when they felt the ship itself shudder as that heavy body struck against it.

As swiftly as their trembling legs would carry them, they raced back to the control room. They reached it just in time to see the huge triped lunge against the side of the ship a second time ... and again held their breaths as it did so once more.

The beast's slow mind evidently realized, then, that it could not so easily overthrow this strange, great thing that had appeared so mysteriously in its jungle clearing. It backed away some little distance, still roaring out in that horrible voice.

Once again the beast bunched its mighty muscles for another attack ... when it seemed to stop in the middle of a roar. It wobbled a bit. Slowly its mighty legs buckled, and it sprawled on the ground. A few spasmodic shudders, a convulsive shiver that ran through the tremendous frame, then it was still.

The boys let out their breaths. They were just beginning to congratulate themselves when the door of the control room opened and their mother's frightened face appeared.

"What was that, Boys? I was taking a nap, but your shooting woke me, then I felt the ship shake as though there was an earthquake or something."

"It was noth...." Jon began, but Jak went up to her and put his arm about her.

"It was just a big animal, Mother. Jon and I killed it."

"An animal? Big enough to jar the ship that way? Where is it?" she gasped.

Jak pointed silently toward the port, and she hurried to look out. At sight of that huge mountain of flesh she cried out, and her face became ever more white and strained.

"What a horrible beast! Are you sure it's dead?"

"Quite sure. It wasn't hard to kill." Jak minimized the danger and made himself grin encouragingly. "I'm going out and hack off some steaks. Bet they'll be good, too."

"You'll do no such thing!" she cried, shocked. "Now you boys listen to your mother. You're not to budge outside the ship. I want you to leave this awful world at once."

The two boys looked at their mother, and suddenly they seemed to feel strength and maturity growing within them. As though the act had been discussed and rehearsed, they both came up and, taking their mother each by an arm, led her out of the control room and back to their living quarters. There they sat her down in her favorite recline seat.

"Look, Mother, you know how much we both love you and want to obey you always," Jak said earnestly. "But we're in a peculiar situation here...."

"On a strange planet, and Pop out of commission," Jon broke in.

"I know Jon and I are still boys," Jak continued, "but we're all the men here right now. We think you've got to begin trusting us to make the decisions."

"Jak's right," Jon chimed in. "We're not going to take any fool chances, but I say we've got to go ahead and do things just as we think Pop would if he was well and in command. As best we can, that is."

Mrs. Carver looked from one son to the other doubtfully for a long minute, then smiled tremulously.

"I keep forgetting you're not my babies any longer," she said slowly. "Mothers do that, you know. You're both almost grown men; I know you have good minds, almost mature minds. The various things you've been through have done that. So I release you from my apron strings. You two take charge, and do whatever you feel necessary."

They threw themselves on their knees, one on either side of her, their arms about her.

"Oh, Mother, we didn't mean it like that!"

"We never felt you had us tied to your apron strings, Mom," Jon added. "We still want to be your boys, even though we do have to act like men—at least until Pop takes charge again."

Her smile now was warm and tender, all hesitancy and most of the fears gone. "Mr. C. and I have tried to make you self-reliant and resourceful, and he'll be as proud of you as I am. You're right—you are the men of the party

and must do whatever you decide should be done. But be careful," she could not help adding.

"We will, Mother."

"We think just as much of us as you do," Jon quipped.

They left her sitting there, then, and went back to the control room. As they came close to the window-ports they peered through eagerly, and were surprised to see the huge carcass of the triped literally covered with strange looking winged, featherless but fur-covered, bird-things. The latter had large, sharp beaks, with which they were tearing great gobs of flesh from the hulk, gulping them down with ravenous relish.

"Scavengers!" Jon exclaimed, his eyes glued to the scene.

"Yes, there go our steaks." Jak's tone was so lugubrious that Jon looked up and laughed. "I had hoped for some fresh meat."

"There'll be plenty later on," Jon consoled his brother. "Probably this one would've been too tough, anyway."

Jak suddenly chuckled. "Yes, like the fox said, the grapes were probably sour."

They grinned companionably at each other, then turned back to watch through the port again. So numerous and so voracious were the scavenger birds that within a few minutes they had even that mammoth carcass stripped of flesh, leaving only the huge bones. One by one, the birds then flew into the forest, the last ones fighting among themselves for the few remaining scraps of stringy flesh or entrails before they, too, took wing.

"Shall we try it now?" Jak asked after the last of the bird-things had gone.

"Might as well. We sure don't want to be cooped up here forever."

They went back to the airlock again, making sure their guns were reloaded and their ammo belts filled. When both boys were in the lock, Jon punched the button that closed the inner door, then opened the outer one. "Safer for Mom to have one of them shut," he exclaimed.

The two stood there a moment, looking all about them. Except for that strange pile of huge bones, now covered thickly with some sort of reddish,

chitinous-covered, ant-like insects, the clearing seemed empty of all life except the peculiar, flower-like grasses.

Jon climbed down to the ground and Jak followed closely. They walked a short distance away, then turned and looked back, scanning carefully in all directions to make sure no enemy was at their backs.

"Let's go over and study that jungle a bit," Jak suggested when they were sure their rear was not, apparently, menaced.

Jon had been looking at the remains of the beast. "I'd like to try to salvage those tusks," he said, and with Jak at his side went up to them. The two boys managed, after considerable work, to get the great fangs out of the jaws. They brushed off the clinging insects, then ran back and placed the tusks inside the lock.

"Thanks, Owl. Now we'll go take a look at your trees. Then I want to examine the tubes and the outside of the ship. But we'd better stick together, at least this first time. So I'll sort of cover you, then when you've had a looksee, we'll go back and you keep guard while I see what shape the boat's in."

"Right. Let's get going."

Once past the seared place, they found that the peculiar, flower-tipped grasses were as stiff as wheat stubble. The grass-blades were knife sharp, but unable to penetrate the heavy, knee-high leather boots the boys wore.

Jak stooped to examine and study them. "The blossoms all seem to have three of these whitish petals," he said as he rose at last, "and that yellowish bulb in the center will be the seed pod."

When they started on again, they found walking difficult until they learned the trick of scuffling along without trying to raise their feet above the tops of the grasses each step. Then it was easier, particularly since the gravity here was about five per cent less than that of Terra, so they weighed less and their strength consequently seemed greater.

The trees were closely clustered for the most part, and after studying them for some time Jak said, "They're a lot like some of the pines back home, although not too much like any I ever saw."

"Notice how there're no limbs until you get up thirty feet or so?" Jon asked. "They'd be hard to climb without spurs." Indeed, after anyone did reach the first low limbs he would not be in much better shape for climbing, for the branches were ten to fifteen feet apart all the way up.

"Don't see any fruits, though maybe we're just not where any fruit trees are growing," Jak said after a bit.

"Yes, lots of woods back home don't have any fruit or nut trees in them."

The strange grasses grew only in small, occasional clusters inside the forest, but the ground was so deeply covered with fallen twigs, rotted branches and the needle-like leaves of previous years, that walking was extremely difficult, almost impossible in places.

"There're probably trails somewhere—that triped would've made some sort of path."

"He sure was heavy enough, and if this was part of his regular stamping ground, he undoubtedly used the same route."

"Maybe, but not necessarily. He might've been attracted by our descent. Anyway, we can look for that later. Let's go back now so you can look over the ship while it's still light. We should be ready for a quick take-off, if we run into anything too hot to handle."

Jon looked the surprise he felt. This sudden responsibility was making Jak more practical than he had ever been before ... just the same as it was making him. After all, it was to be expected. Jon knew Jak had an excellent mind—the elder brother had just used it for what Jon had previously felt were unimportant things—not mechanics, math, or such practical interests.

But the way Jak had taken care of Pop; the way he had figured out how to feed him, and the right medicines to use on the bruise on his head—that must have injured Pop inside—and had known how to set his leg—it was a danged good thing, after all, that Jak had spent so much time studying those other subjects. Maybe mechanics and such sciences were not the main things in life, after all. Other things had their uses, too.

Now the two went back to the ship and around to the stern. There, while Jak stayed on the ground on guard, watching in all directions in turn for any possible dangers, Jon surveyed the great driving tubes.

He climbed the metal rungs set into the ship for that purpose, so he could reach each of the tubes. With his glotorch he studied the lining of each tube, crawling partially inside each one in turn.

Finally he backed out of the last one and down to the ground. "They're all in fine shape," he reported happily to his brother. "Can't find a single thin spot in any of them. That new alloy is really something."

Although the older brother did not know too much about such things, he felt a sudden relief at this report, for he felt that Jon did know, and he had real confidence in him. He had long realized the differences in their temperaments, and for several years had known his brother was almost a genius in the mechanical field.

He remembered mentioning this matter to their father one time, and how his eyes had shone with pride as he answered, "Jon's really remarkable. Some day, if he keeps on like this, he's going to be known all over the galaxy because of what he'll do in mechanics."

Nor had Jak been jealous at this high praise of his younger brother. "Jon's just a kid," he had said, "and he's thoughtless rather than conceited. But sometimes he makes me so darned mad."

His father laughed. "Yes, like all kids, he hates the thought of letting anyone get ahead of him. That's particularly true of younger brothers. They feel, within themselves, that they are just as good or better than the older members of the family, and sometimes can't help showing it."

Jak grinned. "I'll bust him one yet, some day, though, if he doesn't watch out."

But he knew, and so did his father, that he never would. For both knew the real love that existed between the two brothers. Jak realized that his swiftly growing brother—now several inches taller and many pounds heavier than he—had a terrific mind. So, as now, he generally respected Jon's ideas, and shrugged away any momentary angers when Jon was particularly "bossy."

Jak followed as Jon walked slowly along the side of the ship, giving it a careful survey, especially toward the bottom, to see if anything on the lower surfaces appeared wrong.

"I'll climb up and give the top a going-over tomorrow," Jon said as they went ahead.

The *Star Rover* was really a space-yacht. It was seventy-two feet long, and about eighteen feet in diameter at its thickest part, which was about a third of the way back from the bow. The front of the ship was bluntly rounded, and contained the control room with its thick, quartzite window-ports, and just outside that room the four bow-retarding tubes, which Jon also carefully examined when the boys reached them.

Just aft of the control room were the living quarters. These consisted of the large, comfortable living room, and two small but compact bunkrooms, the bath-toilet, the kitchen and many ingeniously designed closet and drawer spaces for stowing personal belongings, clothing and supplies.

Beyond these were the storerooms for food, tools and other supplies and equipment. The stern two-fifths of the ship was devoted to the storage of fuel and the various machines that drove the space-yacht and kept it a self-contained world while in space.

Here were the refrigerators and heaters, the air- and water-purifiers, the generators of electricity for light and cooking and for their auxiliary motors, such as the ones controlling the airlock doors and pumps. In the lower part of the hull, under their living and control rooms and storerooms, were hydroponic tanks which not only grew vegetables and greens for their table, but which furnished oxygen to replace that unavoidably lost when the locks were opened.

At the far stem were the driving mechanisms. The latter were the latest development in the atomic-powered field, and were surprisingly small for the tremendous work they did.

Even Jon did not yet fully understand how they operated, although he knew how to run them. He did know they took specially-treated copper, in the form of small nuggets, and utilized the tremendous force locked within their atoms as the propelling medium by which the ship operated.

In some manner these nuggets were vaporized inside the generators, into which they were automatically fed from the storage bins as needed—the power-controls regulating the speed with which they were fed into the generators. This vaporized copper was run through some sort of a modified

cyclotron-type mechanism, where the binding-force of its atoms was liberated. That indescribable power then forced its vaporized particles out through the tubes—using the Newtonian law of action and reaction to propel the ship.

Suddenly Jon turned to his brother. "Hey, I just happened to think. We ought to rig up a siren or something, so Mom can call us if she needs us when we're away from the ship."

"Sounds like a swell idea. Can fix?"

"Sure, nothing to it. We may even have one among the stores. If not, it's just a diaphragm inside a tube, oscillated by electricity. I'll see if we've got one, or else make one and install it."

As they neared the entrance to the ship they saw their mother standing in the opened lock, getting a breath of fresh air, and looking about the clearing with an interested expression.

Jon had just opened his mouth to call to her when suddenly, without warning, without even a change in the light or feeling in the air, rain began coming down in great sheets. The boys, after only a momentary start of surprise, raced for the airlock. Their mother stayed to help them climb in. But by the time they were inside and the outer door was closed, they were wet through to the skin.

"Wow, that's sure some storm! Wonder if it's a regular feature here?"

"I wouldn't know," Jak panted. "Did you get a look at that lightning, and hear the thunder?"

"Didn't take time—I was too busy running." Jon laughed as he tried to wring the worst of the water out of his coveralls before going through the living room to the bunkroom, where they could change to dry garments.

As they came out their mother, now also in dry clothes, met them with a smile. "I think your father is getting better—he moved about quite a bit a while ago, although he didn't completely regain consciousness."

"Wonderful!"

"That's super!"

Later, as the three were eating dinner, Jak suddenly laid down his fork in excitement. "Just happened to think. We didn't see any cities here, so doesn't that make this a prime discovery?"

"That it do, that it do," Jon said delightedly.

"Then that means we have the right to name and claim this system...."

"Unless there are intelligent inhabitants on some of the other planets."

"Seems to me if there were any, they'd be here—this is certain to be the most logical world to support life. What'll we call this system?"

"'Carveria,' of course, stupid. After Pop," Jon answered witheringly.

"That's very thoughtful of you, Son." His mother smiled at him fondly.

"We'll call the sun 'Carveria,' then, and the five planets will be 'Tad,' 'Marci,' 'Jak,' 'Jon,' and 'Rover.'"

"Ouch, how corny can you get?" Jon sniffed. "Since there are five, I know the fifth should be named for the ship, and we can't very well call it 'Star Rover.' But certainly not just 'Rover,' either."

"Why not leave off the last 'r' and just call it 'Rove'?" their mother suggested.

"Good!" "Swell!" the two exclaimed at once.

"That means this one is named after you, Mom. How does it feel to have a whole world named after you?"

"You ought to know," she retorted with a smile that brought out her dimples in the old way. "You've each got one named for you."

"Then let's call this moon 'Diana,' after the ancient goddess of the moon," Jak said.

"Look, Owl, this is Mom's planet. She has the right to name her own moon." Jon's voice was almost a sneer.

"I think 'Diana' is a very nice name, and I'll accept that, although I'm going to make it 'Diane,'" his mother soothed. "That has always been my favorite girl name. If I'd ever had a daughter, I probably would have named her 'Diane.' So it will make a doubly fine name for my moon."

"Haven't time to measure or weigh it now, but I'll bet it's big enough, and close enough, to cause tides," Jon said meditatively.

"What's that got to do with the price of onions in Bermuda?"

"Nothing, just thought it was interesting. Well, bed for me. Need a good rest tonight."

"Why, especially, Son? What do you plan for tomorrow?"

"Just some more exploring, that's all. And we'll be careful," Jon added hastily as he saw the familiar words forming on her lips. "Night, Mom."

4

At breakfast the next morning Jon suddenly stopped eating. "Say, as we were coming down, did you notice a small river or creek just over there to the right? I was pretty busy at the time, but seem to remember something of the sort."

"Yes, there was one near, but don't know just how far. Why?"

The boy grinned. "If there's a stream, there're probably fish. I was thinking we could get some fresh supplies that way."

"You and your fishing! Don't you ever think of anything else?"

"Sure I do, but I notice you always eat your share when I catch any and Mom cooks 'em."

Their mother said quickly, "Some fresh fish would taste good, Boys. If you have time and can catch any, I know we would all appreciate them."

"Look, Jak, you want to explore some more of that jungle, and I want to see if there's any of that stuff Pop was looking for, near here. We can just as well do both while working toward that creek, and I can take my rod along. But first, we've got to set up our marker here in the clearing."

"That's right, I'd almost forgotten your telling us about that. And we don't want to stay too long, either. Didn't you say we have to place one on each planet in order to prove our claim as original discovers?"

"Yes, and one in an orbit about the sun, too." Jon pushed back his chair and rose. "I'll go get one from the storeroom."

"I'll get my specimen cases ready, and see to the guns." Jak, too, rose, then forestalled his mother by turning to her, "I'll feed Father first, and we'll be careful outside. You can call us back with the new siren Jon installed, if you need us."

"All right, Boys." She smiled at them. "Mr. C. seemed to rest well last night, although I do wish he would regain full consciousness. I've plenty of

housekeeping to keep busy while you're gone. Really should do some washing, but that doesn't take long. Just don't stay out too late."

"We won't," they both assured her. "We'll be back long before dark."

The marker which Jon fetched from the storeroom and placed near the inner lockdoor, ready to take outside and set up, was one developed by the scientists and technicians of Terra for just such use.

It consisted of an exceptionally strong broadcasting unit that beamed the message of a tape, continuously, toward Terra. Jon made up the tape while Jak was giving the feeding. It read, "This planetary system was first discovered by Tad Carver, on fourteenth January, 2136. This is the second planet, and has been named 'Marci.'"

Over and over, at five-minute intervals, the sender would broadcast that message on a beam aimed at Terra. The controlling mechanism was a marvelously precise uranium clock, and a small atomic motor with fuel enough for five years gave all the needed power.

By the terms of the Terran Colonial laws, this was supposed to entitle the prime discoverer to certain rights in the system. For one thing, he would receive a one-half per cent share of the value of all minerals, oils, jewels and certain other natural resources later colonists might wrest from those planets, for twenty years following his discovery and the acceptance of his claim.

In this way, the Colonial Board of the World Government of Terra sponsored and assured the far-flung exploration which the development of deep-space travel had made possible. The dangers and expense were so considerable that something well worth while had to be offered to make individuals or companies willing to gamble on the hardships and tremendous costs of exploration.

When the boys left the ship to place the marker, they left both lockdoors open so that the fresh morning air from outside could circulate throughout the ship, replacing the somewhat stuffy, although chemically pure air that their purifiers kept renewed.

"Keep your eyes and ears open, and shut the doors if you think there's any danger," both boys cautioned their mother, after making sure she knew how to work the door controls.

"I will," she promised with a laugh, and couldn't help adding, "Just you be as careful as I'll be."

The boys carried the signal-sender to a distant corner of the clearing, to what Jon said was a good spot. "The book says to dig a hole and plant it with the top projecting three inches above the ground, whenever such a thing is possible."

"You know what to do, so take charge," Jak said simply. When they had dug the hole and placed the sender in it, they shoveled the dirt back, then Jon opened the lid. He started the tape reels and the broadcasting unit, then carefully shut and locked the cover.

In digging, they found the ground here to be damp and soggy, apparently from that terrific downpour of the previous evening. It was almost like a wet clay, although, even to their inexperienced eyes, it seemed to be a very rich type of soil.

"Look how wet it is, even over two feet down," Jon said.

"That was a real rain last night," Jak shook his head slowly, "but somehow I can't believe it made this. Maybe this is the rainy season."

They started toward the jungle, but turned to look back toward the ship. They saw their mother at the open door, and waved to her.

After seeing her answering wave, they plunged into the forest at a point where they saw a trail, left either by the frequent passings of the great triped they had shot, or by other beasts of some type not yet seen. Memory of that gigantic beast, though, made them doubly cautious.

"Sure don't want to meet his relatives," Jon said.

"Especially the mate," Jak added, and could not conceal a shiver.

They had noticed with considerable interest and surprise that those native ant-like scavengers had almost entirely eaten the bones of the triped.

"Apparently we'll not find much in the way of remains on this world," Jak commented as they walked carefully along the trail. "Those scavenger birds and ants sure clean up things in a hurry."

"Except for old vegetation," Jon grunted as he stumbled over a dead branch protruding out into the trail. He was keeping his rifle ready in his hands,

and his keen eyes alert to one side and then the other, rather than downward.

Knowing his younger brother was so carefully on guard, Jak felt free to study and examine the various trees and other plant life near the irregular path they were following. He was almost in a frenzy of delight, constantly darting off the trail a few yards to look at some specimen he had detected, studying it carefully and exclaiming over his find.

"Hey, this one is like an *acer compestris*," he yelped, intently studying the bark with his magnifying glass.

"Spik Englis," Jon scolded. "What is it?"

"A hard maple," Jak's voice was condescending. Then he ran over to another. "This one's almost like a silver poplar. See how its light bark glints where the sunlight hits it?"

He started toward another farther away, but Jon called him back. "Don't get so far from the trail." Reluctantly, Jak retraced his steps, only to be off again a moment later.

"This 'un's got nuts almost like small coconuts." He picked a fallen one from the ground and tossed it to Jon. "See if you can crack it and find out what's inside."

But when Jon had done so, it proved to be dried and half-rotted. They could not get a fresh one from the tree by shaking, and it was too smooth and high to climb without spurs.

Jak quickly filled his knapsacks with first one and then another of the smaller plants, twigs and leaves he was continually finding. Soon Jon was laughing heartily, for his brother now had to discard an older specimen to make room for the new.

"You'll have to make several trips to get anywhere near all of those just around here, Owl," Jon called at last. "You can't take back everything, anyway. Way you're going now, you'd soon have the ship so full of your junk there'd be no place for us. And this is only the first planet, remember?"

"But these are unique," Jak wailed. "Botanists will want to study them."

"Then let them come here," Jon stated practically.

Jak looked at him, and grew shamefaced. "Guess I did go a little nuts," he said. But before long his excitement rose to fever pitch again. "There's so much here that's new and different, yet something like the ones we know. I must take back samples of everything."

"How many different kinds of—oh, say, roses—are there on Terra?"

"Why ... why ... I don't really know. Hundreds, I'm sure. Maybe thousands. What's that got to do with this?"

"Simply trying to make you realize you can't take back samples of 'everything,' as you said."

"Ouch!" Jak laughed good-naturedly then. "You've got me, pal. I'll take it easier."

But he soon forgot his good intentions as he found ever newer and more different plants and trees and mosses. There was such a dissimilarity, yet at the same time so many points of likeness between the plant life of this new world and that of Terra, that the young botanist was in a continual state of excitement.

Jon, meanwhile, although still keeping a sharp watch for any possible dangers, had been noticing the profusion of other life in this jungle. There were a number of different bird forms, although he saw that those he was close enough to examine were fur-covered rather than feathered. Nor did they seem to be songsters, for the only noises he heard were the sighing of the wind through the trees and vines and bushes, and the *swish* of wings as the birds flew past.

They had gone some distance when he stopped short. Off at one side there was movement among the small bushes. A quick sibilant whisper froze Jak in his tracks. Jon raised his gun, his eyes searching quickly. Then two quick shots ... and a threshing in the underbrush. Soon stillness—and the two boys advanced cautiously, both with their guns at the ready. In the bushes they found what Jon had shot—two small tripeds somewhat resembling large jack rabbits.

"Hah, these should be good eating." Jon was in transports as he picked them up, examining them carefully.

"Should be tender, at least, if the flesh is suitable to us." Jak was excited, too. "There's enough for a good meal."

Jon took a piece of cord from his coverall pocket and tied the hind legs together, then slung them over his shoulder. "Let's keep going."

Jak continued finding new and different plants, and Jon kept on guard. Once they saw one of the huge tripeds in the distance, and stopped instantly, being very quiet as they slipped behind the boles of large trees, from which they peered out cautiously. But apparently the great beast had not heard, seen nor smelled them—it finally wandered away—grazing.

"Well, I'll be a tadpole!" Jon exclaimed. "A grass-eater."

But Jak was not so sure. "Lots of meat-eaters also eat a little grass. Those teeth didn't look like the ones of a herbivore. I think I'll keep away from them, anyway."

"You and me both!" Jon was agreeable to the idea.

At last, after nearly two hours, the two boys came to the banks of the stream, which was about a quarter mile wide at this point, and seemed not too deep, at least near the shore. Now it was Jon's turn to become the most excited. He ran to the edge and peered into the shallow depths, then called out delightedly at seeing dozens of darting forms of some type of marine life in the clear waters.

"You watch while I fish," he commanded, dropping his gun and the two hare-like creatures. He took the carrying case from his shoulder, opened it and in moments had his rod, reel and line ready.

"Yippee!" he yelled as he got an immediate strike on his first cast. With true fisherman's skill, he played the now fighting, swiftly darting denizen of the river. Carefully he reeled in his catch, giving line when the fish ran or plunged, reeling in when he felt the least bit of slack, exerting only enough pressure to force the fish-thing in toward him without losing it.

Soon the wriggling creature was in shallow water, and Jon waded out with his landing net. A quick, darting movement with hand and net, and he had his first catch.

He took it carefully from the net and held it aloft, examining and admiring it, while Jak danced about on the shore near him, uttering shrill yelps of

triumph.

They could see that Jon's catch was streamlined almost like a trout or barracuda. It was nearly fifteen inches long, and very slender. There seemed to be no scales—the skin was more like that of an eel or bullhead.

"Fish or snake?" Jak asked.

"Don't know for sure." Jon was still studying it. "Think it's a fish, all right, but it hasn't any fins, and swims with the same wriggles a snake uses. I think it's more eel than snake, though, and I'm quite sure it'll be good eating."

The mouth was large and ran back almost three and a half inches. When Jon pried it open to remove his hook he saw there was a triple row of needle-sharp teeth, so quickly took a pair of pliers from his tool belt, and used these to remove the deeply swallowed hook.

The eel-fish freed, he dropped it into his creel, then cast again. It was apparent these water denizens were unused to lures, for hardly had his spinner touched the surface of the water than he had another strike.

As swiftly as he could reel in and remove one from his hook and cast again, Jon brought in fish after fish. All this time Jak was dancing about, now as excited as his brother at this prospect of fresh food to replace for the time the nourishing but hardly-delectable concentrates and frozen foods on which they had been living for so long.

But when Jon finally was satisfied with the size of his catch, he found that leaving the river was not to be a simple matter of wading ashore. So intent had he been on his fun he had not noticed that his feet were sinking further and further into the bottom.

Only now, as he tried to return to shore, did he find he could not lift his feet. They were firmly embedded in the sand or muck, more than halfway to his knees.

For a long moment he struggled to pull first one foot and then the other from the clinging stuff. Then he realized he must be in a sort of quicksand, and he began to panic.

"Quick, Jak, come help me! I'm caught."

But almost instantly he countermanded that sharply. "No! Stay back. The bottom here's quicksand or something."

Jak had come running at Jon's first cry. At this warning, though, he slid to a halt just short of the water. "How can I help?" he cried anxiously.

"Catch these first." And Jon threw first his rod, then his creel filled with fish.

Jak caught each and tossed them farther back onto the bank. He then looked quickly about, and spied a long, fallen branch at some little distance. He called to his brother, who was still trying desperately to free himself, "Hang on a minute. I'll be right back."

Racing for the branch, he picked it up and brought it back to the water's edge. But when he extended it toward Jon, it was too short by several feet, even though both leaned forward. Jak would have gone into the water with it, but Jon would not let him.

"We'll have to try something else, then." Jak was getting really worried now, for he could see that the water was up to Jon's waist.

"You'll have to make it snappy," Jon spoke as calmly as he could. "I'm sinking deeper all the time."

Again Jak searched swiftly and purposefully about him. He saw something he thought might help and ran swiftly toward one of the smaller trees. With difficulty, because of the scarcity of limbs, he climbed this and soon was hacking, with his machete-like knife, at the long, slender liana or climbing vine that hung downward from it. It took only a few moments to sever the top end, then Jak slid down the trunk and traced the vine to its root, cutting it there. With this long section he ran back to the water's edge.

"Catch," he yelled—but it took several attempts before he could get the unwieldy vine-end near enough for Jon to grasp.

Jak dug his heels into the ground and started pulling. His face grew red, cords stood out in his neck, and his muscles bulged. But quickly the strain proved too great for him. Since he was the lighter and weaker he was being pulled toward the water, rather than freeing his embedded brother.

"I ... can't ... do it," Jak panted, his strength gone, his muscles and limbs aching and trembling.

"Tie your end around a tree. I'll try to work myself out."

Jak did so, and the muscles on Jon's more powerful arms, back and shoulders stood out in ridges as he threw all his splendid young strength into this climactic effort. He pulled, he wriggled about from side to side.

Slow, heartbreaking moments passed as the tug of war continued. Inch by hard-fought inch Jon was withdrawing his imprisoned legs from the sucking, gripping stuff that was so determined not to yield its victim.

But he was still only a boy, and he had neither the strength nor the endurance to continue for long this tremendous struggle. Slowly his efforts grew weaker and less successful. The sand began reclaiming that which it had lost. Before long Jon sank back, and the strain on the vine relaxed.

"Can't ... make it. You've been a great brother...." He tried to smile. "Take care of Mom and Pop ... and break it to them gently."

"Shut up, you dope," Jak yelled, but there was a catch in his voice. "We're not licked yet!"

Desperately his mind raced. He must think of some more effective mode of leverage. If only he knew how to handle the ship! He could bring that here, and with the loading winch in the lock drag his brother loose. But that was out—he didn't know how to handle it.

He thought of going after his mother, but realized quickly that before he got her and brought her back, Jon would be gone.

No, it was strictly up to him—and time was swiftly running out.

5

Jak Carver's eyes searched the edge of the jungle feverishly for any idea—for some means of rescuing his younger brother, embedded in the quicksand of the stream there.

Suddenly he spied a slim but stout-looking tree close to the water's edge ... and a trick the two boys had often played with a small tree in their back yard at home sprang into his mind.

"Got an idea, Jon. Slack off a minute."

For Jon had been trying again and again, as he felt a momentary return of part of his strength, to pull himself free. He had, by this means, barely managed to keep from sinking further, but that was all.

Now, with a quick twist, Jak unfastened the end of the liana from the tree to which he had tied it. "Tie your end about you, just under the arms," he called. Then, placing his end of the vine in his mouth and gripping it firmly with his teeth, he started climbing that slim tree. It was about seven inches in diameter at the base, and some forty to forty-five feet tall.

His brother instantly recognized what he had in mind. So, as Jak climbed, Jon made sure his end was securely fastened about him. Then he grasped the vine firmly with both hands, a few inches in front of his chest.

As Jak climbed ever higher into the tree, the slender sapling bent beneath his weight. He still climbed, but carefully now, on the side nearest the water, so the treetop would bend in that direction. The higher he climbed the tree, the more his weight made it curve downward, so that toward the last, his back was almost parallel to the ground.

Holding with his legs wrapped about the trunk, when he was almost three-quarters of the way up, Jak fastened his end of the liana tautly in place. This was extremely difficult because of his unnatural position, as well as the stiffness of the vine and his having to work with one hand. But without wasting time, he took pains to make sure the knot was tight and secure.

Then he started climbing again, further and further toward the slender top of the now bent tree. But carefully, lest his weight and the bending splinter or snap the treetop as it bent still further.

"Get tight, Jon. Be ready for the yank when I let go."

"All set and line tight. Yell when you drop."

Glancing down to see that the way was clear below him, Jak let his legs go and swung by his arms until he was hanging clear. He yelled sharply and let go—plunging down the fifteen or eighteen feet to the ground.

Disregarding the shock, he scrambled up, and peered closely at the tree, then the vine, then at Jon. The tree was straining to pull back into its accustomed erectness. The liana was taut—but bits of its bark were flecking off. It creaked so alarmingly Jak was afraid it would break.

All the time Jon was wriggling and twisting to help free his feet and legs. And the vine held, as the tree proved its natural strength and desire for an upright position. Slowly but surely Jon's body was pulled from its prison. As he came more nearly free the tree snapped upright so swiftly he was whipped out of the water and a dozen feet onto the sand. He landed, face down, with a terrific jar.

Jak ran up and helped untie the vine. Jon sat up slowly with his brother's help. His face was scuffed where it had slid along the sandy beach, and he slowly, painfully wiped it somewhat clean with his handkerchief. His breath came in gasps from the terrible constriction of the vine about his chest, and from his unusual exertions.

Sympathetically, Jak hovered about until finally Jon's breathing was a bit easier. When his brother started to try to get up, he helped and held him.

"Guess I can make it now." Jon finally broke away and did manage to stand alone, although he still reeled a bit from the fatigue and the terrible ordeal through which he had been.

He walked slowly about, rolling his shoulders and moving his arms and fingers, exercising his cramped muscles. Jak gave him a couple of anti-fatigue pills from his pocket first-aid kit, and Jon swallowed these. Finally, he began collecting his rod and creel.

"They'd danged well better be good to eat," he declared, shaking the offending fish basket.

"It certainly wasn't worth all that narrow escape," Jak said soberly as he took the things from his brother and went over to pick up the little animal carcasses. But when he got there he exclaimed in disgust, "Darn, those ants have eaten them almost all up!"

"We mustn't let Mom know how close I came to not getting back," Jon said as he staggered along the little trail, although as he went his strength and limberness returned somewhat.

"I'll say not. I'll keep my trap shut. One thing's sure, though. There'll be no more fishing trips here."

"Aw, I wouldn't say that," Jon snapped back. "I know enough now to stay on the bank. And if these are good eating, it's too easy a way to get fresh food to waste."

They were just climbing into the lock when again that sudden heavy downpour of rain began.

Jon grinned as he opened the inner door. "Glad to see the rain this time. It'll keep Mom from wondering why my clothes are so wet."

As soon as they had changed to dry clothing, Jon went to clean his "fish," then took them to his mother in the galley. Jak, meanwhile, was in the control room, rearranging and trying to begin the classification of his plant specimens.

When their mother called them to table, the boys sniffed appreciatively at the delicious odor of the nicely browned fish-things.

"They cook nicely, but how do we tell if they're good to eat?" Mrs. Carver asked.

Jak flipped one onto his plate and cut off a tiny portion. "Tell you soon." And he forked the piece into his mouth. With his tongue and teeth he tested it, but did not swallow. "Tastes good," he said a moment later, retrieving the piece with his fork and laying it on the side of his plate. "One more test."

He cut off another small piece and took it into the storeroom, where he placed a piece in one of the cages containing half a dozen white rats. A

couple of them came up immediately, smelled the food, then one of them gobbled it up. Jak watched anxiously for a moment, then gave another rat a piece. It, too, gobbled it up, and then joined the rest who were pressed against the wires begging for more. Jak stood watching for one minute, then two, then three. Satisfied that the meat had done the rodents no harm, he returned to the table.

"It's all right," he said and began eating. "The rats liked it and it didn't seem to hurt them."

The others pitched in then, and soon the entire platterful was reduced to a pile of bones on the three plates.

"How's Father been today," Jak asked. "He was asleep when I glanced at him after getting back."

"He moved about several times, tossing and groaning a bit, and seeming to be trying to touch his broken leg, although..."

"Probably it itches inside the cast," Jak said.

"He didn't regain full consciousness, but I tried spooning some concentrated broth into his mouth, and he was able to swallow a little of it."

"Golly, that's great!" Jak exclaimed in relief. "His drifting out of his coma from time to time shows there is no real damage to his brain, and now he's evidently beginning to come out of the concussion."

"Whatever it is, I feel more sure he'll soon regain consciousness and be all right." Mrs. Carver spoke with quiet confidence.

"Of course he will, Mom. Pop's too tough for a busted leg and a bump on the head to kill him." Jon smiled at her comfortingly.

"As the surface wound heals, the brain tissues beneath will also be healing," Jak said pedantically. "As long as we can keep him fed and otherwise healthy, the concussion will grow less and finally dissipate entirely."

"Doctor Carver, I presume." Jon sniggered, and his brother flushed a bit, then poked him in the ribs.

Jon tried not to wince at that light jab. Luckily their mother had not noticed anything so, as quickly as possible, he said, "Well, Owl, let's hit the sack."

Want to move around this planet tomorrow and get our pics and info, then take a look at the others."

Jak started to protest, but caught his brother's almost imperceptible but frantic signal, and changed his words. "Maybe Jon's right at that, and we should get an early start. 'Night, Mother."

"Good night, Boys." She responded to their kisses, and soon the two were in their bunkroom, with the door closed.

Jak turned swiftly on his brother. "What's the big idea, making us go to bed so early, and why that funny look you gave me?"

"I had to get out of there." Jon winced as he began taking off his shirt, and Jak crammed his fist into his mouth to keep from crying out as he saw the great, angry red welts and the terrible black-and-blue splotches on Jon's torso.

"Great guns! What happened?"

"That vine must have really hurt when it pulled me loose from that quicksand. I didn't notice it particularly, though, until you poked me in the ribs."

Jak quickly dragged his large first-aid kit from its place in the wall cupboard, and opened it. "Lie down on the bunk, and I'll fix you up," he said as he took out tubes of unguents, bottles of antiseptic and rolls of bandages and plasters from the kit. "Golly, kid, I had no idea you were in that shape, or I'd have done this before."

Jon gritted his teeth as the other gently felt to see if any ribs were broken, and later as Jak applied the healing lotions and sometimes smarting antiseptics. But he could not entirely restrain his exclamations of pain, although he muffled them with his pillow lest their mother hear and come to investigate. He knew his brother was being sympathetically gentle, and when at last it was done, Jon did feel easier. The burning had largely stopped, and some of the ache was gone.

"I'd better give you some barbit so you'll sleep sounder." Jak shook two small pills from a bottle. "The calmer you sleep, the less you'll mess up those dressings, and the quicker you'll heal."

He got a glass of water and Jon took the pills and washed them down. "You do have your uses now and then," he growled, but the grateful look in his eyes belied the ungraciousness of his words—and Jak was well content.

In the morning much of the soreness and discoloration was gone, and there was no sign of inflammation or pus. After Jak had again tended to the abrasions and friction sores, the two boys dressed and went in to breakfast.

Their mother was in good spirits. "Mr. C.'s breathing seems much easier than it was," she announced with delight.

They all went in to see him, and while Jak was redressing the now almost healed head wound, Jon looked on happily.

"Won't be long now." He hugged his mother joyfully.

"I hope not," she sighed. "He does seem to be getting better, though."

"We're lucky we still have him, Mother." Jak's voice was serious. "If that rock had even touched him, it would have been the end. His leg looks OK—no signs of swelling or inflammation."

Breakfast was quiet, and as soon as they finished Jon rose purposefully. "I'll take us up now, and we'll cruise around and see what we can see. Have to take lots of recordings and pictures, you know."

"Are you sure you understand all that has to be done?" His mother's voice was anxious.

"Sure, Mom. It tells all about it on the papers the Colonial Board furnished. All we have to do is follow their instructions. You coming, Jak?"

"Right with you." His brother hastily drank the rest of his coffee and rose, wiping his mouth. "Be sure you strap down at the signal, Mother, if you aren't coming with us."

She flashed him a smile. "I will. Meanwhile, I'll clear the table—if I have time?" She looked questioningly at Jon.

"Sure, it'll take ten—fifteen minutes to get ready, and I'll give you a couple of one-minute warnings."

When all was ready, Jak strapped himself down in the co-pilot's seat, the book of instructions in his hand. Jon touched the stud of the buzzer, waited a full minute then punched two buzzes. Then he nodded at his brother.

"Close fuel dump valves," Jak said, referring to the manual.

"Valves closed."

"Switch on fuel pumps."

"Pumps on."

"Switch on generators."

"Gens on."

"Open all oil valves."

"Oil open."

"Check heaters."

"Heaters on."

"Check refrigerators."

"Frigs on."

"Fire tube one and balance."

Jon snapped a switch. A dull rumbling began and the ship seemed to strain as the first tube started functioning, although at minimum strength. He carefully watched a dial to see that it was working smoothly.

Finally, "Tube one firing."

In like manner tubes four, two and then three were started and tested, and finally reported firing evenly. The ship seemed more than ever straining, as though anxious to get into the air and into free space—but remained on the ground.

"Up landing props."

Jon touched another stud, and they could feel the motor lifting the landing props into their slots in the hull.

"Take off."

The roar deepened as Jon increased the amounts of fuel being fed into the tubes. The ship lifted effortlessly, easily, into the air.

"Check acceleration pressure."

"Normal to speed."

"Check altimeter."

"One thousand seven hundred."

"Level off."

A moment of maneuvering, then Jon reported, "Ship level at twenty-four hundred, traveling parallel to ground surface."

"Check rocket balance."

"All tubes on balance."

"Switch on auto-pilot."

"Auto on, but keeping ready to switch back to manual if necessary."

Jak loosened his straps and went to look out of the port, but Jon kept his gaze fastened on the lookout plate before him, his hands resting lightly on the controls, although they were not connected now.

Beneath them the land was sliding by, as the ship cruised at the slow speed, for it, of just under a thousand miles an hour. The boys saw the same sort of jungle forests, the same occasional clearings. From time to time the glint of water revealed rivers or lakes, the latter seldom more than a mile or so in width or length.

After nearly an hour, they were flying above a huge plain, covered with some sort of grass or grain. They had been above this for some minutes when Jon uttered an exclamation and Jak came up quickly to see what his brother had spotted in the magnifier-screen.

"Look down there, Owl," the younger brother was excited. "Thousands of cattle!"

"Whew! Most like those old buffalo-herds we read the old pioneers saw on the western plains of Noramer. Hey, those things are tripeds, too, like the big one we shot, and the rabbits."

"Yes, I see. Must be the usual thing here. But those down there are smaller, like cows. Wonder if they're good to eat, or give milk?"

"Don't know, but we sure want to report this."

He took several pictures with the recording camera, then made notations in the data book. The two continued watching until the tremendous herd was out of sight behind them, and they were flying once more above a great forest. They had gone almost two thousand miles when they saw ahead and downward the beginnings of what was either an ocean or a great sea. As they drew closer, they still could not see its further shore.

"I don't remember this from before, do you?" Jon looked perplexed.

"Yes, I think this must be the one we saw part of from the north—that is, I assume it was north, as we were near the icecap. But I didn't realize it was so...."

"Hey, look down there! That proves I was right." Jon pointed triumphantly toward his visiplat. "See those high-water marks along the shore? That means this moon is big enough to cause tides, same as Luna does to Terra."

"What good, really, are tides?"

"Why," superciliously, "they're one of the most useful things God has given man. They ... they...." Jon stopped, flushed, then laughed. "Darned if I know what they're good for. Of course, if they're high enough, men can make tide-motors and produce power, but now that we've got atomics, we don't need those."

"I suppose we should record them, though." Jak was tactful enough not to laugh.

"Yes, write it down."

They were over an hour passing above this ocean, and had begun to wonder if it was greater in extent than Terra's Pacific. But finally they made out in the distance the dim blueness of the farther shore.

"That's some ocean all right. Shows there's lots of water here on Two."

"With those heavy rains there'd almost have to be. This'll be of special interest to colonists—means not only plenty of water, but if that stream was

any example, there'll be lots of fish down there to start a big food industry later."

About two hundred miles past the eastern shore of the ocean, they saw the blue of mountains in the decreasing distance. Soon Jon had to rise higher and higher to clear them safely. Some of the individual peaks seemed to be nearly five miles high, and one or two of them, almost at the range of visibility, the boys estimated to be even taller.

"Probably lots of metals here," Jon commented. "I'll swing back and over them again, and let 'Annie' get to work."

"Yes, this list says to report on metallic ores. Say, doesn't it seem funny to you that there are no people on a world as capable as this of supporting life? Wonder why?"

"No telling. Pop says lots of Earthlike planets don't have any inhabitants capable of any sort of civilization. But that means more ready-made worlds for Terrans to colonize."

Jon made their ship circle above the mountains while the boys took readings with the spectro-analyzer. Then they started on again. After almost another hour, when they were over one of the few desert places they had seen, Jon suddenly leaned forward with a little intake of breath that his brother noticed.

"What's up?"

"Not sure. But listen to 'Annie' click. From the reading, I think there must be some of that metal Pop was so positive about, down there somewhere."

"The stuff for a new fuel?"

"Yes. We don't know it'd be any good as fuel, but its atomic weight seems to be so high Pop was all excited when the spectrogram of this sun showed it. He said he felt sure we'd find it on at least one of these planets."

"It'll take a lot of time to locate it exactly, won't it?"

"Not too much, with the new gadgets they have for locating metal ores." Jon tried not to sound impatient with his brother's ignorance. "We've got one that lets us cruise around in the air and spot it fairly close, then land and find the exact place quite easily."

"What sort of gadget?"

Jon shrugged. "Don't know exactly how they work, but I can use one. Something like a spectroscope that works without first having to heat the metals into gas. Plus something like those old Geiger counters they used to trace radioactives. Plus some other ideas the technies put into them. It tells about them in one of our reelbooks there. You go get ours—I think it's in Bin 14, in the storeroom. Looks like a small black suitcase with carrying straps. Meanwhile, I'll get ready to set us down."

"I'll hurry so's to be back to read the routine for you."

While his brother was gone, Jon activated the bow retarders—after snapping off the stern tubes. Then he sent the ship into a curve that would bring them back nearer the place where he wanted to land. But only part of his mind was doing that—the rest was wondering why there had to be so much fuss and detail in landing and taking-off with a ship. Why couldn't it be fixed so one man could navigate and pilot without all this bother? It ought not to be too difficult....

Jak was soon back with the recorder, and Jon showed him how to read it. Soon they located what seemed to be the center of that strange disturbance, and with Jak's help, Jon set the ship down on the sand, fairly close to where they thought that hoped-for metal or its ore might be found.

When the two boys went into the living room, they told their mother what they had landed for, and that they were going out to look for the source of this excitement.

"Is it really necessary?" she asked anxiously. "Mr. C. didn't say anything to me about any such thing. Haven't we got fuel enough to get home on?"

"Sure, Mom," Jon hastened to explain. "But Pop thought this new stuff would be a lot more powerful than the fuel we're using. Said it ought to give us far greater cruising range with lots less storage space. If we found something of the sort, it would be a great contribution to space travel."

"That's right," Jak added, "and if we do find such a thing here, miners will soon be flocking after it, and that'll mean *beaucoup* credits for us."

"Well," doubtfully, "I guess you know best. Your father seems to be growing better, and lets me feed him, even though he hasn't ever seemed to

regain full consciousness. If you are sure this is what he'd do if awake, I suppose it is what you should do."

"Looks like a funny place for ore," Jon said as the two boys left the ship and started at a fast pace in the direction "Annie" had pointed out as the center of activity. "I'd have expected it to be in the mountains, not in a desert like this."

"Yes, I was wondering about that." The elder brother shook his head slowly. "But you can tell there's something here. What is it we're really looking for? Oh, I know it's metal or ore of some kind," he added hurriedly as he saw Jon start a retort. "What I mean is, is it ore or natural nuggets, and is it radioactive, or what?"

Jon grinned as he trotted along. "Don't really know much more than you. I know how to detect it, and I'll know it if we find it. But to tell ahead of time, I haven't the minnow of an idea."

They had actually gone less than a quarter of a mile when the heat of the sun, reflected from the hot, white, desert sand, became almost unbearable. Finally Jak stopped, wiping the pouring perspiration from his face and neck. "We can't take much of this. Better go back and get our suits."

"Yes, guess you're right." Jon was also working his handkerchief overtime. "The refrigs in them will keep us cooler, even if they're harder to walk in."

"And the suit-goggles will protect us better from the actinic rays of this sun," Jak said. "We're so close—only sixty-five or seventy millions, you said?—that the solar rays are lots stronger than those we get back on Terra, even in the deserts."

"Sure, those jungle trees protected us before, so we didn't notice them."

Their mother heard them as they returned and came to see what the trouble was. When they explained, as they were putting on their suits, she again warned them to be careful.

Then she added, somewhat hastily, "It's just a mother's instinct to keep warning her children to be careful. I know you boys always are—the fact

that you came back rather than take chances shows this. Please don't feel badly that I keep nagging at you."

"Heck, Mom, we know you aren't nagging." Jon hugged her. "If you ever quit warning us, that's when we'd really get worried."

Their suits on and the refrigerators working, the pair began retracing their steps. Jon led the way, since he was carrying the detector. They went in a decreasing spiral to locate the center, then made a beeline for that spot.

But after almost a mile the signal seemed to grow weaker, and they stopped for a conference.

"Must have passed it," Jon said over his suit-radio as his puzzled eyes studied the meters on the finder.

"Try going back thirty or forty yards to the right, then back toward the left," Jak suggested.

Soon Jon shouted and started off in a new direction, but more slowly, and Jak ran quarteringly toward him.

Inside half a mile Jon lost the beam again, and once more they quartered to find it. In narrower and narrower circles they searched.

Suddenly Jak stumbled and fell to the ground. As he started to rise, Jon heard his excited yell coming through his earphones.

6

At his brother's eager cry, Jon ran over toward where the older boy was stooping down, examining carefully something almost completely embedded in the sand. He saw Jak rise, take his shovel from the carrying straps on his suit's back, and start uncovering whatever it was he had stumbled over.

As Jon came up, he uttered an exclamation of surprise. "Why ... what ... that's a metal plate. What is it, Jak?"

"Don't ... know ... yet," the elder panted as he worked even more feverishly with his shovel. Jon quickly laid down the detector, which was clicking excitedly, to unsling his own shovel and begin digging.

In a few moments they had completely bared the metal plate, and could now see it was about ten by four feet, and hinged on one side.

"Looks like a trap door."

"Sure does. Lend a hand—let's see if we can open it."

The crack along the edge was not wide enough for the gloved fingers of their suits, so Jon inserted his shovel tip in the crack as a prize. Jak did the same and after many attempts—for it was much heavier even than they expected—they managed to lift the edge a bit.

"Can you hold it alone a sec?" Jak asked.

"Try," Jon threw his whole weight on the end of the shovel handle, while Jak quickly found a small stone and wedged it in the opening. Then, with Jon moving his shovel farther and farther back along the edge, Jak pushed ever larger stones closely behind him, and they finally managed to get the cover high enough so they were able to tip it back, but only after considerable straining and puffing.

"What do you suppose is down there?" Jon hopped about, digging at the sand with his boot tip.

"Don't know, but it has sure been a long time since anyone used this." Jak spoke slowly.

"How can you tell?"

"Because of all the sand that's sifted in here, silly. You don't think this is just a box of sand, do you? It could have been here thousands and thousands of years."

"You mean ... there were people living here then?"

The elder boy shook his head. "Maybe yes, maybe no. It could have been folks who merely visited here. Well, what do we do now?"

Jon picked up his shovel. "You're the one that's silly now. We clear it out and look."

For some time the two made the sand fly, then Jak's shovel struck metal, and feverishly the two concentrated on that spot. Another few minutes and they could see it was a large metal chest that almost filled this covered, metal-lined pit. When they finally had the top of the box completely exposed, they found its cover fastened with a simple hasp, which was quickly opened. Then they lifted this second lid.

Inside, the box was completely filled with thousands of small cubes of some sort of glistening metal. Jak started to reach for one but Jon struck his hand away.

"Listen to that detector—it has gone crazy," he yelled. "That stuff's deadly radioactive, I bet." He started to scramble out of the hole after slamming down the lid. "We get out of here, but fast. Then we talk about it."

Jak had sense enough to heed his brother's warning, and lost no time in following. Some little distance away, the two stopped to debate what they were to do.

"You know what I think?" Jon's eyes gleamed. "I think this was a fuel cache left by people who used to make trips around the galaxy, and not something left by people who once lived here."

"You're nuts. Who—and when—and why didn't they ever come to Terra, if they had space-flight? If they came this close, wouldn't they have gone there, too?"

"Not necessarily—space is so big and Sol is relatively small, you know. But maybe they did get to Earth, at that." Jon grew more thoughtful. "Remember our reading about all the strange things people reported seeing, hundreds...."

"You mean those old 'flying saucer' reports a couple of centuries ago?"

"Yes, them. And even before that, there'd been reports of strange airships and things. Why, there was one—that's almost four hundred years ago—of a Dutch sea captain who saw something in the air above the Indi Ocean and reported it in his log. Even made a sketch of it, that was almost exactly like those made later by people who said they'd seen it."

"*Mmmm*," Jak had been thinking back, "then maybe the Bible story of Ezekiel's 'wheel within a wheel' he saw in the air, was one?"

"Sure. Earth people for centuries have seen all sorts of unknown things."

"Then maybe your idea isn't so wild, after all."

"The question now is," Jon ignored the apology, "how do we take some of this stuff back to the ship, and how do we test it to see if it's fuel—or don't we?"

"That's more your line than mine. What do you suggest?"

Jon thought seriously for several minutes, then brightened. "You stay here so I can see where I'm going, and I'll go get the ship and bring it here. Then we'll try one of those cubes in the generator."

"You ... you think it's safe?"

"What would get you, out here in this desert?"

"I didn't mean that, and you know it. I meant, do you think it's safe to try this stuff that way?"

"Oh, that? Sure." Jon threw down the extra things he was carrying, and started away at a trot.

When he reached the ship and was inside the airlocks, he called to his mother as he was sitting down at the controls.

"What's the matter, Jon?" she asked as she came in and saw him working at the controls. "And where's Jak?"

"We found something out there too heavy to carry, and Jak's watching it while I bring up the ship. Strap down."

She sank into the co-pilot's seat and fastened the broad belt about her even as Jon was activating the generator and tubes. Raggedly, since he was trying to handle the controls and read the directions at the same time—it simply didn't occur to him to ask his mother to read them to him—the boy finally got the ship into the air.

"What is it you've found?" His mother could contain her curiosity no longer.

"Something we think is that new fuel Pop talked about, but it's radioactive and we didn't dare try to carry it without special equipment," he told her absently as the ship began lowering.

He maneuvered it to a bumpy landing close to his brother, whom they could now see through the port, excitedly waving his arms at them. "We think it's something some other people left here as a cache, a long, long time ago," Jon explained as he put his controls in neutral, his voice an excited squeak.

"Some other Earth people?" she asked incredulously. "You mean we aren't the first ones here, after all?"

"No, we don't think it was Terrans," he said as he unstrapped. But before he could get out of the seat, they heard the lockdoor mechanisms working, and knew Jak was coming inboard, so the two stayed in the control room. Jon answered his mother's anxious questions as best he could.

Jak soon came running in and the two boys held a quick council, almost ignoring their mother in the excitement of trying to figure a safe way of bringing some of the fuel-stuff aboard and trying it out.

But at last she made herself heard. "I think you should wait and let Mr. C. decide about this," she said with determination.

"How is he—awake?"

"About the same—still unconscious."

"Then don't you see, Mom, that there's no telling when he'll wake up, and we don't want to wait that long?"

"I still say you mustn't take the chance of blowing us all to Kingdom Come before you can have his advice and help in deciding what to do with that

new, untried and dangerous metal," she declared so firmly that they could not ignore her. "Now you listen to your mother. This is once when I'm setting my foot down. I will *not* let you do it!"

Nor could their pleas move her.

"All right, Mother," Jak finally conceded defeat with—if the truth must be told—an inner sense of relief. He, too, had been more than a little afraid of that untried stuff. But Jon had seemed so sure, while he knew very little about it. "We'll leave it here while we go set our markers on the other planets."

"Unless Pop wakes up before we're finished here," Jon added sullenly, somewhat humiliated because he felt his mother was treating him like a little boy instead of the man and scientist he now considered himself to be. "When he does, though, you'll see he'll say we should have tried it."

His mother, understanding well how he felt, but still worried over the possibilities for danger her anxious mind insisted on painting, patted his shoulder. "In that case, Son," she said softly, "I'll apologize."

"We'd better go out and shut that trap door and mark the place some way before we leave." Jak tried to lighten the tension.

"I'll take measurements of where we are, and that'll do just as well." Jon's voice still held that injured tone.

Mrs. Carver kept her voice level, but her eyes caught and held those of her younger son. "I'm sorry if I seem too stubborn about this, Jon; but I just don't like the idea of you boys trying to handle, alone, something you don't know anything about, especially since you yourselves admit that it's highly dangerous."

Jon's petulance slowly disappeared, and finally he grinned and kissed her. "You're right, as always, Mom. I'm getting too big for my coveralls. I'll calculate courses to One, and to and around the sun, and we'll let this ride until Pop wakes up."

While Jak and Mrs. Carver busied themselves at other tasks, Jon sweated over the complicated math of the new courses. He knew how important this was, especially the plan he had in mind for placing the marker in its orbit about the sun. He knew their very lives depended on the correctness of his

calculations. So he did them slowly, carefully, and checked them closely to make sure he had done them right and made no mistakes.

But when he was finished, he put the sheets of calculations in a drawer, took more paper and figured the courses over a second time. That solution he also put in the drawer, and figured it the third time, without consulting what he had done before.

When he had completed this third computation he took out the other two sets and compared the three. All came out exactly the same ... and he gasped in relief and sank back, trembling with thankfulness, in the pilot's seat. They must be right.

While Mrs. Carver and her sons were eating lunch they heard a weak call from the bunkroom, and ran in to find their father fully awake. He seemed surprised at his condition, but Jak explained swiftly what had happened, and Jon told briefly what they had done and were planning to do next.

"That's good; that's very good," he said drowsily, and before Jon could say anything about finding that new metal, his father had again sunk into sleep—or unconsciousness; not even Jak could tell which.

"Well," Jon tried to be brave about his disappointment, "I guess we'll just have to go ahead. But isn't it swell that Pop woke up fully?"

"It certainly is." Mrs. Carver had tears of joy in her eyes. "Now we know he'll soon be all right."

The trip to Planet One—"Tad"—was neither long nor eventful, once they got started. They found, as expected, that the small world—smaller than Sol's Mercury—was so close to the sun that it was fearfully hot, even on the equator, or "intermediate" zone.

Despite the refrigerators on the ship, it was becoming hot inside, and all stripped as far as decency allowed. The planet had no real atmosphere, but many of the metals—indeed, the very rocks, themselves—were so largely molten, especially on the eternally sunward side, that there was a fog of gasses about the surface. These gaseous emanations were in a state of

motion much like that of Earthly cyclones, constantly swirling and blowing with terrific velocity.

The boys carefully examined their spectro-analyzer, but "Annie" showed none of that strange fuel-metal they were so keen to locate in its natural state. "Maybe we found all there is here," Jak suggested.

"Perhaps, but somehow I can't feel that way." Jon's voice was worried. "I must have slipped somewhere. Don't see how just one boxful could have shown up so clearly from as many light years away as we first discovered it."

Despite the conditions the young planet mappers found here on One, the Colonial law required that a sending beacon be set up on ALL planets, or else in an orbit about them. They decided to place theirs on top of one of the highest of the small mountains that comprised the twilight zone.

Jon made up the tape for this planet's signal-marker, while Jak brought it from the storeroom. When the tape was installed and running, the sender was placed in the lock between the inner and outer doors, and the boys returned to the control room.

Jon directed the ship toward the range of mountains and when he neared them Jak—from his co-pilot's seat—worked the remote controls and the outer lockdoor swung open. Then he activated the "distant hands"—the handling mechanism that was an integral part of the airlock's equipment, for handling materials into and out of the ship.

Watching through his special visiplat—really a sort of two-way television—Jak made the grips pick up the signal-sender box, ready to deposit it on the hard, hot ground outside when Jon would swoop down over the pre-selected mountaintop.

"Move it outside," Jon called, and Jak did so. "Set it down." Jon yelled, and as soon as he was sure Jak had placed the sender solidly, sent the space-yacht rising higher and away from the planet. Then Jak closed the outer door; turned in his co-pilot's seat, and tuned in their receiver. Soon they caught the message and knew everything was jetting fair.

"Nice going, Owl," Jon applauded.

"Aw, you're just saying that because it's true," Jak grunted, and Jon turned his attention once more to his controls and the new course he had plotted for their swing around the sun of this system, now less than thirty million miles away.

"How close d'you go?" Jak was more interested than fearful, having confidence in his brother's skill.

"We have to follow a course so that when the sender is dumped, it will take up a closed orbit—the more nearly circular the better—around the sun. Also, we'll have to have speed enough so we won't get fried to a crisp at the near-point, which figures to be about ten million miles."

"Isn't that pretty close?" their mother, who had slipped into the control room quietly just after Jak had placed the sender, tried not to sound too frightened.

"Relax, Mother, the kid knows what he's doing," Jak tried to calm her.

"I've figured this three times, Mom," Jon said earnestly. "Got the same answer each time, so I *know* we can do it."

"Well," still doubtfully, "I guess you do know what you are doing, but that seems awfully close." She struggled with herself and finally managed a weak smile. "I promised to let you boys make the decisions. I'll go lie down in my bunk so I won't know what's happening until it's all over."

"You do that, Mother. I'm not worrying. Jon really knows his stuff," Jak assured her brightly. But as soon as she had left the control room, he turned worried eyes to his brother. "I ... I hope you actually do, Chubby." His voice quavered a bit.

Jon grinned mockingly. "There's one sure thing. If I'm wrong, we'll never know it. But I've studied this a lot since I knew it was up to me. I know the technique and, as I said before, I've computed our course three times and come up with the same figures each time. And we have to set it as close as possible. Now, either hit your bunk or set your seat to recline. We're up to better than two G's already, and I'm building to five."

"Yes, I feel us getting heavier. I'll stay with you." Jak made sure his straps were in place, then tilted his seat.

Jon cranked his own to recline, the control panel automatically slanting to keep it in the same relative position. His arms were resting on movable slides, and the controls he would have to manipulate on this dangerous orbit were all beneath his hands and fingers.

Closer and closer they drove to the sun with ever mounting speed. Their gallant little ship's refrigerators were full on; all shutters in place. Their only view of the outside was through one visiplat whose aperture was closed until only a tiny slit was open. But it was enough, although Jon was forced to keep building up layer after layer of protective, colored plastic to make the intense, blinding light of the swiftly approaching sun bearable.

Clearly visible now were the tremendous streamers of matter the sun was throwing up as prominences. Jon was able to see huge sunspots occurring here and there about the surface of that mighty furnace—tremendous cyclonic storms of atomic disintegration. So interested was he in this first close view of a sun that he almost forgot the reason for this dangerous trip.

Almost—but not quite, for his mind was well-trained to remember the things that had to be recalled, young as he was. So his eyes glanced often at the distance gauge. Soon he yelled at Jak, "Get ready to throw out the sender."

Jak struggled to place his hands on the controls, a thing he had not had the foresight to do before Jon started building up that tremendous acceleration. His muscles strained. Sweat broke out on him even worse than that the heat from the sun brought. His breathing became gasps. There seemed to be a constricting band about his chest. His eyes felt as though the balls were being pushed down into his head. He just couldn't possibly move a muscle under this terrible pressure.

Still he exerted every force of will and of muscle. Slowly, painfully, he stretched out his fingertips a fraction of an inch. He dug them into the fabric of the arm rest and pulled the palm of his hand along. Then he forced the rest of his arm to follow his fingers and hand. Over and over, straining to do what had to be done. Then victory at last—his hand and arm were on the sliding arms. Now it was easier, and soon his fingers were on the controls.

"S-say when," he panted then.

"Open the outer door now ... we're almost there," Jon commanded, watching his controls intently. "We're going ... so fast ... won't have ... much time."

"You're sure ... sender'll keep ... correct orbit?"

"Sure," Jon's voice was confident. "If we don't dump ... exactly on zero ... it'll just change shape ... of orbit a little ... that's all."

"Door open," Jak reported a moment later.

"Lift sender, but don't eject yet."

"Right."

More minutes while the heat increased, and even through that tiny aperture and the covering shields, the blinding light was coming in so fiercely Jon was tempted to close it entirely. Then, with a snort of disgust at his stupidity, he did close it—and breathed a sigh of relief as that piercing beam died. He didn't need to see. There was no reason to look. Even if there was, it was too late now to do anything about it. If his calculations were correct, the ship would get away safely. If his figures were wrong ... he shuddered. Well, they'd never know it, that was for sure.

He made himself forget that dire possibility and kept his eyes glued to his indicators.

"Almost there."

"Ready." Jak tensed his hand and fingers above the controls. He hoped he could do it when the time came. But this awful heat ... this horrible acceleration pressure....

"Drop it!" Jon yelled suddenly.

Jak tensed hand and fingers and tried to depress the button. It seemed he couldn't move. He gritted his teeth, and again called upon his inner strength, his will. From that hidden depth he found that extra measure of energy necessary to curve his fingertip downward.

His eyes, peering into the shielded intercom visiplat, saw those distant hands—the servo-mechanism in the lock—swing the box out through the opened doorway. When he could no longer see it, because of the angle at which his visiplat was set, he touched and depressed the second button.

Now, if the mechanism was still functioning in spite of that terrific heat, its arms were opening and the box slipping away.

He withdrew the handling arms, and as they came into sight again he saw with satisfaction that they were empty. He locked them into their cradle, then closed the outer lockdoor.

"Done," he reported thankfully ... and let himself go. Unconsciousness claimed him at once. Why suffer, had been his thought, when he could so easily sleep until this intolerable pressure was gone.

So quickly did he slip away he did not realize that Jon, too, after a final quick glance at his board, and knowing that everything necessary had been done, had also relaxed into unconsciousness. Did not know, or care, that their ship was now speeding around and away from the sun. Did not realize that all four of the Carvers were now unconscious.

But their blackout did not last too long. In a few hours, during which the auto-pilot took them smoothly and accurately away from that titanic furnace, safety distance was attained and the frightful acceleration began to ease.

By the time they were traveling at a little less than two gravities, Jon stirred. His memory cells began functioning once more, and slowly he awakened. As soon as he realized where he was, and why, he glanced at his various telltales.

"We made it!" he yelled triumphantly. Then, as he heard no reply from his brother, he quickly raised his seat to upright, and turned to look at Jak. The latter was still lying down, his face white and strained.

Quickly, anxiously, Jon released himself and sprang across to his brother's side. He rubbed Jak's wrists and temples. Soon the flush of returning blood showed, and the elder sighed and opened his eyes.

"We made it!" Jon cried again as he pushed Jak's seat into erect. "Everything went off shark-y."

Jak struggled into full consciousness, then began loosening his straps. "Mother and Father?" he exclaimed. "Did they come through all right?"

7

At Jak's question, Jon started. "Haven't looked yet. Let's go see."

The two raced into the living room and into their parents' bunkroom. Mrs. Carver was just opening her eyes, and seeing the boys' anxious looks, struggled to sit up. They helped her, and Jak turned quickly to look at his father. To his relief, the latter's pulse was no weaker, and his breathing was regular.

"We got through all safely," Jon assured his mother, and she threw her arms about him and broke into tears. "Hey, no need of crying now; it's all over."

"I know." She reached down for a corner of her apron and wiped her eyes. "Just relief, I guess. Is Mr. C. all right?" she asked Jak.

"Didn't seem to hurt him a bit."

Indeed, just then there was a mutter from Mr. Carver's lips, and his eyelids fluttered open.

The three gathered closely beside him, and were tremendously heartened at the look of sane awareness in his eyes.

"Hullo?" as though surprised to find himself in bed and the others gathered about him. "Did I oversleep?"

Jak reached out and took his father's hand. "No, Father. You've been a little ill and unconscious, that's all. But you're almost well now. A bit more rest and you'll be all OK."

The invalid looked surprised, then doubtfully at his wife, who quickly stooped and kissed him. "Jak's right, Mr. C. You get some more sleep so you'll get strong quicker."

Dutifully he closed his eyes and immediately his regular breathing told the three he was asleep once more.

Quietly Jak drew the others out of the bunkroom and closed the door. Then his eyes shone and he grabbed his mother and danced her about, while Jon

"tried to get into the act."

"He's almost well; he's almost well!" Jak chanted.

Jon yelled in honest praise. "You did a grand job, Owl." But his voice broke into a boyish treble with the excitement.

After several minutes of jubilation, Jon went back into the control room and began figuring their course to Planet Three.

He turned on the receiver and pointed the directional antenna. Soon the broadcast of their solar signal came in. This one about the sun had most worried him, but he could read it clearly: "This solar system was first discovered and charted by Tad Carver of Terra, on fourteenth January, 2136. It has been named 'Carveria,' and the five planets and seven satellites are being charted and named. Details will be filed with the Terran Colonial Board."

Finally Jon finished his astrogation, then went back into the living quarters. "Ready to set course to Three, Folks. Strap down while I change course."

"How long'll it take?"

"Just under a couple of days at two G's."

"Ouch! Do we have to go that fast?" Jak complained.

"You want to get there, don't you?" Jon turned away indifferently, while Mrs. Carver smiled at Jak and shrugged.

During the balance of that "day" Jon stayed in the control room. When either of the others looked in, he was studying intently. Right after breakfast the next morning he put in a long session at the computer and his drawing board, then after lunch went into the storeroom. After a while he came out with his arms filled with wires, cells, relays and other oddments, which he carried into the control room.

The others, busy with their own work and chores, paid no special attention to what Jon was doing. Seeing him busy like this had become so commonplace they seldom bothered even asking what he was doing when he did not volunteer the information.

But as they approached Planet Three early the following morning, under negative acceleration, all three were in the control room, peering intently

into the visiplates.

What would they find there? Would there be people of some sort? Cities? Jungles, deserts, ice fields?

All three minds were busy with such conjectures as they came closer in. Their instruments had already told them Three possessed an atmosphere containing water vapor, so they knew it could not be entirely untenable, unless the air contained poisonous gases. But what real conditions they would discover there remained to be seen.

They had already found, charted and photographed the two small moons that circled the planet. One of these was fairly large—about nine hundred miles in diameter, and the other much smaller, about a hundred and fifty. Three, itself, was about five thousand miles through.

"There are clouds down there," Jak called suddenly as they approached ever nearer at constantly decreasing speed.

"Yes, I see them."

"And there's a big ocean!" Their mother was equally excited.

"Three's only about thirty million farther away than Two, although on the opposite side of the sun right now. So there shouldn't be too much difference, except Three'll be colder," Jon stated. "We're about a hundred miles up now, so I'm throwing us into a descending spiral."

"There's a big mountain range, and some of the peaks are snow-covered," Jak called out a few minutes later.

"I see them. We're down to about twenty miles now, and I'm setting a crisscross orbit for two or three revolutions to get a better view and take our first pictures. Mom, if you can tear yourself away, I'm hungry."

She stepped back from the screen, laughing. "You're always hungry." Then she glanced at her wrist-chronom and gasped in dismay. "No wonder—it's over an hour past lunchtime!"

"We'll yell if anything especially interesting shows up," Jak called as she was leaving.

By circling the planet from east to west they kept to the daylight side most of the time, and as the hours passed they were able to get most of their

pictures and reports on the geography, climate and other conditions. Their spectro-analyzer showed considerable mineral deposits in many of the places over which they passed.

They saw plenty of vegetation and Jon exclaimed about its coloring.

"Must be fall here," Jak explained. "Unless, of course, those plants don't contain chlorophyll, which I doubt."

But nowhere did they see anything that looked like the works of intelligent beings. Like Planet Two, there was no sign of people anywhere.

When they became so tired they could no longer keep awake, Jon set the ship into a higher, safer orbit, and they all went to bed. Their father had awakened only once during the day, and then only for a few minutes, nor had his wife allowed him to talk, greatly as the boys, especially, desired it.

After breakfast the next morning Jon maneuvered the ship down closer to the surface and they completed exploring the planet, taking their pictures and recordings. Jak made tests and reported the atmosphere not poisonous, although so scant they would have to wear suits most of the time when outdoors.

"It's lots better than Mars, but not near as dense as Terra or Two back there," he told Jon. "Temp's below freezing, but I imagine it'll get warmer when the sun's nearer noon here."

"Humans can adapt themselves to living here, then." Jon's voice was joyful. "They've already colonized planets worse than this, as far as temperature and air are concerned."

"Yes, the human animal seems to be marvelously adaptable to almost any conditions not actually poisonous," Jak said admiringly. "There's even a colony of people from the High Andes of Souamer living on Mars now, without domes."

"They could transport those Andean Indians to Mars direct because they were used to living in the rarefied atmosphere of the high mountains, eh?"

"That's right. Those Indians would have suffocated at sea level back on Terra. Indeed, they seldom went down the mountains below ten thousand feet because of the discomfort. On Mars, they had some difficulty at first, but I understand the second generation born there are perfectly at home."

Jon's blue eyes had been watching his detectors, even while his ears had been listening to Jak's explanations. So far he had not discovered any of that strange fuel-metal—if it was fuel—they had found on Two. He spoke of this now to his brother. "Wonder if those people didn't leave any caches here on Three, or what?"

"Maybe they didn't like cold weather." Jak grinned. "More likely, though, either we haven't come close enough to detect it, or else they may only have made a cache on one planet in a system."

"That's probably it. I've been watching for it all the way in, and 'Annie' didn't chirp at all. Well, do we land and see what the joint is like?"

"Don't know about you, Chubby, but I sure want to. How about closer to the equator? Ought to be warmer there, and more comfortable. I want to study that plant life."

"OK by me—if you don't try to load the boat with your specimens." Jon laughed, and Jak joined in sheepishly.

"I promise not to go hog-wild like I did last time."

"Going to land, Mom. Strap down," Jon called into the intercom.

Jak reached for the sheet of landing instructions, but Jon shook his head. "Don't think we'll need those. Tighten your belt, here we go."

"Hey, what gives?" Jak's eyes widened as he saw his brother throw in one switch and then take his hands off the controls, although his eyes were alertly watching his many dials and lights, and his body was tensely ready for emergencies.

Jon did not answer, and Jak watched in the plate as the ground below appeared to rush closer each second. It almost seemed to him they were not slowing as fast as was usual on landings, but he was not unduly worried—he trusted Jon to know what he was doing ... even if he didn't!

But apparently Jon was not satisfied—for when the ship was only a few hundred yards above ground, he suddenly worked frantically at his controls, and the nose of the little yacht came up sharply and she zoomed into the upper air with a push from her stern tubes.

Thirty-some miles up, Jon set the ship into a circular orbit, then got out of his pilot's seat and began tinkering with some of the controls.

"What's wrong?" Jak asked. "How come you went down without following the manual, and then came up again?"

But Jon was tight-lipped and uncommunicative. Their mother's voice came over the intercom, asking why they had not landed, and Jon answered her question.

"Just a slight miscalculation of height, Mom, so I came up to try again," he answered. "Stay strapped down—I'll be going down again in a minute."

Soon he was back in his seat, scanning his various instruments, then again Jak saw him throw that one switch. Once more the little ship began settling toward the ground beneath, without any handling of the controls.

This time the landing was smooth, soft and even. Still without any move by Jon, Jak could feel the various generators and engines stop, the landing props go down, and finally the board show a clear green "neutral" condition.

"How ... how come?" Jak gasped, and this time Jon chose to answer.

"Just rigged up a series of photo-electric cells and relays, so now all I have to do is throw one switch and it takes care of all the little details of landing, just as this other one does of take-offs." Jon tried to make it sound like an offhand comment. "My height-to-descent-speed ratio was off a bit, and that was what I had to fix."

"But ... but that's something brand new, isn't it? I never heard of such a thing before." Jak still could hardly believe what he had just witnessed.

"Oh, it wasn't so much a much." Jon looked down as he guessed that his brother would soon realize what a remarkable thing he had done.

"Boy, you're good!" Jak applauded, and as their mother came into the control room, he almost shouted, "Jon's gone and...."

"Landed so we could go out a bit and make a fuller report in our log," Jon cut in sharply, with a warning look at his senior. "How's Pop?"

"Been moving about some, although he hasn't wakened fully yet today. His breathing is much easier. He still makes noises—but then, he always did sort of snore when he slept."

The boys went with her into the bunkroom to look at their father before they started outside. There was a flush of color on his skin, although it was paler than its usual state. When Jak examined the side of his patient's head he could see that it was practically healed. Also, the broken leg seemed in fine shape, as seen through the clear plastic of the cast.

"He'll be waking up for good any day now, I'm sure," he said thankfully.

"Gosh, I hope so," Jon said. "I feel like a fish out of water without my Pop."

"You seem to be doing fine, anyway," his mother cheered him. "And so is Jak," she hastened to add, fearful her elder son might think her prejudiced.

The boys went out to get ready for their outside trip.

"What's the big idea, not letting me tell Mother about your new dinkus for landings and take-offs?" Jak railed.

"Aw, she wouldn't understand it, and it'd worry her for fear it wouldn't work." Jon was clearly uncomfortable about the praise he could not help seeing in his brother's eyes. "I'll tell Pop, when he wakes up. Come on, I'll race you into our suits."

The boys donned their spacesuits, and examined each other to make sure they were "tight." They saw to it that their guns and bandoliers were fully loaded; that they had with them what tools and other equipment they felt might be needed. Then they opened the lockdoors and went outside.

They started off in a predetermined direction, having made plans to go about five miles. Then they would swing in a circle around the ship. If they saw anything they thought exceptionally interesting, they would make short side trips, and if necessary, complete their circle on another day. In any event, they had promised their mother to be back by dark.

The first leg of their journey was completed without any excitement, although Jak was continually finding new plants he wanted to collect for

future study.

"Nix, Owl, not this trip," Jon kept protesting. "You promised, remember?"

"Oh, all right, killjoy. But there's so much here I want to find out about."

"Yes, so much you couldn't even make a dent in it in a lifetime. Want us to leave you behind to do it?"

"You just try that, and I'll knock your teeth loose."

"You and what platoon of space marines?" Jon jeered good-naturedly, knowing that with his greater size and strength Jak could not make good his threat—even if he had really wanted to.

Bickering in more or less friendly fashion, they covered their first five miles, then turned to the left and started circling. About a mile of this and they entered a fairly large wood. The trees here were so strange the boys looked about them with a growing excitement.

Unconsciously, they drew closer together, and finally Jon voiced what was in both their minds.

"I'm beginning to get scared, Jak. Ought we to keep trying to go this way?"

"I'm not sure," slowly. "I'm getting a feeling there's something here that seems to be unfriendly—perhaps dangerous. But there isn't a thing we can see—not even animal life."

"Maybe it's only because this forest is so unlike either those on Terra or the ones on Two." But Jon gripped his rifle more tightly, and his thumb unlocked the safety catch.

The two boys finally came to a dead halt in a small clearing perhaps a hundred feet in diameter, and examined more closely the few trees and bushes about them. The ground on which they were now standing was bare and sandy, although beneath the trees it had been more like black loam.

"This sand must be why there's practically no vegetation here," Jak said. He dug into the ground a bit, and found it to be sand as deep as he went.

Rising, he looked even more closely at the trees about the edge of the clearing.

Not one of them was the straight, slim type with which they were familiar. These were ungainly and appeared stunted, although many were actually close to thirty feet tall. Even so, they looked too large in diameter for their height. None of them had more than five or six twigless, leafless limbs, and those were almost as large in diameter as the trunks from which they grew. These branches twisted and curved, although in most cases the curve was upward, so that the leafless limbs often ended at a higher point than the main trunk of the tree.

Suddenly Jak began laughing—but with a high-pitched, mirthless laughter. As Jon looked at him in surprise, the elder tried to calm himself.

"I know what makes them look so scary," he finally said between gasps. "It's that weird look. But remember those pictures we've seen of the Zona and Newmex deserts in Noramer, back home? Remember the Josha trees growing there? They're as alien-looking as anything on Terra, and these look something like them."

Jon, too, began grinning as remembrance came. "'Most let ourselves get scared over nothing, didn't we? Come on, let's travel." And he started forward.

Yet the strangeness persisted, and before the boys had passed through the fringe of those tortured trees on the other side of that wood they started to get that queazy feeling again, in spite of their realization of what caused it. They began going more slowly, cautiously; ready for a quick turn and run, yet both inwardly hating themselves for the fear, and each determined not to let the other know he was afraid.

But it was with a distinct sense of relief that they saw the end of that forest ahead of them. Unconsciously they hurried their steps until they were almost trotting.

For the balance of their trip Jon was strangely acquiescent as Jak became more and more engrossed in the strange plant life of this world Three. He knew that this was Jak's dish, and he was perfectly willing to defer to the elder's knowledge and desire to learn. His main concern was to keep his brother from overloading himself with specimens, or from loitering too much.

Jak had been especially studying the soil here, Jon noticed, and finally he asked about it. "Notice one peculiar thing about this planet?"

"What's on your mind?"

"The total absence, as far as we've seen, of any sort or type of protoplasmic life," Jak reported.

"Hey, that's right, though I hadn't thought of it before. Our examination from the air, I remember now, showed no animals, birds or people. Plenty of vegetation, though."

"Yes, it has everything in that line. I wonder, though...." He paused, and he grew thoughtful.

"Wonder what?"

"How those plants can grow, without any worms or ants or anything to loosen and irrigate the soil, and no animals or birds to make fertilizers, or bees or butterflies or anything to carry pollen?"

Jon shrugged. "Wouldn't have the foggiest. That's your line, not mine. But they must do it some way—there's sure lots of plant life here."

But Jak was still shaking his head in puzzlement as they finally returned to the ship.

It was quite dark outside when the boys went into the control room after dinner. Jon went over to the window-ports, while Jak began working with his plant specimens.

"Jak, come here," Jon called after a moment or two. The elder, prompted by the curious urgency in his brother's voice, left his specimens and ran to the other's side.

"What're those things?"

Jak stared through the port in amazement. Outside, drifting across the clearing, were nearly a dozen large, spherical things like ghostly white balloons. They must have been almost a yard in diameter, and by straining their eyes the boys could see tentacles or tendrils of some sort depending from the bottom surfaces.

"Gosh, never saw anything like those. Let's go out and see what they are."

"Let's not and say we did," Jon retorted. "I want to find out more about them first." He went over to the control panel and switched on the searchlight, as well as the pilot's visi-screen. Looking into the latter, he was able to direct the light so it shone on a couple of the floating balls.

Jak was studying the plants—for so he believed them to be—more carefully, now that they were lighted. But after a moment he yelped excitedly, "Hey, they're deflating. Must be the light does it."

Jon was watching them in his screen. "Yes, I see now. What causes it?"

"I don't know," Jak answered sadly and absently. "But I sure want to know. How's about covering me while I go out and see if I can get one?"

"Well, maybe in your suit you'd be safe."

Once suited up, Jak went outside and across the short distance to where the balls seems to be slightly closer together. He tried first one way and then another to catch one, but at his lightest touch they burst and deflated. After several unsuccessful attempts, though, he called excitedly through his suit-sender.

"Jon, you read me?"

"Coming through."

"I'm going to try fanning one toward the air-analyzer. I want to see if we can get an idea of what's inside. I've got a screwy hunch."

"Right, I'll switch the light away from them, up into the air."

Carefully Jak herded one of the globes near the ship, and was finally successful in getting it close to the hull-vent of the air-analyzer. When it was almost touching the ship's side, he reached out and touched it, and it promptly broke.

"Get anything?" he yelled.

"Yes, gas of some sort. Taking the reading now. It seems to be mostly nitrogen."

"Hah, that's it, then! I'm coming in."

When Jak was back inside, Jon helped him remove his helmet, then demanded curiously, "What's it all about, Owl?"

"I'm not positive, of course, but I bet those things take the place of bees for pollinating, and also furnish the fertilizer for the ground when they burst and their nitrogen gets into the soil some way."

8

Later that evening Jon Carver sat for nearly an hour, studying intently from one of his reelbooks, and the frown on his face grew deeper and deeper.

Jak had been working over their father. He had given him a careful sponge bath, then fed him another intravenous dosage of the combined liquid protein, salt, sugar and glucose. Even though their mother had been able to spoon-feed her husband small amounts of food each day, the young hoped-to-be doctor felt additional nourishment was necessary.

When he finished his task and started to seek a comfortable seat in the living quarters of the space yacht, to relax with a little reading of his own, he noticed his brother's intent look and worried face.

"What's the matter, Jon?"

"Eh?" The younger boy looked up, startled, from his deep study. Then, as Jak repeated the question, he answered unhappily, "I just don't know enough, Owl. I can't figure out why Pop found such strong spectroscopic lines of that new element while we were billions of miles away, and yet we can't find any traces of it anywhere on these planets, except what we found in that cache."

"Maybe it's in the sun."

"I tried that when we were out there, but 'Annie' didn't even peep."

The elder brother studied the problem a moment.

"Could it be so strong that even the little bit we found would have shown those lines?"

"Maybe," doubtfully, "but I don't think so. Tomorrow morning, when the sun comes up, I'm going to try to take a new reading from here. I tried to read Two, but couldn't get anything. However, I'm not so hot with the regular spectrograph, and that's why I'm boning up on it."

"Is this important?" Their mother had laid her sewing in her lap to listen to them, trying to follow and understand what her sons were talking about.

"Pop thought it was, Mom," Jon explained. "One of the things men have been looking for ever since they first started dreaming of rockets and spaceships, was the best possible fuel. We knew the one we're using now isn't the ultimate, but it's the best they've been able to get so far. Pop thought perhaps this new stuff might be it—if we could find it, and *if* we could learn how to use it."

"Why can't we use it if you find it?" Jak wanted to know.

"There are so many problems. Maybe it would be so radioactive we wouldn't be able to handle it or keep it in the storage bins without endangering the people on the ship. Maybe the exciters and convertors wouldn't handle it without a lot of new experimenting and new designs we wouldn't have the scientific or technical know-how to make. Or it might be that instead of getting a steady stream of power as we do with our present activated-copper fuel, the stuff would want to blow up all at once. If the metal's as powerful as I think it is, it might cause an explosion that would make man's biggest H- or C-bomb look like a firecracker."

"Then don't you go experimenting with it and blow us all up," his mother said sharply.

Jon grinned at her. "You needn't worry about that, Mom, now that I've had a chance to learn how little I know. Although I would've gone off half-cocked that day you stopped me—for which I'm grateful, even though I was sore at you for a while then. But I'm sure going to study it as soon as we get the other markers set and can get back to Two."

"By that time Father will be well again," Jak said.

"Isn't it wonderful that he really is coming around all right? Seems to be taking an awful long time for him to recover fully, though."

"I'm sure he'll be his own keen self again soon ... although he'll have to stay in bed until that leg is strong enough to stand on again."

"Well, let's hit the sack, so we can get a good start in the morning. 'Night, Mom."

During their journeys over the surface of Planet Three the boys conscientiously tended the machines and recorders that gave them the data on land and water conditions, the proportions of each, the approximate amounts of metallic ores their analyzers showed, the information on weather, temperature and humidity. They took numerous pictures as required by law—their mother often helping in this, after Jak had taught her how to operate the cameras. These pictures Jak developed and printed as he had time, and mounted them in their data book for the Colonial Board to study when they got back. They also mapped and recorded the size and distances of Three's two moons.

Jak named these "Zinnia" and "Begonia," much to Jon's sarcastic and openly-expressed derision.

"This'll make a swell home for people who like cold weather." Jak tried to change the subject.

"Yes, just as Two will for those who like it hotter." Jon's eyes shone. "Pop sure picked a winner when he decided to explore this system. Even with just these two worlds he has a prize."

"If they accept our work as proof. Wonder what the fourth planet will be like?" Jak continued in a different tone.

"Cold. Lots colder, probably, than Mars."

"Then it won't do us any good?"

"Depends on what's on it in the way of metals that can be mined. Maybe we'll find something there. Might be natural gems or jewels, too."

"And anyway, cold never stopped man."

"That's right," Jon said admiringly. "They have mines on Pluto, even—although they're mostly worked by automatics while the men stay warm in their bubble-cities."

As the *Star Rover* approached closer to the distant, smaller planet they had named "Jon," their instruments showed it to have a diameter of about 4400 miles, and a density of about 4.6, a little lighter than Terra. This meant the gravity would be a bit weaker, and they would weigh less than on their home planet. Four was almost a quarter of a billion miles from the sun, and would be very cold, as Jon had said.

While their ship drove in closer, the boys' mother came into the control room. All three Carvers stared excitedly into their visiplates, watching their rapid approach to this new world. Would they find anything of value there, or was it simply a barren wasteland of ice and frozen air and rocks, far too cold and forbidding for men even to bother trying to explore it?

When Jak, eyes still glued to the telescopic sights of his spectro-analyzer, voiced something of this, Jon drawled, "You know better than that, Owl. We said just yesterday that there's no place, no matter how bad, that man won't explore to see if there's anything he can possibly use. They'll follow us here, don't worry."

After cruising about the surface for some time, recording their data and taking the needed pictures, they saw a fairly level valley, ice-covered and bare, and Jon set the ship down there. By now he was becoming an expert astrogator and pilot, and with his new controls they could hardly feel the jar of the ship's landing.

"How's the temp outside?"

Jak was examining the gauges. "About a hundred below, and not a bit of moisture, naturally. Going to try going out?"

"I don't...." their mother started to speak against it, but made herself stop. Her boys were showing such resourcefulness and unexpected habits of caution that she felt she must let them decide things for themselves, even though her motherly instinct was always to hold them back from possible dangers.

"Sure we're going out for a bit," Jon answered his brother, then faced their mother. "It'll be all OK, Mom," he said affectionately. "We'll wear our suits, of course, with the heaters on. We won't go far, because the moment we feel any cold we'll run back. But I want to see what it's like out there, and if there's any sort of life. We're supposed to report...."

"Life? Here?" incredulously.

It was Jak who answered this. "Sure, Mother, there can be life-forms anywhere. Oh, not necessarily nor even probably anything we know on Terra. But there should be some sort of moss or lichen in the plant line."

"Yes, it has been learned from experience there's some sort of life almost everywhere," Jon chimed in.

"Even though most of it's so different from the basic protoplasm-type we're used to that it's hard to realize it's really life at all," Jak continued. "But then, remember back on Terra, the vast difference between animal and vegetable life—so totally unlike each other. I second Jon's plan to go out. I'd really like to see what's out there."

She sighed as if in recognition of the fact that these boys of hers were fast becoming reliable, self-sufficient men. They were not her babies any more. She was proud, of course—but she couldn't resist the motherly impulse to warn, "Well, be careful, anyway."

"Sure, we will."

Jon locked all the controls in neutral, and the two boys went to put on their suits. Knowing, as they did, the vital necessity of making sure they were "tight" and fully equipped, they examined and inspected their own and each other's spacesuits carefully before they opened the inner lockdoor.

Once outside, they stood on the icy ground for several minutes to make sure their heaters were working capably enough to keep them—and especially their feet—warm. Finding they were as completely comfortable as anyone ever can be inside that sort of a suit, they started off across the frozen plain, headed for the near-distant hills on the side of the valley closest to the ship.

Jak examined the ground about them intently as they walked, hoping to find some sort of plant life, while Jon kept his eyes mostly on the portable analyzer he carried, hoping they might discover valuable deposits of inorganics. Was there any of that unknown fuel-metal here, he wondered anxiously. Their big analyzer had not shown it as they were coming in on the survey or landing spiral, but that did not necessarily mean the portable wouldn't show it on closer approach, or that there might not be some on a portion of the surface they had not yet covered from above.

Their trips about and above the surface had, however, shown traces of iron, manganese, gold, silver, copper and several other metals, although not strongly enough to indicate great deposits. But Jon knew experience had shown over the years that one of the inefficiencies of such analyzers was that they would not show the *depth* of a deposit. Many times, when only a

slight trace had been detected while flying above the surface, prospectors on the ground had found veritable bonanzas, once they started mining.

Even though the gravity was about eight per cent lighter than on Terra, the boys found walking not too easy. The terrain was mostly rough, although there were many spots of slick, glare ice. Too, there were many hillocks, and cracks and crevasses between the slippery places. So, even though they had added caulks to their metallic suit-boots, walking was unsafe and hard. By the time they reached the base of the first low hills they were winded and glad to rest a few minutes.

"Not a thing so far," Jak panted into his suit-mike. "I can't see even a bit of color—just this white glare."

"'Annie' hasn't let out a peep, either. Guess this is a dead 'un all right."

"At least this district looks it."

"Let's climb a ways, and if we don't find anything there, go back to the ship and try somewhere else."

"I'll buy a chunk of that."

They started up the hill before them. The climbing was difficult because of the ice and because in most places the side of the hill was not a gradual slope, but a starkly steep climb. It was evident there had been no gradual "weathering" here, to produce rounded edges and rolling slopes, although there were occasional smooth places. These, though, the boys knew could not be climbed at all without special equipment which they did not carry.

"This isn't frozen-water ice, is it?" Jon asked as they panted upward.

"No, silly. There can be no water vapor here, any more than there is on Neptune or Pluto back home. This is mostly frozen carbon dioxide."

"Well, it's just as cold and just as hard to climb as polar ice."

They climbed the quarter mile to the crest of the first hill and peered eagerly over its top. In front and slightly below was another valley—not as deep as the one in which their ship lay, but even larger. From their higher position the floor of this new valley seemed quite smooth.

"But that can be just an optical illusion," Jak answered Jon's statement, adding, "the glare of white would make it look smoother from a height."

Jon ignored the tone of superiority. "Good thing our suits have tinted lenses. Do we go down?"

"Natch." Jak had already started. "Off there to the right and part way down are some darker places. I want to look at them."

"Could lichens grow here?"

"Some could, possibly, though not exactly like the kind we'd find on Earth. If there's life here, it's probably a type that can convert energy directly from the elements in the ground or ice, instead of using photo-synthesis or other methods of obtaining nourishment we know about."

Half-sliding, half-climbing, they made their difficult way to the little patch of gray-greenness, which Jak examined with growing delight.

"Hey, that's gneiss."

"What's nice about ... Oh!" Jon grew red-faced at having been caught that way. "You and your education!" he snorted.

"See how brittle it is," Jak ignored the interruption as he touched a stem, only to have it snap off like a slim glass fibre. "Can't tell without a more thorough microscopic examination, but I'll bet this is some sort of silicon-based life—crystalline instead of being like the gneissic rocks back home."

Jon, meanwhile, had been surveying the valley with his binoculars. Suddenly he gave a gasp, and focussed his glasses more steadily on something that had caught his eye.

For some minutes he studied it, then called excitedly, "Hey, Owl, give a look over there. See, beside that spire of rock," he pointed as his brother rose and unlimbered his own pair of binoculars. "There's movement of some sort there, though it's very, very slow, on that sort of pyramid a yard or so high."

For long moments the two studied the spot through their high-powered glasses, then Jak said slowly but with mounting excitement, "I think you're right, Chubby, and that we've got to see."

In their excitement, the two started off faster and more carelessly than was safe. They found out that fact when both, almost at the same time, lost their footing and fell, coasting down the remainder of the hill. Faster and faster

they slid, shaken and becoming bruised, although luckily neither broke any bones.

At the bottom they picked themselves up and started on again. Both walked more gingerly now, and Jak limped a bit from a twisted ankle. Yet they were so eager to see what this strange movement might be, they soon forgot their bruises and hurried once more.

It was a good half mile across the valley floor to their destination. But there, sure enough, they found *life*!

Strange, unearthly life it was, but they soon discovered that it had reproduction, growth and movement—the three main criteria of life-forms.

"Crystalline, by golly!" Jon yelled.

Jak was squatting beside the growing thing. It was somewhat pyramidal, yet the sides were not smooth. Rather, they were many-faceted, like the pieces of rock crystal with which the boys were familiar. It was a grayish-white color—with just enough of the gray in it so it had been visible from a distance, against the white background. But now, as the boys were on the sunward side of one of the pyramids, for there were many of them about, they could see that the light reflected from it was kaleidoscopic coloration at times.

Jak reached out a gloved hand and rapped on the pyramid ... and it gave forth a tinkling sound, then collapsed into a thousand tiny shards.

"You ... you broke it."

"Yes, I'm sorry. I didn't mean to kill it. Had no idea it was so fragile." Jak rose, moved over to another pyramid, and squatted beside it, examining it closely, but careful not to touch it. Jon sank onto his heels beside him.

For a few seconds as they watched there was no change. But suddenly they heard a small, clear *ping*, and a new crystal sprang into existence near the base. Almost at once there was a repetition of the sound and another appeared further up on the adjoining side of the structure—or creature. As the boys continued watching this was repeated over and over—with each tiny sound a new facet came into being somewhere on the pyramid. Before their very eyes the crystal-being was growing.

"Boy, that's something!" Jon exclaimed admiringly.

"Yes, it's a life-form, all right," Jak said more seriously, without taking his eyes from it. "It's all new to us, but I'll bet there's silicon of some sort beneath this carbon-dioxide ice, and that this thing gets its nourishment from that."

"What makes it keep growing?"

"What makes a man or an animal or plant grow when it eats?"

"Oh!" Then, "Do you suppose it has any mentality?"

Jak was silent a moment, mulling that over. Then he looked at his brother, a crease of concentration on his forehead. "I feel quite sure that it probably has, but of a sort we wouldn't be able to understand, even if we could get in contact with its so-called 'mind.' Even reading that, I doubt very much if we'd be able to understand its way of thinking, reasoning, or the motivations by which it lives." He went back to studying the strange crystallization.

"Ummm, probably you're right," Jon agreed after some thought. A moment later he asked, "Is it good for anything? I mean, can man use it for something?"

Jak wrenched his gaze away from that astounding growth to look up in shocked disgust. "Is that all you think about in the face of such a marvel as this—whether it's worth anything or not? Here we've found an entirely new type of life, and...."

"Hey, keep your suit tight, Owl. We have to report this, you know, and I'm just trying to find out what to write down."

"Oh!" Jak spoke slowly, his voice now admitting the lightness of that point of view. "I can't, offhand, see any practical value, especially considering how easily these crystals are broken. But I know geologists—and possibly chemists—will be intensely interested in studying them. There's a lot they can learn here, I'm sure. We'll naturally report all that, you're right, and the location of this valley."

"Think they may occur all over the planet?"

"No telling, but probably if they can find the right sort of soil nourishment. We didn't see any while coming down, but they might've been there and we missed them, not expecting anything like this."

"We didn't see any other life-forms, either, that we could recognize. Maybe these're the dominant species here."

Jak rose to his feet and looked all about him. There were hundreds of the pyramids to be seen, some towering a dozen or more feet high and as large across each base line; others very small—babies, he thought with a grin.

Again he watched one of the smaller ones intently, noticing how it grew. Jon walked about, looking at the different structures of that mysterious, growing crystal.

Suddenly he stiffened, straining, listening. Then he called, "Hey, Jak, you hear anything?"

"Huh?" his brother tore his gaze from the crystallization he was watching. "Hear what?"

"Turn up the power of your suit receiver. There. There it is again.... Hey, sounds like our siren!"

"Yes, I heard it then. Mother must be in trouble, or something."

Jak's last words were flung back across his shoulder as he ran as fast as he could across the icy wastes of the valley floor. Nor was Jon far behind. In fact, after a few strides the younger, but longer-legged boy was beside him, then forged ahead.

"Hurry, Owl! Mom wouldn't signal unless it was urgent."

"Maybe Father's worse."

They tried to conserve their breath after that for running and climbing. Once Jon broke the silence. "Turn your oxygen a little higher, Jak," he said as he twisted the small lever at his own shoulder to increase the flow of the strength-giving energy.

They were panting and winded by the time they reached the top of the hill. But they disregarded fatigue in the face of their mother's probable danger—or their father's.

Jon looked quickly to one side and then the other. As Jak topped the ridge he saw his brother run some twenty feet or so to where he had spotted a fairly smooth downward slope. Down this the younger boy launched

himself feet first, sliding on his suit's back. Jak instantly realized the reason, and threw himself after his brother.

In less than a tenth of the time it would have taken them to climb down, the boys were at the foot of the hill. They struggled to their feet and started off toward the ship. Both were again shaken and sorely bruised from their rough slide, but they trotted on. Mother had called—nothing else mattered.

As they came closer to the ship they saw her reason for summoning them.

All about the outer lockdoor were those strange crystalline structures, growing swiftly. As the two boys came still closer, they could see that streamers of the crystals had already reached the lower edge and were trying to force their way through the almost imperceptible crack.

"They'll never get ... through there," Jon panted as he raced the last few feet.

"Don't see ... how they can ... but watch 'em." Jak waded into the alien, growing things. His gloved fists smashed right and left as he spoke. Jon was already doing the same.

But whether these crystal-beings were of a different type from those that Jak had broken in the distant valley, or just what was the reason, the boys now found it more difficult to break these crystals down.

"These aren't ... like those ... back there." Jon had now seen that these crystals did not always grow in pyramidal shape.

"No, they grow ... new crystals ... wherever needed." Jak had been concentrating on the tendrils, or chains of crystals that were reaching, always reaching, toward the lockdoor, while Jon had been trying to break the bases of the pyramids from which these arms sprang.

Although the crystals were still fairly easy to break—especially the tentacles, which were only a thin string—new ones replaced them so swiftly, and their numbers increased so constantly, that it seemed almost a losing battle.

"These're growing lots faster than the others." Jon gritted his teeth as he now tried crushing the bases with his heavy metallic boots, hoping thus to make it harder for the crystal-beings to reach the door.

For minutes the two boys fought in desperation; then Jon grunted in disgust at his thoughtlessness, and yanked out his flame-gun. "Never thought of this," he yelled as he trained it on the crystal-beings. The terrifically hot flame washed off them in coruscating showers—but did no damage.

"Try bullets," Jak unlimbered his gun from his back, and started firing it into the base of the crystals nearest the lockdoor.

The heavy bullets shattered the crystals easily, and soon the boys could begin to see that they were clearing the way.

"You keep firing while I open the door and climb in," Jon yelled. "Then you climb in while I'm going to the control room and I'll lift ship."

"Right," Jak replied and fired even faster as Jon touched the outer mechanism-stud that opened the lock.

Hardly had it begun opening, however, than they heard the sound of another gun being fired through the opening. They looked up in surprise and saw it was their mother, shooting a shotgun. Jon scrambled up into the lock.

"Good work, Mom, but get back in. I'm lifting ship."

He dashed through the inner doorway and into the control room. He threw the switch and *Star Rover* shuddered as its tubes roared into life. Jon punched on the intercom visiplat that scanned the interior of the lock, and saw his mother pulling Jak into the ship, then closing the outer door. Quickly Jon put the ship into a slow cruising orbit and switched on the auto-pilot. Remembering the open doors and the bitter outside cold, he glanced to see that the automatic heaters were taking care of the inside temperature, then ran back toward the lock.

There he found his brother desperately trying to warm their mother's unsuited body, now growing blue from that terrible cold.

"Help me carry her into bed." Jak rose and grasped her arms, but Jon pushed him aside. Stooping, he picked her up bodily. He ran, staggering a bit, with her into the bunkroom. Jak was right behind, and pulled some extra blankets from a drawer. Then, while he was piling covers about her, Jon dashed into the galley.

He drew hot water from the tap and quickly made a cupful of instant tea, then ran back with it to the bunkroom.

Some minutes later they saw with satisfaction that their mother's color was growing more natural, and her body tremors were slowing from the combined warmth of the extra blankets and hot drink. Only then did the boys stop to help each other out of their suits.

"Thanks for the help, Mother, but don't you know enough to wear a suit in weather as cold as this?" Jak's worry made his voice sharp.

"Yes, who's always fussing about us being careful?" Jon added. "Then pull a stunt like this."

Their mother looked up at them, and the old impish grin they had seen so seldom of late came onto her face.

"You've got me, Chums," she drawled. "From now on I reckon I'll keep my big mouth shut."

Jon howled, and Jak added in the same sort of drawl, "Well, now, I wouldn't go for to say it was 'big,'" and ducked as she slapped out at him.

Soon the two boys sobered down. "We'd better go examine the lock and make sure no crystals got in," Jak said.

"Yow, I forgot about that!" Jon sprang forward. "We sure don't want any of them in or on the ship."

9

Despite Jon's desire to get away from this unfriendly world that bore his name, he was careful to see that the signal-marker was set out and functioning, and that the ship's log contained as complete a record of the resources and data on the planet as was required by the Terran Colonial Board. The same was true of Four's four moons. Jak checked all the work, nor did they leave Four until both boys were satisfied it was complete. Their mother was a great help in taking the numerous photographs needed, having become quite competent in handling the cameras. She was so relieved at the steady progress of her husband's convalescence that she put extra enthusiasm into her photography. The family still felt that Mr. Carver should be kept as quiet as possible and away from any mental strain in connection with the ship and the planet mapping and, in his weakness, he seemed content to leave it that way for the time being. He asked few questions and accepted the reassuring answers contentedly.

Nor, even though Jon wished to get back to friendly Two as soon as he could, did he forget they still had to visit Planet Five, and scout and record that.

So, as soon as they were completely done on Four, he lost no time setting course for Five. Once on the way, he announced his names for the four moons of Four, and now it was Jak's turn to scoff.

"Well, if you can name yours after flowers, I don't see why you've got any kick coming because I name mine after fish," Jon asserted. "I leave it to you, Mom—aren't Tuna, Betta, Sturgeon and Porpoise nice names?"

"I think they are fine, just as I think Zinnia and Begonia are aptly named," she said diplomatically.

The two boys made faces at each other, then Jon turned back to his computations. "I'm not as good at figuring these things out as Pop is, but I think Five is about a half billion miles from the sun. It's almost three hundred and fifty million miles from here, since it's further around the sun. But we'll cut across on a direct route."

To his surprise, Jak came up and clapped him on the shoulder. "You're doing a grand job of astrogating, Chubby. I'm really proud of you." His voice was sincere and appreciative.

"Yes," their mother came over and kissed Jon, rumpling his hair affectionately. "I've been unexpectedly relieved that you've managed to get us to each world so surely and to land us so gently. Though maybe I shouldn't have been so surprised, at that." She laughed gaily as her younger son flushed from this unexpected praise.

"Aw, you guys are just saying that because I'm so wonderful." Jon tried to joke, but they could tell how deeply he felt their compliments.

A day later, when Jon announced they were approaching this outermost planet, the other two joined him in the control room, and all were soon deeply engrossed in the sight revealed to them in their visiplates.

An hour or so later Jon was examining their spectro-analyzer, when he let out a yelp of excitement. "Hey, that fuel-stuff's showing up. It must've come from Five." And a moment later, "Listen to 'Annie' rattle. It sure is there—but plenty."

They clustered about him, and even though they could not tell anything from the lines on the spectrograph that he pointed out, they could hear the machine chattering, and they grew excited from his exultation.

"Miners can use the same type of automatics they use on Pluto to get it, can't they?" Jak asked.

"I'd imagine so, although I really don't know anything about it. However, if we find we can use it, there's where they can get it, and that's the important thing."

Their plates again showed only the blinding whiteness of ice they knew was frozen carbon dioxide rather than frozen water or snow. For even more than on Four, there could be no water here, even in its most frozen form.

They cruised about above the surface, watching their instruments to find and record any metallic ore deposits, especially the new one. The terrain

was so forbidding, so desolate, that even the irrepressible Jon felt no desire to land on it, or to go outside.

Again their mother took most of the needed photographs, while the boys recorded all the other data of geography, size and conditions generally. Finally, Jon set the ship down on a fairly level plateau close to what they figured was the equator.

"Well, here we are and that's all I care about," Jon announced with a shiver. "We'll use the distant hands to put out the marker. Then we'll see if we can find the location of that fuel deposit."

Jak agreed. "I wouldn't go out there for a million credits." He shuddered as he looked out the port while the others crowded about to view that forbidding scene. "Maybe we should, but I sure wouldn't get any fun out of it."

"Doubt if our suits would be able to keep us warm, even with the heaters at max."

"No," their mother said sharply, although they could detect the relief in her voice that they had already made the decision. "This is one time I would have set my foot down, and not allowed it. This place gives me the creeps."

While Jon was making up the tape, Jak carried a signal-sender into the lock and placed it beneath those "distant hands." Jon came in and installed the tape, then started the mechanism running.

They returned to the control room, and Jak, whistling unmelodiously between his teeth, operated the controls that opened the outer door, then used the lifting servo-mechanism to set the signal-sender outside on the icy ground. When the outer door was closed, he nodded to Jon and the latter lifted the ship again.

"I'm going up a couple of miles, then circle about to look for deposits of that fuel metal. Meanwhile, as we go we can get the rest of our dope, and then scoot out of here."

Jak again took his place at the recorders, while his mother was at the cameras. Jon set the ship into a quartering circle, and when he had located the direction in which the analyzer showed the strongest indications of the enigmatic metal, swung into that course.

They had gone less than five hundred miles when they noticed a reddish glow in the distance. As they came closer, they saw that ahead and below was a terrific, whirling mass of colored gas.

"Wow, look at that storm!" Jak yelled. But he could not help adding, "Did you ever see anything more beautiful?"

"Better get well above it, hadn't you?" Mrs. Carver asked anxiously. "It looks dangerous."

"I'm sure not going through it." Jon was already lifting the ship. "But 'Annie' says the stuffs right close."

At five miles high he leveled off and put the ship into a narrowing spiral. From that vantage point they could see that the storm was a purely localized affair, perhaps some twenty miles in diameter.

"Wonder what makes those colors?" Jak called from the telescopic-visiplate into which he was staring.

"Suppose it could be a volcano?"

"Could there be volcanic action on so cold a planet?" their mother asked in astonishment.

"I don't see how there could be," Jak answered slowly, "but that certainly looks like flames of some sort down there."

"Maybe the experts from Terra can figure it out from our color pictures," Mrs. Carver said. "I'm taking a lot of extra ones, with the variable focus lens."

"That new metal's down there, though, whatever it is. Since those other people mined it, our miners'll figure out how to get it, too, I'll bet."

"I wonder." Jak was suddenly diffident. "Don't laugh now, but do you suppose maybe those flames could be some sort of life, and that they're feeding on the metal, which you said was highly radioactive?"

"Now who's nutty?" Jon asked witheringly, while Mrs. Carver gasped at the daring concept.

"Well, some of those flames are coming higher and aiming for us." Jak tried to defend his position. "We'd have said such crystal-creatures as we found

on Four were impossible, but we know they aren't and that they have some sort of—well, intelligence, from the way they tried to get our ship. So why not flame-beings?"

"Do you think they're dangerous?" Their mother's voice held a frightened note as she saw in her plate those swiftly approaching flames.

"Don't see how they could possibly hurt the ship, or us." Jon tried to speak calmly ... but he tilted the nose and the space-yacht was soon nearly ten miles high, although it still continued circling.

"Man, oh man, they're certainly beautiful!" Jak was enthralled as those bright, shining tongues of flame grew taller and taller. "There ... there does seem to be a ... a purpose in the way they act, though." His tone changed to a more anxious one.

The flames were now high above the storm of fire that constituted the main ... body? Now these tongues broke loose, and as they continued rising toward the ship they became more spherical in shape—were no longer simply extensions of the planet-based fires. And as they rose ever higher and faster, they seemed to the anxious watchers to be really thinking, intelligent entities.

"Let's move away from here," their mother pleaded. "I'm getting the feeling that they are actually pursuing us—and for no good purpose, either."

Jon touched the controls, and the ship began rising more swiftly.

"No, don't leave; I want to study ..." Jak began, but Jon interrupted him.

"So would I like to know more about them, but if Mom wants to leave, away we go." Yet there was an undercurrent of relief in his voice.

But as if guessing his intention, the flames hurtled after them at such tremendous speed that before the ship had barely begun accelerating, they were almost up to it.

"Hang on tight!" Jon yelled, and increased the acceleration. Soon the ship had left the flames behind. Peering in their telescopic plates, the three could see the flames, reluctantly and as if baffled, return at last to their home below.

"All gone, Mother. We're safe now," Jak said comfortingly.

"Thank you, God," she said devoutly and sank limply into a seat. "I was afraid for awhile...."

"So was I," Jon's teeth began chattering and his body shaking so hard that he, too, was glad he was sitting down. Now that it was all over the shock of that strangeness—that utter alienness—was hitting him. Nor was Jak in much better shape, in spite of his expressed desire to stay and study the enigmatic flame-life.

It was many minutes before the trio were able to discuss the matter calmly, and to realize they had been in actual danger.

"I see now we sure would have been, if Jon hadn't zoomed us out of there so fast," Jak said.

Finally, Mrs. Carver shook herself. "I'll go get lunch. It must be time, hungry as I feel."

"Me, too," Jon laughed. "But then, I'm always hungry."

As soon as the three had finished eating, Mrs. Carver and Jak went to sit with the invalid and watch hopefully for those semi-conscious moments which were becoming more and more frequent. Jon went back to check his course back to Planet Two, and to lounge later in the pilot's seat, studying from one of his reelbooks.

"There must be," he told himself, "some way of handling that fuel, and of storing and using it. The fact that it was cached there on Two shows that. But then, those folks who used it were so evidently far advanced in science."

A bit later the thought intruded, "Hey, if that stuff's so powerful now, after all the untold time it was stored there, what was it like when it was new?"

An hour or so later he heard his name called, urgently. He sprang up and ran into the other room, to see his brother beckoning him from the doorway of their parents' bunkroom. As he came up Jon saw his mother inside, bending over the bunk.

"Is—is Pop worse? He'd been so much better!" Jon's heart was clogging his speech.

"No, he seems to be waking up fully." Jak turned a radiant face toward him, then immediately knelt by his father's side.

Jon knelt, too, his eyes fastened on the still figure in the bed. But even as he watched, the eyelids slowly fluttered a bit, then a hand was raised to the forehead. Mr. Carver's head turned from side to side, restlessly, and then his eyes opened. They seemed to be studying each of the three watchers in turn, as well as the room in which he was lying.

"What—" The voice was low, and they strained to hear, "What happened to me?"

His wife answered quietly, "Don't you worry about that now, Dear. You were hurt and have been unconscious for some time. But now you're getting well, and I'll tell you all about it when you wake up again. Go back to sleep now. You are getting stronger that way."

Mr. Carver seemed to be weighing that advice, then to accept it. "All right," he said with an affectionate smile. He closed his eyes, and soon the rhythmic breathing told the three anxious watchers he was asleep once more.

Jon let out his breath in a happy sigh, there were tears of joy in his mother's eyes. Jak exclaimed delightedly, "He's getting well, Mother! In another day or so he'll be all right. Naturally he'd not remember clearly for a while. My textreels say people with concussions seldom do. But as soon as he gets a little stronger and those damaged places in his brain completely restore themselves, he will, you'll see."

Jon chimed in quickly, although he was not too sure of what he was saying. "It's just the shock, like Jak says. He'll snap out of it in a day or so."

She wiped her eyes on the corner of her apron, and smiled tremulously, "Of course, Boys. I ... I guess I've just been so nervous and—and he was so much more like himself this time."

"No wonder," Jon laid his hand gently on her arm. "You've been under a terrible strain, too, what with Pop sick and us boys roaming around on alien planets. But we'll be back on Two where it's more like home, and there's only a little more to be done before we can start back for Terra. Anyway, we did and are doing what had to be done to prove up Pop's claim, and we've

beaten Slik Bogin, supposing he's out here trying to cheat us out of this system."

The boys went into the control room. "We'll have to figure out where to lay out our townsite, and which planet is best to put it on."

"I vote for Two," Jak said after only a moment's hesitation. "It seems the most homelike to me, and we can stand the climate so much better there. Won't have to work in suits all the time."

"Yes, that's where I wanted it, too. This is a funny system, in a way, though—there's a much greater difference in the distances between the planets than we usually find."

"Why's that?"

"Ask some astronomer, not me. Has something to do with sizes and densities, I believe—but I'm not sure even of that. Maybe it's because this sun is larger and denser than the others we've studied. I know it's almost a quarter bigger than Sol."

"Two's almost as far away from this sun as Terra is from Sol, didn't you say?"

"Yes, Earth's average is about ninety-three million miles, while Two is about eighty-seven, which accounts for its warmth. Then, near as I can figure it out, Three is lots closer than Mars, yet Five is only a little further than Jupe, and Four is between them."

"Didn't I read somewhere that Sol's Asteroid Belt is really a broken-up planet?"

"Some people think so. But nobody knows for sure, yet."

"All right, then, let's put our city on Two." Jak grinned. "Has the mastermind decided where to put it?"

"Not the exact location." Jon flushed, but grinned back. "Colonial says it must be fairly close to good water, good soil, forests, and mineral deposits, however."

He sobered and looked at his brother appealingly. "Golly, Owl, any chance of Pop getting entirely well before we have to start it, so that it wouldn't

hurt him to do brainwork? I don't know very much about city planning, you know—only the specs Colonial furnished."

"He could—but maybe he won't. I don't know enough to say how soon. I've been fooled before—thought it would only take a few days, when he was first hurt."

"What about his being so ... so ... dreamy?"

"You mean, when he woke up just now? That's really nothing to worry about, honestly, just as I told you and Mother. I looked it up again, and the text says amnesiacs often act that way just before they recover full consciousness."

Jon let out his breath in relief.

"But listen," Jak changed back to the old subject. "If you don't know anything about city planning—and I don't either—how're we going to know what has to be done to satisfy the Colonial Board?"

"It's all in the papers they gave us that have to be filled out to file when we get back. It tells how much area to cover, how far apart the streets are to be and how wide, and how to mark out everything."

"Gosh, that sounds like a complicated deal. It'll take us an awful long time, won't it?"

"Not as much as you'd think. If we work really hard, I figure we should be able to do it in a couple of weeks. We just have to sketch in the bare outline, not fill it all in.

"Get out the papers, then, and we'll study them."

"Right!"

10

With the various members of the Carver family busy with their studies and chores, it did not seem long before they came close to Planet Two, and Jon laid down his reelbooks to begin preparing for the landing. He had started down through the atmosphere when Jak suddenly roused.

"Why don't we circle around a bit and look for our townsite from the air?"

"I was going to," Jon said with a disarming grin. "Just wanted to get close enough to see well before I called you in to help me decide."

At about five miles above the surface he leveled off and began circumnavigating the globe. He headed in the direction which—he remembered from their previous visit—would soonest bring them near the great ocean, but zigzagged from north to south as they proceeded eastward.

"Watch for rivers," he said. "A fairly large one that empties into the ocean."

They were about halfway between the north pole and the equator, as they had already decided the climate in that latitude would probably be the most suitably average. In a few minutes of traveling they saw ahead a plain that looked to be just what they wanted.

Quickly Jon maneuvered the ship downward and soon landed. They were just a few miles from the edge of the great ocean, at the mouth of a large river of considerable width and length, and not far from an extensive forest to the south, and a range of considerable mountains to the west.

"Hope the soil here is good," Jon commented as he locked the controls in neutral. "Everything else seems perfect."

"The spec-anal shows minerals fairly close, in those hills to the west." Jak was eagerly peering from the port.

"I know—that's why I chose this spot. Nice flat land here; good river; close to the ocean, mountains less than a hundred miles away, and not too high. Ideal place, I'd say.

"That it is; that it is. Going out today?"

"Why not? Nearly four hours till sundown. We can start planning today, then get busy in the morning."

Their mother came in just then. "I thought it felt like we had landed. What are you going to do here?" She glanced out of the port.

"We have to lay out a townsite," Jon answered, and at her astonished look he explained.

She shook her head admiringly and with surprise. "You boys continually amaze me—you seem able to do anything."

Jon shrugged that off. "When a thing has to be done, a fellow usually can figure out some way of doing it."

"Besides," Jak grinned, "we're like those old chaps on Terra who used to say, 'The difficult we do immediately, the impossible takes a little longer.'"

They left her then and hurried out of the ship. Even though they felt there could be no possible danger here—they could see for miles in every direction, and noticed nothing moving—they were wise enough from experience to carry their rifles slung across their backs, and wore bandoliers of ammunition. In addition, both carried tools and what equipment they felt might be needed.

Once outside, they ran first to the river, and tested the water. It was fresh and clear, and they knew this would be a good source of water for their proposed city ... until men might pollute it with their garbage and wastes.

"Got your pedometer?" Jak asked.

"Sure, right here in my coverall pocket. Why?"

"I suggest this would be a good place for the center of the town's north side...."

"Yes, here by the riverbank would...."

"I'll go east and you go west a half mile each, then we'll set our corner stakes."

"Then we'll both walk south a mile and set those, and have the four corners done. Sometimes, Owl, I have to give you credit for having brains."

"Wish I could say the same about you." Jak reached out and gave his brother a friendly shove. "Get going, Stupe. And when we start south, be sure you keep your line straight."

"Look who's yelping. Mine'll be as plumb as yours—probably more so, because I'm a better plumber than you are."

Jon started his pacing, while Jak went in the opposite direction after a pretended "*grrr*" at Jon's horrible pun.

When they returned to the ship, as the sun was going down, they felt they had made a good beginning. But as they went into the control room to talk alone, away from their mother's hearing—lest they worry her—they were not too cheerful.

"You know anything about surveying?" Jak slumped into a seat.

"Nope, not a sardine's worth." Jon paced forth and back in the little room. "That's what has me worried. How're we going to place those other marking stakes in exactly the right spots."

"Guess we'll have to measure them some other way."

"How?"

"Darned if I know. You're the mechanically—minded one—I thought you could figure it out."

Jon continued his pacing, his young forehead creased with thought. Finally, just as their mother called them to dinner, he looked up excitedly. "Hey, it'll be easy, after all!"

"How?" Jak was as excited as his brother as they went in to the living quarters and sat down at the table.

"A light plastic line that won't stretch, exactly measured, and fastened to two metal pin-stakes. We'll make two sets, and...."

"I get it. One the length of the blocks, the other the width of the streets."

"Right. Stick a pin in the ground, measure out the line, then plant one of our regular stakes."

"Then give a yank, pull the pin out and haul it in. Then use the other set to measure the street...."

"Yes, just keep going. Hey, I believe with that system we could each work alone, so I'll make two sets."

Jak thought all this over swiftly for several minutes, working fork and knife and jaws meanwhile. Finally, between mouthfuls, he said slowly, "I can't see a flaw in it—as long as we're mighty careful. Do you think it'll pass inspection?"

"If we take our time and make sure we're right, I don't see why not."

"What're you boys talking about now?" Their mother set a refilled dish of steaming Chlorella stew on the table, and resumed her own seat.

They explained, and told her the necessity of what they had to do in order to prove up their father's claim on this system, when they returned to Terra and appeared before the Colonial Board with their proofs of prime discovery.

The worried look came back into her eyes. "I always understood that surveying was a mighty exact profession. Do you really think you can measure it exactly enough to take the place of a regular survey?"

"I think we can make it close enough so that when Pop wakes up and shows us how to do the final survey with the instruments, we can save a lot of time, at least," Jon assured her. "That's what we're thinking and planning about now."

"You see, Mother," Jak broke in, "if we have the stakes all set, all we'll have to do is to make the sights on each one, after Father teaches us how to use the transit. Then, if we should be off anywhere, we can fix them easily."

"Yes, it'll cut down the time a lot," Jon went on, "and now we're so near done, I want to get everything finished so we can go back to Terra immejit."

"Why? Getting homesick?" his mother teased.

"Not so much that, but we want to get our claim before the Board. Anything can happen when such distances and time are concerned...."

"And we just don't want anything to spoil Father's chances of having this valuable claim verified."

"I see." She smiled now in relief, and again her eyes showed the pride she felt in her two manly boys who were daily proving themselves more than

equal to the unusual situations in which they found themselves. "Your father woke up again while you were out, and...."

"He did?" It was a duet of happy excitement.

"... and while he still didn't seem to realize what had happened, he acted even more as though he recognized me. He let me feed him some broth, then went back to sleep again very contentedly."

"Golly, that's great!" Jon reached out and patted her hand.

The three chatted together with more freedom and animation than they had known since the terrible accident first occurred. It seemed as though their worst troubles were over. For Tad Carver was so reliable, so confident in himself, so trust-inspiring—even beyond their natural love for him—that they felt everything would just have to work out right, once he was again in command.

As soon as they had finished eating, the boys hurried to the storeroom and found some metal rods.

"Cut me four lengths about fifteen inches each," Jon ordered as he went to the workbench. He cleared a space, then began getting the tools he wanted, and hooking up the induction furnace.

"You'll need eight for two full sets, won't you?"

"I got to thinking we'd better make only one set for now. If it works out all OK, then we can make the other."

By the time Jak had the pieces cut, Jon was ready to heat one end of each in the furnace, then bend it into a small eye. The other end he sharpened on the emery wheel.

"Now measure out pieces of that plastic rope," he ordered, pointing to a reel of small-diametered but very strong line. "Figure about six inches extra on each...."

"Look, Chum, you tend to your job and give me credit for brains enough to know that much." Jak's tone was almost cross, for sometimes this younger brother got on his nerves, since Jon did occasionally get quite "bossy."

But the elder quickly subdued that feeling—helped by the surprised and somewhat hurt look in Jon's eyes. He knew so well that Jon was merely

trying—as he himself was learning to do—to see that neither made any mistakes in this important work they were attempting to do in their father's absence. Father was always cautioning them to take pains with whatever they were doing, and they usually accepted his warning and advice—as they did their mother's—without any more grumbling than boys ordinarily make about such "fussing."

But now each of them—and both of them together—had to be, and did try to be, extra painstaking in all the things their father would have cautioned them about, and they checked and rechecked each other constantly. So Jak said nothing more, and quietly helped Jon complete the stakes-and-line sets. After all, he admitted honestly, there were undoubtedly times when he got just as "bossy" as Jon did.

Soon the two sets of pins and line were done. Each of the boys measured each once—twice—to make doubly sure their work was right. Then they cut up and sharpened a number of wooden stakes from some inch-by-inch strips they found in the storeroom.

The next morning they started out early. Each carried a bundle of the marking stakes, and Jon had a small sledge in one hand. In addition, they had their rifles slung across their backs.

"Working together to begin with," Jon said at breakfast, "we can start the eastward leg from the southwest corner, and run it a ways, then come back and start the northward one from this same corner."

"Yes, if we get that first corner square and right, there's less chance of the other three being wrong—they'll more or less check themselves."

They soon found they could work at quite a swift pace, and at lunch time Jon cried, "At this rate we'll have time to go back and re-check everything, and still get done within our two weeks."

"Yes, if we don't run into any trouble, this seems to be working out fine. Much better than I'd have given you credit for being able to figure out, Chubby."

"Catfish to you, Brother!" Jon grinned. "Hey, that reminds me. I want to see if that river's got any fish in it—and no...." He caught himself and stopped, but Jak knew what he meant. Their mother still didn't know about that

quicksand Jon had been almost trapped in, and they didn't want her to learn of it.

"I suppose it would be worth knowing," Jak had hastened to say, almost as if interrupting. "For once your eternal love of fishing will have its good points—as well as getting us fresh food. What about the ocean?"

"I'll try that, too, if I have time. Surf-fishing won't tell us too much about the deeper sea, and I haven't any heavy tackle for anything very big if we happen to run into it. But probably, close to shore like I'll have to fish, we wouldn't catch anything my lines and hooks won't handle."

"If *you* can handle 'em," Jak said with a grin.

"Don't you worry about me," Jon retorted. "I can pull in anything I can get my hook on."

"Except a sunken ship," Jak jibed, and Jon's face grew red. That incident, when he and his father had been fishing off the coast of Southern California, back in Terra's Pacific Ocean, was still a tender subject with him. He had had to cut his line that time, because they could not loosen his hooks, and he had lost a favorite spinner and leader and half his best line.

That first week passed uneventfully. The boys worked hard, from shortly after sunup to almost sundown. So hard, in fact, that their mother finally protested after noticing that they were so weary that they slumped in their chairs at the table and could hardly eat each evening when they returned to the ship.

"Now you boys listen to your mother," she commanded one night at dinner. "I'm just as anxious to get back to Earth as you are, but there's no sense killing yourselves to save a day or two. From now on, you are to start an hour later, and quit an hour earlier."

Jak managed a weak grin. "Guess you're right, Mother. But we are coming along fine."

"Sure, we've almost completed outlining the site. We'll have to take tomorrow off anyway, to go to the forest out there and cut some more stakes," Jon added.

"It'll make a nice vacation. I'm really fed up with so much sameness of hard work."

"Yes, it's been a steady grind, no fooling, but we wanted to get it done as quick as we can, so Pop can check it."

For their father had been waking up several times every day, their mother reported. True, he had only been conscious for short periods, and was still too weak to be bothered with any of their problems. But, she told the boys, he was able to eat something each time he awoke, and his mind was clear again. She was preparing easily eaten and digested foods that would bring back his strength quickly.

Jak asked anxiously whether his father had mentioned how the leg felt, and Mrs. Carver told him, "He says it doesn't pain any, although sometimes it itches beneath the cast."

Later on, just as they finished eating, Jak suggested, "Take your tackle along tomorrow, Chubby, and we'll chop where the woods meet the river."

"Why, t'anks, pal, you're a good kid." Jon made a fake pass at his brother, who jumped up from the table and yanked the other's chair backward, starting a small scuffle which their mother wisely did not try to stop, knowing that, tired as they were, it would last only a few seconds and would be good for them.

When the boys returned from their expedition the next night, with arms and backs loaded with bundles of stakes, and Jon's creel well-filled with Two's fish-things, she met them anxiously at the lockdoor.

"Did you boys hear or see the ship that passed over us this morning?"

"Ship?"

"No, we didn't see nor hear a thing. Sure it was a ship?"

"No, I didn't see it, either. I thought I heard one, and ran to look out, but couldn't locate anything. Maybe it was just my imagination."

"Spaceship or airship?" Jon asked.

"I couldn't tell you that, either, except that if I did really hear one, it must have been a spaceship to disappear so quickly."

"Unless it was a fast jet—they're just as hard to spot."

They discussed the affair for some time, but could come to no conclusions. If it was a ship, why hadn't it stopped or signalled? And if it wasn't one, what had she heard? Or had she actually heard anything?

Two evenings later the two boys had completed outlining their city site, and were just climbing into the *Star Rover* when they heard their mother's voice.

"That you, boys? Come in here. Hurry!"

At the urgency in her voice, they ran quickly and found her sitting at the side of their father's bunk. As they got closer they saw his head turn toward them, and recognition in the wide-open eyes.

"Hello, fellows!" His voice was weak but happy.

"Father!"

"Oh, Pop, you're awake at last!"

The two almost fought to be closest to him, but their mother moved a little and both sank to their haunches beside the bunk, each with one of their father's somewhat emaciated hands in theirs.

"Your mother tells me I've been sick quite a long time, and that you chaps have been carrying on. I'm grateful, and proud."

"You should see the way Mom has caught on to doing things," Jon said quickly. "She does almost all our photographing now."

"And Jon has developed into a real astrogator," Jak said.

"Yes, Pop, but you wouldn't be as well as you are today if it hadn't been for Owl knowing how to set your leg and make a cast for it, and giving you the proper medicines and intravenous feedings." Jon turned to smile at his brother, who grew red in the face and tried to stop the compliments, but the younger boy rushed on. "He's really a whiz as a doctor. Knew exactly what to do for you. How's your leg feel?"

"Fine, thanks to you, Jak."

"Oh, it wasn't so much—and I didn't *know*. I had to study a lot to find out...."

"Anyway, I'm still alive and that shows you did a fine job." Mr. Carver lifted a weak hand to caress his elder son's face. Then he turned toward Jon. "I've had fine reports of you, too, Son. Your mother says...."

"Yes, if anyone deserves praise around here, it's Jon," Jak broke in. "He has done all the piloting and figuring courses, and he even invented a one-man control so he can land and take off without all the trouble and preciseness needed before. Why, he...."

"Nix Owl!" Jon was the one to be embarrassed now. "That can wait until Pop's stronger. The main thing is to report now, so he can tell us what to do next."

"Where are we—and what has been done so far?" Mr. Carver asked. "Your mother hasn't—or wouldn't—tell me, except that we've reached the new system, and are landed on one of the worlds there."

"We're on Planet Two, and we're laying out the city site that the Board requires. This sun has five planets, and Two and Three are perfectly habitable for humans, but no natives above animal level," Jak began.

"There're seven moons—one at Two, two at Three, and four at Four," Jon took up the tale. "We've visited all the planets, and have set out the automatic signal-senders, with tapes giving you the credit for the prime discovery."

"They named the sun 'Carveria' after you, Mr. C." His wife leaned forward, eyes shining with pride, and an arm across the shoulders of each of the boys. "They named the planets after one of us, each, and the fifth one after the ship, and we've given names to the moons, too."

"This world will be swell for people who like it warm, and Three will be just as good for those who prefer colder weather. Both are a lot like Terra at different seasons and sections, and both are rich in soil, water, forests, metals and...."

"And we discovered a cache of that new fuel-metal you thought you saw in the spectroscope," Jon broke in, his voice bubbling with eagerness. "Right here on Two. Not a mine or a vein, understand, but a cache, in a metal box buried in the sand. Must have been some people a long, long time ago, because from the sand drifted inside the box it apparently hadn't been touched for thousands of years. And it showed up on Five...."

"But it's guarded by some sort of sentient flames," Jak burst in.

His father's face lighted up. "Have you tried it yet?" he asked Jon.

"No," The boy's face showed disappointment. "Mom wouldn't let me take any chances when I first wanted to, but now I'm glad—it's very highly radioactive still, in spite of who knows how many half-life deteriorations. It might've blown us higher than up. Maybe, though, when you get better we can study...."

"If we haven't got a small lead box, you ought to be able to make one," his father broke in. "You could probably handle a small quantity of it that way, to bring it in so we can study it. Maybe, though," as an afterthought, "if it's that strong, you'd better wait for me to help before trying any of it in the generators."

"It's in little cubes, a bit smaller than our copper pellets. That's why I'm so sure it's a fuel, and that it was put here by some sort of people who had advanced space travel a long, long time ago." Jon was still excited. "We figure all we have to do is finish laying out the town here, and then we can start back for Terra and put your claim before the Colonial Board," Jon said. "Of course, we all hoped and expected you'd be well enough to check what we've done...."

"But we tried to follow all the items in the papers the Board gave us," Jak added seriously. "And now you're well, we can make sure...."

"You father is far from well yet," their mother broke in, her voice imperative. "We are all so happy he's awake at last, but I can see he's very weak and that all this excitement has been almost too much for him. You boys say 'Good night' to him now, and then run off and eat your dinner, and let him sleep. Tomorrow evening you can finish your report."

Reluctantly the boys obeyed, and went into their living quarters and to the table.

"Golly, I should think Pop'd want to hear all about it now." Jon frowned with disappointment.

"He does, don't you worry," Jak tried to cheer him. "I should have watched him more closely to see we didn't excite or overtire him, but I was just as happy and eager as you were. He'll be stronger after another good long

sleep, but we've got to be careful not to expect too much of him for some time yet."

"Yes, I know you're right." Then Jon's face lighted with relief. "But it's sure swell to have him awake so we can talk to him and he can take charge of things again. You did a grand job, Owl, bringing him through."

"That's another thing, you big bum. You go handing out praise like that again, and I'll bust you one."

"Oh, yeh, and who was the parrotfish talking up so big the few little things I did?"

Jak came over and threw his arm across the shoulders of his taller but younger brother. "Both of us were so carried away by our enthusiasm we forgot to belittle each other," he said sagely. "Maybe we do sort of like each other, after all."

Jon pushed him away with rough tenderness, but his eyes were suspiciously moist. His words, though, were an attempted snort, as he picked up his knife and fork.

"What do you want—the next waltz?"

11

When the boys came into the dinette the next morning, their mother was humming happily as she prepared breakfast, and greeted them with a cheery smile.

"Pop awake yet?" Jon asked as he saw her mood.

"No, but he's sleeping so sweetly I know he's all right," she answered.

They sat down and began eating. After finishing, Jak said, "Well, we might as well go out and work some more on our townsite."

"Call us when Pop wakes up, will you, please?" Jon took a last sip of his juice-concentrate.

"That'd be silly." Jak frowned. "We know he can't come and help us, so why should we run several miles back here when we can see him when we get back?"

Jon opened his mouth to reply, his eyes flashing almost angrily, but their mother interrupted quickly with a question, "Boys, just why do you have to lay out such a site?"

"The Board requires it," Jon answered shortly.

"In the early days of exploration," Jak explained more patiently, "some of the space crews used to make their reports after merely flying above the surface of the planets of a new system. In fact, some of them didn't even go that close, and merely made up sketchy reports."

"Then when colonists got there," Jon, who had simmered down by now, took up the explanation, "they often found conditions very different, and many times quite dangerous to them."

"Yes, sometimes there were even intelligent inhabitants who hadn't been reported, so their planets couldn't be used for colonization. So the Board made this new ruling," Jak continued. "Now we have to have so many photos taken from various heights and at different places all over the surface of each planet, and each moon more than one hundred miles in

diameter. And we have to lay out a townsite on the most Earthlike planet, mostly to show we actually have been there and spent some time there...."

"And it really doesn't make any difference whether the people who'll come here to live use it or not...."

"But we think they will use ours because we selected a place close to a river and the ocean, close to forests and fairly near minerals."

"Yes, you have done a wonderful job, I know that much about it."

"Well, we'll go out and re-check our lines," Jon said. "I've been studying and experimenting with the theodolite, and I can...."

"What is that?" she asked.

"What's what? Oh, the 'theodolite'? That's the surveyor's telescope. I've learned enough about it so I can tell if our lines have been run straight, and as we were so careful in measuring the distances, I'm quite sure they're fairly accurate."

"Yes," Jak chimed in, "I'll bet none of them are more than an inch off, if that."

"Optimist," Jon scoffed. "I'd take that bet away from you, only it'd be cheating an infant."

Jak started a retort, then thought better of it, and shut up.

They left the ship soon, Jon carrying the surveying instrument over his shoulder, and Jak the marker-pole. Arrived at the nearest corner of their townsite, Jon set the instrument down, while Jak went on to the next stake.

By means of the graduated circle attached just below the telescope, and the plumb line suspended from it, Jon adjusted the collapsible legs until he felt sure it was correctly focussed. Then, as Jak went ahead from stake to stake, Jon took sights to make sure each marker was centered on his cross hairs. The ones that were not, he indicated by hand signals, and Jak reset them to left or right, until Jon was satisfied.

They completed all of one side before lunch, then returned to the ship. They found their mother had opened both lockdoors while they were gone, and fresh, crisp, though warm, air was circulating through the ship, blowing out

the old chemically pure yet "stale-feeling" air their purifiers had been re-circulating for so long.

Their father was awake, but still so weak he was making no attempt to sit fully up in bed, although his wife had slipped an extra pillow beneath his head.

"Ho, fellows!" he greeted the boys as they came into the bunkroom. "How's the job coming?"

"Just fine, Pop."

"We have the townsite all laid out, and now we're checking to make sure the lines are straight," Jak told him.

He frowned a bit. "How did you manage it? Neither of you is a surveyor. Or have you learned how to do that, too?"

"I think I've figured out the theodolite well enough to tell if our lines are straight, and that's what we're using now," Jon continued. "I can't measure distances with it, though."

Jak explained more in detail how they had measured the blocks and street widths, and rechecked them all.

"I can't see why it won't pass," their father said when they finished. "Probably no one will ever check it, unless they actually use the site when the colonists come. It shows we were landed here long enough to do the work, and that's the important thing. What about the rest of the mapping?"

"I'll go get the papers." Jon ran out, to return in a few minutes with the book of reports, and the rolls of film and prints they had made on all the planets and satellites. "You can check these as you feel up to it, Pop, and anything that looks wrong we can go back and re-check or do over."

Mr. Carver riffled quickly through the pages, and saw that each question had been answered; each measurement given an answer—though whether correct or not, of course, he could not know. All the information required had been supplied, at least.

He gave the boys his old-time grin, even as he was shaking his head in wonder. "You chaps certainly have done a job. Looks like I'll have to take the backseat from now...."

"No!" The two boys were shocked by that.

"Not on your life, Pop! We maybe did fairly well, but we need you, just the same."

"I'll say we do," Jak chimed in. "There's so much yet you can teach us. Why, we've only begun learning most of the things we want to know."

Mr. Carver smiled up at his sons. "I'm always glad to tell you anything I can, Fellows. It's good to see you growing up, though." He turned his head to face Jon more directly. "What's that about a new system you rigged up so you can land and take off with only one switch?"

Jon explained, and the two were soon deep in technical talk of electronic relays and cells, and automatic switch-overs. Finally, Mrs. Carver came in with a tray of lunch for her husband, and told the boys their food was on the table.

"All right, you chaps, go and eat," Mr. Carver said. "I'll take another nap while you're out this afternoon. Then maybe I'll feel up to talking some more this evening, and going over these reports with you."

The second day later the boys finished their re-checking, and came back to the ship in midafternoon. Their father was again awake, and they went in to see him.

"We're all done here, Pop, so what say we go back to that fuel-metal cache and see about getting the stuff aboard?" Jon asked.

"I guess from all you've said that's the most important thing now," he agreed after a moment's consideration. "Only thing is, I've been wondering if you couldn't move me into the control room, and fix a couch for me there?"

"Sure, that's easy," Jak told him.

But Jon frowned in thought. "Yes, we can do it, but we'll have to figure out first how to fasten the cot down and then make some arrangement so you can stand any acceleration we may have to use."

"How about fixing the co-pilot's seat into a bunk?"

"Hey, that's the ticket!" Jon brightened. He ran out and soon was helping his mother gather blankets, sheets and pillows, and going with Jak to bring an extra mattress from the storeroom.

They set the seat to recline, and then while Mrs. Carver was making up the bed, the boys carried their father—a much lighter load now than when he had first been hurt—and put him in his new bed.

"Say, this is all right!" Mr. Carver exclaimed after Jon had lowered the copilot's visiplat so his father could look into it without distortion or neck-craneing. "All the comforts of home." He grinned at his wife.

She stooped and kissed him. "Be sure and let us know any time you get too tired, though, Mr. C."

"I will, Honey," he assured her. "But actually, I'm so comfortable I don't see why I can't stay here as well as in bed, until the leg's strong enough to start getting up."

Everything else ready, he watched anxiously, then admiringly, as Jon started the tubes firing, balanced them and took them off with the throwing of his one switch. In his visiplat the elder man watched with intense interest the scenery over which they were passing—Jon had set course so they would go completely around this world of Two until they came to that desert. Mr. Carver made many enthusiastic comments about this splendid planet that now bore his wife's name.

"Yes, and Three's just as nice, only colder," Jon reported eagerly. "Folks who like cold weather can live there without too much trouble at all."

"It's funny, though," Jak declared with a frown, "that there's no protoplasmic life there at all. That we could find," he hastened to add.

"Lots of vegetation, though," Jon added. "That means the soil will be good for growing things, doesn't it?"

"It certainly sounds like it." His father smiled. "The colonists may have to adapt their Earth-seeds to fit, and probably bring their own worms and bees and so on. But they should be able to farm there. From your surveys, it appears there are plenty of minerals so they can start mines and factories of all kinds right away. Yes, this looks like a pretty good solar system."

"You bet, Pop. You sure picked a winner in this one," Jon's eyes gleamed with satisfaction.

"I had an idea, from the spectroscopic examinations we made 'way back there near Sirius, that we'd find it fairly good here. But, to be honest, I didn't dare hope it would be this good. To tell the truth, I was really more interested in that line which seemed to indicate that fuel-stuff, than I was in new planets for colonization, although we needed those, too, to make the trip pay off."

Before long they came above the beginning of that well-remembered desert, and Jon slowed and circled, preparatory to landing.

Jon kept his eyes upon his instruments, and when he saw they were close to the actual latitude and longitude, he killed the speed to their slowest cruising range, and their height to a few hundred yards. When he knew he was almost at the exact spot, he stared intently into his pilot's magnifying visiplat, at the same time keeping his fingers tautly on the landing switch.

Soon, in his plate, he saw the top of that cache cover in the nearing distance. He circled until he judged he could land close to it, then closed the switch.

Softly, easily, the space-yacht came in to a landing on the hard packed sand, and Jon shut off the power and put everything in neutral.

His father had wisely kept silent during this maneuvering, but now he let out his breath in a *whoosh*.

"That's the neatest landing I ever saw," he told Jon admiringly. "That gadget of yours will make you a young fortune when we get it back to Terra."

Jon actually blushed with pleased embarrassment. "Aw, it's...."

But Jak interrupted him almost fiercely. "Don't go playing coy, Chubby. You know darned well it's wonderful."

"Sure I do." Jon laughed then, and the rest joined in. "But you'd have tromped on me if I'd been the one to say so." He turned quickly to face his father. "What do we do about this?"

"Ummm. My suggestion would be for you to put on your suit and go out and open those covers you told me about. Give me the analyzer first, and I'll

study the stuff's emanations when you get it uncovered."

"I'll go out," Jak offered quickly. "You and Jon had better study it together. I don't know anything about it, but the kid does, and he'll be the one to handle it until you're well."

"Better take the jack—that cover's heavy, remember?" Jon said, and Jak ran out.

"I'd never have been able to do anything if it hadn't been for Jak's wiser advice," Jon said honestly as he brought the analyzer to his father from the instrument rack. "I'm apt to go off half-cocked, you know."

Mr. Carver looked fondly up at his wife, who moved quickly to his side, and put her hand against his cheek. "A couple of grand fellows you raised, Darling," he said softly.

"We raised, you mean, Mr. C." She smiled down at him. "They fight all the time, but when it comes to the pinch, they work together and I know they really love and admire each other very much."

Jon chuckled and spoke into the mike. "The folks are taking our good names in vain, Owl."

"Yes, I heard them," came back the elder boy's voice from the speaker. "If they only knew what we really think of each other," and then followed his attempt at a sneering laugh.

In their visiplates those inside the ship could see Jak, in his spacesuit, trotting awkwardly across the sand toward the cache. He carried the jack, and when he got there, used it to raise the heavy cover and throw it back. He jumped into the hole and took the cover off the smaller box. Then scrambled quickly out and ran some distance away.

"Shall I come back now, or wait here to cover it again?" he asked over his suit-sender.

"Maybe you'd better wait out there a few minutes," his father replied into the mike attached to his seat. "If we can't figure out something in a fairly short time, I'll tell you and you can recap the boxes and come back."

He busied himself adjusting the analyzer, and he and young Jon studied the lines carefully for quite a time. Finally the father roused.

"This is going to take a lot of study and work," he told his younger son. Then he spoke into the mike. "Better come back in, Jak." He turned his head again to face the boy with him. "Did you find a lead box, Jon?"

"No, sir, we haven't anything like that in our stores," Jon answered. "But there is quite a roll of lead foil. Can we do anything with that?"

"How much is there? And how thick is it?"

"The foil's twenty inches wide and about twelve feet long," Jon reported as he came back after a quick run to the storeroom to measure the foil. He had delayed a moment or two at the lock to help Jak out of his suit. "It's a thirty-second thick."

"*Hmmm*. That's not so good. Let's see. If we quadrupled it, that would give us an eighth ... no, that's not enough. Better take a piece and fold it to at least eight thicknesses, then go wrap it around a piece of that metal and bring it into the lock."

"That's not too much protection if the stuff's so strong, is it, Pop?"

"Well, double that, then. But I think it'll keep the rays off you long enough to bring it in—especially since you'll be in your suit, and if you put on lead-lined gloves."

"All right." Jon started out, then turned back. "What about the rest of it when we leave? Do we take it all with us?"

"No," slowly. "I doubt if anyone else would find it and steal it before we get back. On the other hand, the more we can take back with us, supposing we learn how to use it and it's as good as we think, the more we could get for it on Terra to give another immediate stake to come back."

"I have a thought, Pop. Why not just weld-fasten the whole big box it's now in to the outside of the ship, and make a small box that'll hold some to bring into the ship to experiment with?" Jon's eyes blazed eagerly.

"That's a thought!" Jak exclaimed, while their father answered more slowly, "Yes, I believe that could be done safely, especially if we put it back near the stern. Is the ship close enough so the lock servo-mechs can bring in the big box?"

"I don't think so," Jon answered after a searching look out of the port-window. "But with our suits on, Jak and I could carry it, couldn't we?"

"We've been close to the stuff several times for about as long as it'd take," Jak added, "and it doesn't seem to have hurt us any."

"Kind of a large box, isn't it?" Mr. Carver asked quizzically. "Might be sort of heavy."

The boys flushed, and Jon picked up his slide rule and did some quick figuring. Then he announced, crestfallen, "Great mackerel, I sure went off half-baked that time. OK, I'll take the ship up and bring it down closer."

"That's mighty delicate maneuvering." His mother looked at him in astonishment. "Sure you can do it?"

Jon shrugged. "If I can't the first time, I'll try again."

His father had to smile at the boy's confidence in himself, but he merely said, "This I've got to watch."

Assured everyone was safely strapped in, Jon started the tubes firing, raised the ship into the air—watching his plate closely as he circled about—then came down again ... right beside and not over five feet from port-lock to box.

"That's perfect," his father cried delightedly, watching in his plate. "You're sure getting to be an expert pilot, Son."

"And you're getting too excited and too tired from all this, Tad," Mrs. Carver said determinedly. "We'll have no more of it today. You boys go into the living room, and you, Mr. C., relax and take a nap. We can't have you getting sick again."

The boys started to protest but their father grinned. "Our mistress' voice, Boys. And she's right, I was trying too much. We're not in that big a hurry. Jon, you and Jak go make a box to hold our specimen."

They left him, and in moments he was asleep from exhaustion.

In the storeroom, Jon found some pieces of one-inch oak, and Jak and he made up their box, finishing just as their mother called them to dinner. It was a six-inch cube, sturdily fastened with plenty of screws; strong enough

to hold solid osmium. The lead foil was carefully fitted into the interior, and was now twelve layers thick—three-eighths of an inch.

"That ought to do it," Jak said, and Jon agreed.

"Let's go out and fill it after we eat." Jak was all eagerness.

Jon shook his head. "Not unless Pop says to. Now that he's awake, I just don't like to make decisions."

Jak grinned. "You're right, of course. Guess we got too big-headed, having to do things ourselves while he was unconscious."

"Yes, we're still pretty inexperienced, and I'm glad we don't have to figure things out now."

"Still, we can't go back to depending too much on him," Jak said thoughtfully. "That way, we'll never get the habit of thinking for ourselves, and deciding—and that would be bad. But about this, I agree fully," he added quickly as he saw his brother about to protest.

"Even if I don't know much about it, I can see that this stuff's dangerous to monkey with."

Their father awakened later, much refreshed by his nap. After the boys had explained and exhibited their new box, he agreed it would be all right for them to go out and get a single piece of the metal.

"Leave it in the lock, though," he added. "Then, in the morning, maybe I'll feel like helping Jon study and experiment with it."

The two boys ran to get into their suits, and soon were outside, carrying their lead-lined box. They jumped into the large cache box after lifting off the lid, and took the top from the inner one. They set the carrier beside it, then ran back to the ship. With the "distant hands," Jon flipped a nugget into the small box, and set it aside on the sand. Using the same servo-mechanism, he closed both covers. Then he brought the little box back and deposited it on the floor of the lock.

The two boys took off their suits and hung them in the wall closet, then went into the control room.

"You were right, Pop. We sure couldn't have handled the big box at all." Jon grinned, still panting. "Even the little one is really heavy with just one

nugget in it."

His father grinned back. "I had an idea, but thought I'd let you learn the hard way. Now maybe you'll remember it longer."

"Anyway, we got it in the lock, and tomorrow, if you feel up to it, we can start experimenting."

"Just how big are the pellets?"

"A little over half the size of our treated-copper ones," Jon told him.

"We'll have to cut it before we try working with it."

Jak, having disposed of the used plastic from their suits, had come into the control room and was listening interestedly, as was their mother, who was hovering near, not quite sure she liked the idea of her menfolks fussing with this unknown but admittedly dangerous metal.

"That means we'll have to make and install a smaller injector, too, doesn't it?" Jon asked. And when his father nodded, he added "I'll see about making it."

"Later, when we've found out whether we can use the stuff. Right now we'd all better get some sleep. I'm bushed, and I imagine you chaps are, too. How about you, Marci?" Mr. Carver turned to his wife.

"Well, I could use some sleep," she admitted.

"Right, Pop. Good night. 'Night, Mom."

12

Early the next morning the boys were clamoring to get started, but their mother would not let them go into the control room.

"Now you listen to Mother," she protested, using a favorite phrase of hers. "Your father hasn't made any sign yet. You wait until he's awake and has had something to eat. I know how anxious you are to do all these things, but you must remember he isn't strong yet, and we must not let him overdo. He is as much a child about such things as you two are, but someone has to watch him."

The boys laughed rather shamefacedly. "It's just we get so interested in things, Mom," Jon apologized.

"Yes, I know. But if you will look in your dictionary, you will find a word called 'moderation.'" She smiled.

"Never heard of it." Jon grinned as he went to get a reelbook on radioactives, and began studying. Jak, too, went back to studying and trying to classify the various specimens he had obtained from the two worlds. However, they soon remembered their usual duties—and whisked through their various chores about the ship, then went back to their absorbing occupations.

They had been at these nearly an hour when they heard their father's voice. Dropping everything, they sprang toward the control room, and found him wide-awake and looking much better. Mrs. Carver came running in, and they were told, "Feel fine. This is a wonderful bed. Seem to be much stronger today, too."

"That's wonderful, Mr. C. I'll go get you some breakfast."

Jon ran for a basin of water and towels, and he and Jak helped their father with his toilet.

"While you're eating, Pop, how about me cutting off that piece of the new metal so we can start studying it?"

"How big a piece were you figuring on?" Mr. Carver asked with that quizzical look.

Jon flushed and mentally changed the size he had planned to get. "About a gram?" he asked.

"I'd say more like a few milligrams." His father grinned. "That's plenty for our initial studies and analyses, and shouldn't hurt us any if we're careful and wear insulation."

"But that's only a pin-head size."

"Well?" again quizzically.

Jon flushed once more. "Yes, that's big enough to test, I realize now. It's a good thing I waited for you to help me. I'd probably have burned myself but bad. Actually," he smiled now, "I was figuring on about a quarter of a pellet."

His father frowned. "You should have known better than that, Jon. I thought I'd taught you something about being careful, and the dangers of rashness or impulsiveness. Especially around anything as dangerous as this stuff undoubtedly is."

"You did, sir, and I'm sorry. But I forget sometimes, when I get too enthusiastic."

"Well," philosophically, "you'll probably learn as you grow older ... if you live that long!" But again there was that disarming grin, which Jon repaid in kind before leaving to get his tools and go after the mite of new metal. This time, he did not neglect his precautions. He wore his suit, and put on a pair of extra-thick, lead-impregnated gloves.

Carefully he lifted a pellet from the box, wrapped it in several layers of lead foil left after making the box. He carried it so into the storeroom, locked it in a vice, and with a fine hacksaw cut off a tiny bit. Still wrapped carefully in the lead foil, he carried the remainder of the pellet back to the box in the lock, closed the lid and then took the sample inside. He took off his suit and donned a lead-impregnated, hooded gown and the leaded gloves.

"Good," his father said when Jon told what he had done. "I think I feel well enough to sit up a bit. Suppose you crank this seat halfway up, then I can watch better while you make the tests."

"Just be sure you don't get too tired," Jon said solicitously as he raised the seat and locked it at half-recline. He had brought in another of the leaded-gowns, and he slipped this over his father's head, arms and upper torso, arranging the balance of it down over his blanket-enwrapped legs.

Then, acting on his father's various instructions, he took the particle from its wrappings and began his tests. He measured the amount of radioactivity, and together they computed its half-life.

"Wow! That sure is high-pressure stuff," Jon exclaimed when they had completed the various tests which they had the equipment to make.

His father silently motioned him to set the seat back to full recline and lay there, concentrating, for some time before he spoke.

"Yes," he said at last, "it's even higher in the scale than I thought. Lots higher than Curium, even now. And no telling, by any tests we can make, what it was originally, before its many half-life reductions that must have taken place over the long time it has undoubtedly been lying out there. Probably way above anything known, even theoretically, to Terran scientists."

"Can we use it?" Jon was quivering with excitement.

"If we can figure out a way to do so safely, so it doesn't want to disintegrate all at once, I think we've really got a fuel—a super fuel. But we'll have to go at it mighty slow and easy. That stuff could blow us higher than up, if used wrongly."

"Yes, I know. But after our scientists first liberated atomic energy for their bombs, many people said they couldn't control a hydrogen bomb, but they did. And later the thorium bomb. And then they got our activated copper. So I'm betting they can figure this out."

Both fell silent, although there were a dozen eager questions the boy wanted so much to ask. But he did not interrupt his father's line of thought, even though long, long minutes dragged away while the elder still pondered the problem.

At last, after more than a quarter of an hour, Tad Carver stirred and looked up. "This is going to take a long time to figure out," he said slowly. "I'm not too much on atomics, myself, and neither are you. Now you run along and

do whatever else you have to do. It's a cinch we won't be able to try this stuff right away—if we try it at all."

The disappointment on Jon's face was plain, but he restrained any protests, knowing his father was right, and not wishing to call down on himself another verbal chastisement like that recent one.

"What about the rest of the stuff?" he asked instead. "Shall I get the box out of the cache and weld it onto the hull, as we thought we might do?"

"I don't see why not. We want to take it back to Terra with us, whether we figure out how to use it, or decide the job's too big for us and turn it over to the scientists there to handle."

"Right." Jon went over to the controls of the handling arms in the lock. Watching in the special visiplat, he opened the outer lockdoor, extended the "hands" and guided them down into the cache, after using them to lift the lid off the larger pit-box.

Carefully he manipulated them to grasp the inner box by its lower end-edges, and experimentally lift it an inch or so. Finding that it balanced, he slowly made the servo-mechanism lift the heavy container from its ages old resting place and up onto the "top" surface of the ship, near the stern. Making sure it was securely held there, he put on his suit, gathered up his welding outfit, and went outside and climbed onto the hull.

Going to where the box rested, he began the task of welding its bottom back-edge onto the metal hull. Then he released the grip of the handlers and, leaving them dangling in the air, welded the other three bottom edges.

Finished, he turned off his torch, rose to his feet and started back. But after a step or two he stopped and thought.

"Pop," he said into his suit-radio, "do you hear me?"

"Yes, Jon," the answer came back at once into his earphones. "What is it?"

"I was just wondering if it wouldn't be a good idea to spot-weld a few places along the edges of the cover, too, so there'd be less chances of its coming open. It'd be easy to open it later."

"How's it fastened now?"

"Just a simple hasp."

"Better touch it in a few places, then, to make sure."

"Right."

When this was done, Jon returned inside the ship, and saw to it that all the equipment was put back in place and carefully locked. Only then did he doff his suit and return to the control room.

"Well, that's done. What now?"

"Anything else you need to do here on this planet?"

"No-o-o, not that I know of. Why?"

"I was thinking that if everything has been taken care of, we might as well start back to Terra. No use staying any longer than is necessary."

"I ... I think we've done everything. Have you checked the record book and the pictures?"

"No, not fully. And I probably should, before we take off, at that. But I think I'd better have another nap or rest now, so I'll go over them after a while. Put them on the table here, so I can reach them."

"Right, sir. You take plenty of time to rest. If Jak's not too busy to go with me, I think I'll go fishing in the river, out there by the edge of the desert. Maybe we can get quite a haul to take with us, for fresh food on the trip."

"Good idea. Your mother said they were delicious."

When the two boys returned with full creels late that afternoon, they went at once to see how their father was getting along. He was awake, and studying the records they had made.

"Hi, fellows! Everything seems to be in fine shape. You chaps certainly did a job while I was *non compos*. Get any fish?"

"Lots of them. They sure bite swell here. Maybe because no one has ever fished them before, and they have no idea of lures and hooks."

"Then let's just rest and eat and sleep, and plan to take off in the morning, eh?"

"You bet. I'll sure be glad to get back home again," Jak declared. "This chasing around is fun, but I'm homesick for Terra, I guess."

"Me, too, kind of. Besides, I want to get some more schooling at one of our atomic institutes," Jon added more slowly.

"Going to give up inter-stellar exploration, Son?" his father asked drily.

"No, sir. But I figured we'd have to stay on Terra for a year or so while you get everything straightened out about this discovery, and get the ship ready for the next trip. So while you're doing that, I might as well be trying to learn something more."

"We will, and you should. And I presume," he turned to face Jak, "you want to study medicine?"

"That, and other things," the elder boy responded soberly. "If we can afford it, sir, I'd like to get several top men in various branches to give me some special coaching, instead of going to a school. That would get me started straight, and they could recommend good books for me to be studying while we're on our future trips."

Their father looked up at his wife with a smile. "What's happened to our babies, Marci?"

"They've just grown up, Mr. C.—but we have some pretty wonderful men in their place." Her eyes shone. "It was pretty hard, at first, after you got hurt and they had to take charge of everything, to realize that they had grown away from us. But I soon found that they hadn't, really," she continued hastily as the boys gave cries of dismay. "They have matured wonderfully, but we have not lost our boys at all."

"Well I should say not!" Jak cried hotly.

"We're still kids, not men," Jon declared. "Why, there's still so much to learn—and experience to gain—we've barely started growing up."

"You can keep learning back on Terra," their mother said. "As for me, I'm glad we're going to be there a year or more. I want to live in a house again, on land I know."

"Then we'd all better get to bed," their father said with his old-time roguish smile. "Otherwise we'll all be too fagged out to take off for home tomorrow."

As soon as breakfast was finished the next morning the Carvers all assembled in the control room for the start back to Terra.

Jon had already made the astrogational calculations for their trip, having worked on them off and on during many evenings of the past several weeks.

But just as they were all strapping down, his father stopped Jon with a sudden exclamation. "Wait, Son! I think we'd better go back close enough to all the planets and the sun to make sure all the signals are working right. That's one of the most important things the Colonial Board will check."

"Oh, I'm sure they're OK, Pop. We listened to each one after we'd placed it."

"But cases have been known where a sender failed—especially those on extremely hot or exceptionally cold planets. I'm not doubting that you handled them all right—it's just that I think it worth the time and effort to check them and make sure while we're still out here."

"All right, you're the captain." Jon opened the drawer in the control desk and hunted out the sheets on which he had figured his former flight plans to the various planets.

"We won't need to land if the signals are working," his father said. "Just get us close enough in line so we can receive the messages."

"In that case, we can fly almost by sight, merely taking into consideration the direction and speed of the planets." Jon shoved his papers back into the drawer. "Let's see ... we'll make the best time going to One, then the Sun, then Three, Four and Five, and then circling about and heading for home."

"Fine! Get going."

"Strap down, everybody."

A quick glance to see that they were all secure, then Jon closed the master switch of his new interlocking controls. Smoothly, with increasing acceleration, the *Star Rover* lifted upward through the atmosphere on the planet Marci—Carveria Two.

Ever more swiftly it flew, and a special sort of gladness was in each heart at the thought that soon they would be once more speeding toward their home on far distant Terra.

Traveling about the universe, seeing new suns, new planets, new and interesting—even though alien, and sometimes dangerous—forms of life of various kinds, all this was a constant source of interest and delight. Still there was within each of them, even Tad Carver, a love of and a longing for the planet that had given them birth. Men had always found it so—it was probable that men born on Terra always would. Probable, too, that men born on other planets would always long for a return to *their* mother world.

It took a special type of person to become a colonist on another and alien planet. Much the same type of pioneer as those great-grandparents, many times removed, who had made the terrible journey across the western plains and mountains of Noramer to conquer the great, wealth-producing West, and *their* forefathers and mothers who had braved the perilous and unknown oceans to come from the Old to the New World in Colonial days, to search for freedom and opportunity.

It had been found that, even among those willing to make the sacrifices and uprootings necessary to become colonists on other worlds, there were always a few who realized they could not stand it, after all. These unfortunate people usually returned to Terra—if they had the funds to do so. Nor did it seem to matter how much this new planet was like Earth, nor how great the opportunities for gaining wealth and prestige. It was that inner feeling of always *remembering* that they were so far from home and everything and everyone they had formerly known and loved.

Tad Carver was a true "son of wanderlust." He had the itching foot; the urge to travel; the zest for new places, new scenes, new outlooks. But even he, after a certain time away, felt that indefinable yet exceedingly strong *must* to return to his home world for a while.

The boys were young, which meant they were eager for new experiences, whether on their own or other worlds. They had not yet come to an age where Terra meant a great deal to them. Life was so thrilling, so interesting—there was so much to see and do. Yet even they did feel nostalgia after too long an absence.

It was Marci Carver who felt it most—this longing, this *need* for the old home. While it is true that her great love for her husband and sons made "home" for her any place in the universe where they might be, yet she had no real interest in exploration, no great desire or even curiosity to see other

lands or other worlds. The deeps of space brought such an *awe* to her that they almost made her afraid. No, if her menfolk had been satisfied there, she would never have dreamed of leaving Earth. She would have been perfectly content to live in one town or city all her life—in the same house, even. She did not have the pioneer spirit; did not in the least desire new scenes. Her home and her man and boys—these were all she asked of life.

Yet she did have the rare knack of making any place where she might be, home. She could make a mansion or a hovel—or this spaceship—seem such a perfect home to her men that they were perfectly happy and contented with their living quarters. It was not a matter of furnishings or their arrangement—not just material things like pictures, books, pillows or other knickknacks placed just so. Rather it was the "spirit of home" with which she impregnated every place in which her family might be living at the moment.

The boys had not yet noticed this consciously—they were so filled with the joy of living and doing and learning that they had not yet stopped to think about such matters. But Tad Carver recognized it, and loved his wife all the more because of her ability.

He often remarked of her, "put her in even a hotel room for ten minutes, and she'll make it home for me." He sometimes felt moments of guilt that he made her chase around so much, instead of letting her stay in one place—and remaining with her there. But he could not stay put—and he knew she would not want to remain any place without him.

That was why he had arranged things so she and the boys could travel with him. And, until he had been hurt and she, with the boys, had had to take over his duties, she had seldom left the ship while on other planets, although she always looked out through port or visiplat in the various places where they had gone, with the keen interest in anything new that made her such a delightful traveling companion.

So now all four felt that eagerness to be done with this matter of last-minute re-checkings, so they could be on their way back to Terra. It made the time pass swiftly—yet made it so draggingly prolonged, it seemed they would never reach their destination.

The ship soon reached an acceleration of two Earth gravities, and Jon asked, "Is this fast enough, Pop, or can you stand more?"

"You might step it up to three G's for an hour. There's no use loafing around here longer than necessary to make the curve so we can come fairly near each planet on the line between it and Terra."

"And that'll get us up to cruising speed quicker when we do start the straight stretch for home," Jon said, and turned back to his controls to apply another notch of speed.

It was not long before they approached Planet One—"Tad." Jon had plotted a course that would take them to within about thirty thousand miles of the little, hot planet, on the Earthward side. As they flashed past it, their receiver clearly picked up the broadcast of their signal-unit.

"That one's all right," their father said in a pleased voice, and Jon looked up and back from his calculations on the orbit to circle them about the sun, to grin his pleasure at the approval.

"Jak put it on top of a peak in the intermediate zone," he explained. "The weather—if you can call it weather—there is more nearly normal than either on the sunward or the spaceward side."

An hour later Jak struggled up from his chair, staggering beneath the triple weight of his body at that acceleration. Seeing him, Jon called, "Wait, Owl, I'm just about to reduce to two G's." And in a moment the older boy found it easier to get the sandwiches and bottles of nourishing broth their mother had prepared before take-off, and distribute them to the others. Gratefully, they all ate and drank.

"After we circle the sun and are en route to Three, I'll cut down to one gravity while we have a real meal," Jon promised.

"Aw, let's not slow down just for ..." Jak began.

"It won't cut our speed, just our acceleration, which means 'constantly added' speed," his father explained good-naturedly. "As soon as we've passed them all and are heading for home, we'll cut to one gravity for the greater part of the trip, but our speed will have been built up tremendously."

"Oh, sure, I know that, but I forgot for the minute."

As they circled toward the sun Mr. Carver studied it carefully in his visiplute. "Just about the same type of sun as Sol," he said after a while.

"That's what I figured, only that it's about one quarter larger and heavier," Jon told him. "I was hoping you'd be well enough before we left to check it for me."

"How close did you set your signal-sender orbit here?"

"Ten million miles."

"*Ten million!*" The man gasped, then laughed in relief as he thought the boy was just trying to spoof him. "Oh, come off it, Jon. How far out were you, really?"

"Unless my figures are all wrong," Jon's voice held a hurt note, "it was really only ten million miles. You can check my calculations. The book says quote said orbit to be as nearly circular and as close to the discovered sun as possible unquote, so I sent us in on a van Sicklenberg throw-out orbit apexing at ten million."

"Boy, that was really taking a chance. You don't need to repeat it for my benefit."

"I wasn't planning to, sir." Jon grinned now. "We'll go around at about twenty million this time, but the same type of orbit as before."

"That's better. Well, I think I'll go back to sleep. All of us should, I suggest."

"Mother has already dropped off," Jak said softly, glancing toward the recline seat in which she lay. "Switch on the auto, Chubby, then douse the glotubes. 'Night, Father."

And soon the little ship was speeding across the interplanetary wastes, guided only by the automatic pilot, while inside four weary people slept peacefully, knowing the mechanisms would guide them safely and surely to their distant, plotted destination.

For, outside of a possible recurrence of the accident that had caused Mr. Carver's injury—and that was a billions-to-one chance that could not possibly strike them again—what was there to fear away out here?

Nevertheless, it was the sudden ringing of an alarm bell that woke them all into instant, wondering wakefulness.

13

"What in the world?" Jon's eyes snapped open and immediately began scanning the various telltales on the panel, while from the other three came a chorused, "What's wrong?"

"Something out here using atomic energy." Jon's surprised voice made them raise their seats quickly to upright, so they could better see for themselves.

Mr. Carver hastily adjusted his visiplate to maximum magnification, and began searching the heavens surrounding them. "A ship, you think?"

"Yes, and quite close." And a moment later, aided more surely by his more complex instruments, Jon cried, "There it is! RA 11; square 17 on the plate."

His father's flying fingers found the object, then narrowed his focus of vision and stepped up the magnification. His eyes grew large, then hard and tense, as he studied the close-up image. "Slik Bogin's ship—I'd know that anywhere!" he exclaimed, and the boys looked at him in puzzled concern.

"Then I must have been right, that day I thought I heard a ship," Mrs. Carver declared.

"You must have been," Jak agreed.

"But what's Bogin doing out here?" Jon asked with a touch of fear in his voice.

"Nothing good, you can bet." His father's voice was grimmer than any of them had ever heard it before. "Any time you run across that pirate, you can lay mighty big odds there's skullduggery afoot."

"Great catfish! He's trying to beat us out of this system."

"I'll lay a thousand to one he is, if he thinks he can get away with it."

"What can we do about it, Father?" There was now a trace of a tremor in Jak's voice. "Jon and I have worked so hard to map these planets—how can Bogin possibly do the same and still beat us?"

"No telling. He's a slippery cuss, and if he really wants to try claim-jumping, he'll figure out some dirty scheme."

"Can't we get back to Earth ahead of him, Mr. C., and report to the Colonial Board first?" Mrs. Carver was almost in tears.

Her husband gave her a tight-lipped smile. "We'll sure try, Honey." His forehead creased with a frown of concentration for some minutes, then he faced Jon, who was watching him from the pilot's seat.

"Bogin's headed in the opposite direction, so no use chasing him to see what he's doing. Besides, I've heard his ship is armed, and we aren't, except for our rifles, which are absolutely no good in space. I say, continue our course, checking our signals, then beat it for home. After all, we don't *know* for sure that Bogin's trying anything—and our best bet is to finish our job as though nothing had happened, but not waste any time doing it ... just in case."

"Right, Pop. As near as I could tell, we have twice his speed, and we don't need to worry. We have all the data and pictures to prove we're the Prime Discoverers, and we didn't hear any signals to show he's put out any senders."

But there was an uneasy and unhappy silence as the little space-yacht continued to eat up the millions of miles.

Tad Carver had intended having his younger son slow down near Planet Three and go into an orbit close enough so he could get a good generalized view of this other Earthlike, though colder, planet. But now he would not do so. Speed and time were essential in getting back to Terra. He would try to keep his worries from the others as much as possible, but there was a deep foreboding in his mind.

Only too well he knew the various types of men who braved the spaceways, and that many of them were out and out criminals. And this Slik Bogin was the most ruthless pirate and cutthroat of them all, from reports. There were so many, many crimes charged against him ... though it was true that none had ever been proven. Yet such was the man's evil reputation that all honest spacemen hated him, even as they were somewhat in fear of him.

Mr. Carver was sure that the man's spacer was almost a warship in her armament. Nor did he doubt that the master criminal would not hesitate to

use his heavy rays to blast out of existence anyone he felt was a menace to his nefarious plans.

And this new system the Carvers had discovered was a prize well worth stealing, if possible. Although Mr. Carver had not seen these splendid worlds with his own eyes, he had carefully studied the boys' concise and complete reports, and their many detailed pictures, so he knew what a rich treasure they had struck in finding this sun and its planets and moons.

It would make him and his family rich beyond their fondest dreams ... and he would be worse than flat broke if they lost out on getting their claim approved.

For Mr. Carver had not told even his wife that all their possessions, including their ship, were mortgaged for every credit he could secure, to enable them to make this costly journey. It was true he had won great wealth on his previous trips into space—but several of his largest investments on Terra had gone sour, and this was a last desperate chance to recoup his fortune in one intensive campaign.

As they neared the point in their trajectory that brought them to the Earthward side of Planet Three, Jon began tuning his receiver and turning his directional antenna-loops, so he could pick up the continuous message of their sender. Soon he began hearing words, and tuned more closely, stepping up his power. The four sat erect, expectant.

Then their faces blanched and their fists tightened as they heard the words:

"This sun and system of five planets, of which this is the third, were discovered and surveyed by Michael Bogin and his crew, on the tenth day of January in the Terran year of 2136."

Over and over the message was repeated, while the Carvers stared at each other in horrified surprise and consternation.

But Mr. Carver rallied quickly. "He has changed the tape in your senders, boys. We'll probably find the same on Four and Five, and he's on his way to Two now to do the same."

"But he'll not be able to change the one we set out around the sun, will he, Pop?" Jon's voice quavered and broke into a boyish soprano. "He can't get

in as close as we did, and still slow down enough to retrieve such a small thing, can he?"

"I don't see how he could. But he has some darned good technies in that pirate crew of his. They'll figure out some way to destroy ours and substitute one of their own, I'll bet. Well, this changes the picture. Now we know what he's up to, so we'll just have to get to Terra ahead of him, and lay our facts before the Board first."

"They'll take our word against his, won't they, especially since we have such complete records and so many photographs?" Jak asked, hoping to be reassured.

"There's no telling," Mr. Carver spoke slowly, shaking his head. "If Bogin is trying to get this claim—and now we know he is—he'll work out some way of getting pictures and records, too. We can only hope."

"And pray," their mother added determinedly.

"We'll make out some way," Jon tried to cheer them all. "Meanwhile, I suggest I cut to one G and that Mom fixes us some grub. We have to eat."

"That's a good idea," his father agreed, and Jon manipulated his controls. They all felt the sudden relief of once more being their accustomed weight. Mrs. Carver unstrapped herself and left for the galley. Jak also unstrapped, saying, "I'll go help Mom."

"Ask her to make a pile of sandwiches, too, and to bring plenty of drinks so we can eat later without slowing our acceleration," his father called, then added, "Don't let your mother talk about this. Get her mind on something else and keep it there."

"Right, Father."

"This is serious, Jon," Mr. Carver said when the two were alone in the control room. "I don't like to worry any of you any more than's necessary, but our chances aren't too good, now that those signals have been changed."

"We've got some hope left, though, haven't we?" came the anxious inquiry.

"I see two fairly good ones—but it all depends on so many factors," Mr. Carver answered after a moment of thought. "We've got to try to get back first and report and show them our records and pictures—which are very

detailed, thanks to you two boys. Second, we've got to hope someone back there caught our original signals, and then noticed the change—if they could tell they came from the same system."

"How are you making out under this acceleration?"

"All right. I don't seem to be any weaker ... but then, what with all the excitement and disappointment, there may be a relapse. But that's not important...." Then, hearing his son's gasp of dismay, he continued rapidly and grimly, "No, Jon, really. I mean that, and I want you to keep it in mind at all times on the rest of this trip. I'm expendable, if we can prove our case. Not that I intend to die," he hastened to add with a grin as Jon started to protest. "But I'd rather take longer to get well and know that you all are provided for the way you should be."

"If we cut for Terra right away, without waiting to go on to Four and Five, Bogin couldn't possibly build up speed enough to beat us in, could he?" Jon questioned anxiously.

"Not unless his ship's a lot faster than ours. It probably is, because his crew can undoubtedly stand more acceleration, and he'll drive to the limit. But if he stops to change those other signals, I don't see how he can do it. Go ahead, change course, and let's hike for home."

"Right. Let's see, now. Terra's behind and down from where we are and the way we're heading. I'll set us into a circle while we're figuring out our course."

"Make it just an approximation for now. We can refine it as we go."

"Right." Jon worked swiftly at his computer, then at his controls, and they could feel the gallant little ship begin to strain toward the right.

"Don't try too short a turn," his father warned.

"OK, I'll let up a bit. I was figuring on a two million radius."

"Better make it three for safety."

In time their circling was completed, the new homeward bound course figured. For days the little ship and its anxious crew were on their way. Three times each day their acceleration was stepped up to two Earth-gravities for a period of four hours, then back to one and a quarter for the

same period—four on and four off continually, to give them a rest from the burden of doubled weight, and to make it easier to prepare and eat their meals, and to do what personal and ship's chores had to be done. In between times, as they could, they slept.

Jon had set their receptor and analyzer to react to atomics. It was now fanning out behind them in a cone-shaped funnel of force. He hoped by this to be able to tell if Bogin began overtaking them.

Of course, space was so vast, and the distance to Sol and Terra so great, and their points of trajectory so different, that the pirate ship might be taking an entirely different course, and not come anywhere near them until the two ships were almost home. On the other hand, Jon was taking the most direct route—and he was sure Bogin would undoubtedly do the same—so they were quite apt to converge sooner or later.

And since Jon's receptors covered an ever-larger sphere of space the farther away they reached, he and his father hoped they would be able to tell if and when their enemy began catching up with them.

Meantime, the two studied almost continuously together the problem of that supposedly new fuel-metal they had discovered on the planet Marci—hoping it could be used in their engines. They were sadly handicapped, both because neither was an atomic physicist, and because their little ship—well-stocked and provided with many instruments as it was—did not contain anywhere near all the testing equipment needed for such a delicate and complex and dangerous task.

Yet they learned much.

Jak took over the routine duties of their flight, after some additional instruction on points about which he was not sure. In between times, as the lessened pressure allowed, he studied the new specimens he had collected, saw to it that the ship's hydroponics kept operating correctly, and did whatever he could to relieve his brother and his father of their ordinary duties so they could devote all their waking time to study and experiment.

Their mother attended to her housekeeping, and saw to the comfort and well-being of her menfolk.

Mr. Carver knew, deep within himself, that he was overdoing, considering his illness. His partially-healed broken leg so often pained and throbbed that

he had difficulty concealing his hurt from the sharp eyes of his family. But he loved his wife and sons so greatly that their future well-being was far more important to him than his own, and so he never mentioned these things.

The sturdy little yacht had covered almost half the tremendous distance back to Sol. The Carvers were beginning to let up a bit in their anxiety and fears. Surely, each one felt, they were winning the race.

Then suddenly their alarm rang.

Three of them found themselves on their feet, rushing toward the control panel.

"How close are they, Jon?" their father yelled from his co-pilot's couch.

"*Mmmm*. I've stepped this up about two hundred per cent.... I figure it about half a billion miles."

"Not very far—in space. They must have lots more speed than we do to have caught up with us like that."

"What shall we do?" Mrs. Carver grabbed her husband's arm with trembling fingers.

He turned his head and smiled up at her. "We'll figure out some way to beat them, Honey," he soothed. "There's lots more can be done yet."

"Sure, Mom, they're still a long way behind us." Jon tried to keep the anxiety out of his voice. "And you know the old saying, 'a stern chase is a long chase.'"

"Can't we increase our acceleration and so our speed?" Jak asked.

"Yes, we'll have to do that, at least." Mr. Carver's voice was grim. He looked at Jon. "Step it up to two and a half, as soon as you're all in your seats. We'll stay there more of the time from now on, and we'll change the period to six in and two up."

"How about one and a half for the two hours?"

"We'll try it. If we sleep or nap more while we're at max, we ought to be able to stand it."

"We're still almost ..." Jon figured rapidly at the computer, "... three weeks out of Terra, even at that increased speed."

His father grimaced, while his wife and elder son uttered gasps of dismay. "I know. It'll be tough, but we've *got* to win."

But after a moment he looked first at his wife, than at Jak. "This is an order," he said seriously. "The minute any of you feel you can't take it any more, say so and we'll cut down, even if we do lose speed. I guess I went off half-cocked just now in saying that we had to win. Our health is more important...."

"Except yours, you're trying to say," Jak broke in. "You haven't been sparing yourself any, I notice, and I know enough doctoring to know you're not getting well as fast...."

"Pooh, I'm all right, and I'm used to ship accelerations." Mr. Carver turned his head toward his son and made himself grin. "Even under these three G's, I can still get up and lick you, even with a half-healed leg."

Jon realized at once that his father was warning him not to worry their mother any more, and forced himself to reply, pretending to be shamefaced, "Yes, sir, you could at that. I'll be good."

But the next morning, by the ship's chronoms, after they had fully awakened from a night of tortured sleep, Jon studied his instruments for some time, then reported to his father, "Bogin's still catching up. He's only about four hundred million behind us now."

"But how can he possibly be?" Jak demanded.

"Probably staying on three G's or better all the time," Jon answered.

"Or else he has a different means of propulsion than we have that affects his whole ship and contents, including crew," his father said slowly. "I don't know what it could be. But theoretically there are a lot of different ways of traveling faster than any we've learned how to use yet."

"But how could they, Mr. C.?" his wife gasped. "I don't pretend to know much about such things, but I thought that better fuels merely meant increased efficiency in the use of the engines, not an increase of speed. Isn't it acceleration that makes the speed faster?"

He turned his head with difficulty—at three gravities acceleration their apparent weight was tripled, and his body now "weighed" over five hundred and fifty pounds, instead of its normal one eighty plus!

"You're both right and wrong, Honey," he explained. "The better the fuel, the less we have to carry for the same distance traveled, and that makes our thrust-to-mass ratio less. We can go home faster than we came out here, because some of our fuel is gone and we have less mass. But that's not what I'm talking about. Theoretically, as I said, there are other ways, none of which our scientists have yet figured out how to use, as far as I know. There could be a complete or partial nullification of gravity or of inertia. Or some type of space warp. Or some method of 'cutting through' the other dimensions, so we could go almost instantly from one point in space to another."

Jak gasped. "Why, how's that possible, Father?"

Jon answered quickly. "I can illustrate, I think. Imagine a sheet of paper, with a dot near either end. The normal way to connect them would be a straight line drawn from one to the other—which is analogous to the way we travel in space now. What Pop's talking about would be the same as if we folded the paper so the two dots touched, and moved from one to the other direct."

"That wouldn't be...."

"That's silly."

The two phrases came simultaneously from Jak and his mother.

"It's not silly, Honey. We merely haven't figured out how to do it yet. But theoretical science knows that there are 'folds' in space. We just haven't learned how to use them yet."

"No," Jak snorted, "and I'll bet you never do."

"And I'll bet they will," Jon blazed. "You just don't realize how wonderful science is—in other lines than your own, I mean. You think it's perfectly natural that medical science has made such tremendous advances in the past couple of centuries. Why shouldn't other branches make just as great strides?"

"Because the advances in medicine and surgery have been logical," his brother began hotly, but their father interrupted.

"Whoa now, boys, don't get started on an endless argument. You're both right—and both wrong. I'll admit that the three methods I mentioned are pretty far-fetched. But after all, science is always doing the unexpected and the impossible. There's no telling what they'll do next—not even of telling what they may have done while we've been gone."

"I'd read about that 'simultaneity' thing," Jon stated. "It was a concept about being able to reproduce the exact nucleonic pattern of some other space and thus being able to transfer to it instantly."

"Another idea is of a 'tube' or 'vortex' method of transversing space at almost instantaneous speeds—and many other such," Mr. Carver declared. "But it's a cinch none of us have brains enough to figure out any of them before we reach Terra. And that Bogin's not using any of them, either, since he's so apparently on a straight-line flight like we are. He may have better engines, or better fuel, but to overtake us like he is—now that I've stopped to think about it—can only be done by using greater acceleration than we are, and for a longer time. So while those other ideas are interesting conjectures, they won't help us out of our present predicament."

"That's right, Pop." Jon wrenched his mind back to their immediate problem. "We've got to figure out what we can do *right now* to beat Bogin."

They all lapsed into silence then, partly to think of their problem, and partly because their personal energy was weakened by the tremendous pressures they were undergoing.

Their new schedule was hard on them all—none of them were really rested, even though they now slept or dozed most of the time. But they were keeping more nearly ahead, although when Jon took his next readings, Bogin's ship had crept up another third of a hundred million miles.

"That means he'll catch up with and pass us in about eleven days, and we're still almost twenty out of Terra." Jon could not entirely keep the worry out of his voice.

During the noon respite, according to ship's time, they cut their acceleration to one and a half, and Mrs. Carver prepared a hot meal, and cold lunches for the balance of that day.

While they were eating, there in the control room, Jak suddenly looked up at his father. "I just wondered, sir. How much pressure could a person stand for long periods, if he was unconscious under some kind of an anaesthetic?"

"Why," the elder hesitated, "I don't know exactly. I imagine around five gravities or so, if it was to be for some time, especially if one was in a pressure pack. Why do you ask?"

"I've been doing a lot of thinking, and I remembered reading about a series of experiments a Swedish scientist has been making about putting animals—even people—into an unconscious state. It's in one of my reelbooks. Seems to me I remember its saying he has found he could keep them there for several days at a time without any sign of permanent harm."

"How'd he do it?" Jon dropped his fork to lean forward.

"With a drug he invented. Wait, I'll go get the book." Jak jumped up from the table, but his mother's voice stopped him.

"We're not going to try anything like that," she said worriedly. "Not even to beat Bogin."

Mr. Carver reached out from his recline seat to lay a hand soothingly on his wife's. "Wait, now, Marci, let's find out first what this is all about. Maybe the boy has something, maybe not. But let's examine it before we decide, shall we?"

Her eyes still held the worried look, but she returned the pressure of his hand. "Well, I guess there's no harm in that, Mr. C. But I just don't like taking dangerous chances, that's all."

He smiled at her fondly. "Pioneers always have to take chances, Honey," he said gently. "Men would never have gotten anywhere if they hadn't. But we'll make sure we know all about what we're getting into before we leap, you can bet."

"Besides," Jon tried to reassure her, "even if this stuff would work, Owl hasn't any of that new drug, so we couldn't try it, much as we might want to."

"Oh, that's right. I hadn't thought of that." She smiled with relief.

In a moment Jak came running back with a reelbook. "Here it is. Let's see now." He rapidly scanned through the reel with his finder. "Ah, here it is!"

He read aloud rapidly, and the three listened intently.

"So you see," Jak raised his head triumphantly when he had finished reading, "it's perfectly possible to put us to sleep for a week at a time. And you said the ship was fully automatic," he turned to Jon, "so it doesn't need guiding, and would keep on its course whether we were awake or not."

"Well, it's way past our two hours." Mr. Carver spoke up hastily to prevent his wife from saying anything. "Time we were getting back into stepped-up acceleration again. Strap down, and we can study this later."

"I still don't like the idea," Mrs. Carver said as the four made themselves as comfortable as possible in the recline seats before Jon turned on more acceleration.

14

During the next two or three "waking periods" Jak busied himself studying his reeltext, but this was such a common sight it attracted no special attention. Nor did the others notice that he began disappearing into the ship's storeroom each "up" period, and had to be called repeatedly when the meal was ready, or it was time to strap down again.

He said nothing of what he was doing, nor did any of the others think to ask, for the boys were customarily here and there about the ship, busy at their many tasks and activities.

But at the start of one "up" period Jak went at once to the storeroom and workshop, and when he came back to the table set in the control room he showed his family a large corked test tube filled with a colorless liquid.

"I've got it!" he exclaimed, his eyes shining. "I found all the ingredients needed in our stores and my medical kit, and made up a batch of the cataleptic fluid. We inject four cc's in each of us and...."

"What're you talking about?" Jon demanded.

"The stuff to make us unconscious so we can stand five G's of acceleration." Jak looked up in hurt surprise. "What we were talking about the other...."

"I thought we were going to forget that nonsense," his mother said sharply.

"Wait, now, Marci, let's hear what the boy has," her husband said gently. Then, "Go ahead, Jak, tell us more about this."

"This medicine, injected into our blood streams, puts us into suspended animation for several days, depending on how much of the fluid we use. We first take an injection of glucose and other nutrients, of course, then this stuff puts us into deep sleep—slows our metabolism. You said in such a state we could stand much heavier acceleration, Father. Then with this we can beat Bogin."

"What sort of shape will it leave us in?" his father almost raised up in interest, and held up his hand when his wife would have broken in. "Are

there any after-effects?"

"The book says the doctor never discovered any, especially after he started giving people the nutrient injections first. He has had people under for as much as two weeks. Four cc's will act for about five days, so I thought we could use that much the first time, at least."

"Five G's would certainly put us 'way ahead of Bogin's ship." Jon had jumped up from the table and had been working swiftly with the computer. "Two such five-day periods—three more days on positive acceleration, then seven on negative—ought to give us a controllable velocity somewhere near Sol. We'd have to compute it more exactly, of course, before we take each shot."

Mr. Carver thought slowly and intently, then spoke decisively. "I believe that this is our best bet, if it's sure. We certainly want to get back first, if possible, and according to our present routine, which is all we can stand as we are, Bogin can beat us in. Besides, he will undoubtedly shoot down the *Star Rover* if he catches up with it—and you know what would happen to us!"

"Yes, when I checked today he was only about two hundred million behind."

"Let's try it!" Jak was all eagerness.

"Take it easy, Son. We've got to talk and study this a lot first."

Mr. Carver then turned to his wife, who had sunk back into her seat, biting her lips to keep from crying out, her hands clenched tightly. "Well make as sure as we possibly can before we decide to do anything, Honey, but don't you see the advantage of this if it will work? We must get to Terra first if we can, and this seems to be the only way we know of doing it."

"I see that," she said with a sigh of resignation, "and I know you'll know what you're doing before you do it."

"We sure will." Then Mr. Carver turned back to Jak. "Tell us again all about this stuff, and what the book says."

Jak talked rapidly but concisely for nearly five minutes. Afterwards he showed his father the reel, and his table of components of the mixture. Mr. Carver studied the book carefully for some time, and minutely compared

the formula as given there with the one Jak had used. Then he lay back and thought with intense concentration for nearly a quarter of an hour. Finally he raised his head with determination.

"I think we should try it. It seems safe, from all the evidence here. I have faith enough in Jak's ability to trust him to have made the fluid correctly—his formula checks exactly with the one in the reels. And if it works, we can win out."

Jon rose purposefully. "Right, Pop. Come on, Jak, let's break out the pressure packs and get them hung."

They went into the storeroom, and soon came back, each staggering under the weight and inconvenience of two packs. These they hung from the bulkhead hooks built into the ship for just that purpose, and made sure they were securely anchored.

"How much time after the injection before we blank out, Owl?" Jon asked then.

"A minute or so, I guess. Why?"

"Figuring how long I'll have to handle the controls. A minute is plenty of time, as I can have everything set up, and only have one switch to throw."

Mr. Carver reached out a hand and patted his wife's cheek as she stood by his side. "It's going to work out all right, Honey." His voice was bright and assured. "These boys of ours are really up on their stuff."

"Yes," she agreed. "I know they know what they are doing, and that you are checking them carefully. It is mainly my not knowing that makes me afraid sometimes." She gave him a lopsided smile. "I hate being the weak member of the party."

"You're nothing of the kind!" He grinned as the boys murmured protests which meant the same thing. "You're the best fellow in the gang." And he blew her a kiss as the boys helped him into his pack and saw to it that he was securely and comfortably strapped in. Then they did the same for their mother.

Jak went to his room and came back quickly with his hypodermic needles, and the bottles of glucose and concentrates. He put these beside the test tube with the new fluid.

Carefully he administered the dosage of nutrients to the other three, then lay down on his recline seat and gave himself his own dose. He rested there for a couple of minutes, then rose. Carefully he drew four cubic centimeters of the new, clear fluid into his needle, then approached his mother. "Ready?" he asked, smiling, but with tight lips.

She pushed up her left sleeve. "Ready, Son." And now her voice was soft but steady.

He tipped the needle into the light, carefully expelled a couple of drops to make sure all air was out of the tube. Then quickly and with a sureness he had trouble making his hands achieve, he pushed the slim needle into her arm, and injected the drug. With the ball of his thumb he rubbed the puncture gently for a moment. "Sleep tight, Mother." He smiled and leaned down to kiss her.

"Who's next?" He turned to the others.

"Me, of course." His father bared his arm. "Jon has to be awake last to handle the controls."

Again Jak filled his needle, and as carefully as before he injected the sleeping drug into his father. Then he stepped up to the pilot's pressure pack where his younger brother was adjusting the controls. "Ready, Jon?"

"Just a sec." The boy was still working, pushing in a button there, turning a switch here, stopping to tighten a wire or connection somewhere else. But in a few moments he had finished, and then rested his right hand on the handle of the master switch, ready to push it into contact.

But just as Jak brought the needle close to his arm the younger boy pushed it away. "Wait now, Jak. How about you? Can you make it to your pack and get in and strapped down and settled, and then give yourself the shot before I throw the switch and the five G's take effect?"

"Don't worry about me," his brother said gruffly. "I'll make out some way."

"Not good enough," Jon said positively. "Let's figure this down to seconds. If I don't close the switch before I black out, all this'll be wasted. How about if you inject yourself first? Will you have time enough to give me my shot and then get back into your pack and strap down before you go under?"

Jak thought swiftly a moment. "Your point's well taken, Jon, but you didn't figure it right. Your way, I'd have to give you your injection besides doing all the rest in the same length of time. If I give you yours first, I can get into my pack and give myself mine there. You merely stay awake until I'm done."

"Yes, guess you're right. But fix your pack so you can be sure of getting in without any trouble."

Jak did this, then came back, filled the needle and injected it into Jon's arm.

Swiftly then he ran to his own pack, climbed in and fastened the straps. He filled the needle, plunged it into his arm, and pushed home the plunger.

Jon had been watching his brother, forcing back the drowsiness that sought to engulf him. As he saw Jak's nod that all was done, he turned to his panel. A quick glance about his board with his fast-diminishing senses told him everything was on the green. With his last measure of consciousness, he rammed home the switch.

He settled back into a more comfortable position, and felt himself plunging down into the blackness of unconsciousness.

Jon felt himself coming awake, and his first, startled thought was, "Didn't the stuff work?"

He began to open his eyes ... and noticed at once how stiff his eyelids felt, but he forced them open. He looked at the date-clock and smiled with relief. The five days and several hours had passed, seemingly in an instant.

Now almost fully awake, his eyes sought the various meters, dials, gauges and telltales on his panel. Everything seemed to be working properly. He tuned in his receptor, and applied greater and still greater power. Space behind was blank of atomics.

Smiling thankfully and beginning to loosen his straps, Jon now noticed how dry his mouth was, and that his skin felt dry, too, and feverish. But he had no headache, and his thoughts seemed to be functioning as clearly and swiftly as always.

"Boy, I sure need a bath and drink, and something to eat!" he thought—then realized that the others would be feeling the same way. The others! He turned quickly to look at them. They were all still lying in their packs, somewhat pale, but with a peaceful, unstrained look on their faces.

Jon tried to rise, but reeled back and almost fell as he got onto his feet. He held himself erect a moment, and gradually felt a measure of strength returning.

As soon as he could, he went into the galley. Quickly he prepared a cup of instant broth, and drank it gratefully. Much refreshed, he made more of the consomme, and further enriched it with some anti-fatigue pills dissolved in the steaming liquid. He set four cups of it on a tray and carried them into the control room. His first quick glance showed the others beginning to stir.

"Morning, folks," he called cheerily. "Soup's on."

They opened their eyes slowly, almost uncomprehendingly, but awareness came quickly, and his mother and brother sat up and fumbled at their straps.

"Did we make it?" his father called anxiously. But Jak noticed at once how weak his father's voice sounded, and went across to his side.

"We sure did." Jon smiled broadly. "We were out just a little over five days, and the receptors don't show a thing behind. I woke up just a few minutes ahead of you, and that's one of the first things I looked at. Then I found I was weak and dry, so I went out and made this broth." He passed the cups and, as the others drank gratefully, Jon spoke again.

"I've got to hand it to you, Owl. You sure fixed us fine this time."

"He certainly did." Mr. Carver spoke as forcefully as he could, having already privately warned Jak to say nothing of his weakened condition. He looked solicitously over at his wife. "You all right, Marci?"

"Yes, I feel fine, now that I've had this good consomme Jon was so thoughtful as to make." She smiled with real relief that they had all come through this dangerous experiment so successfully.

Mr. Carver turned to Jon. "It feels like we're only at one gravity."

"Yes, I rigged the automatics so they'd take care of that at the end of one hundred and twenty-five hours," the boy explained. "Probably it was the

relief from pressure that woke us, as well as the wearing off of the stuff Jak gave us." Then he looked at his brother. "How come we're not famished after five days? That little glucose and stuff you gave us wouldn't last that long, would it?"

"No, but the drug not only made us unconscious, but slowed down our metabolism so that we burned up hardly any energy."

There was silence then while the four sipped their broth. Finally Mr. Carver looked up at Jak. "How soon can we go through this again?"

"The book says the doctor gave as high as four doses to people, one right after the other as they woke up, with only a few hours' rest between them."

"*Hmmm*, then we'd better take some time out. We'll all want baths, plenty of your mother's good cooking, and Jon and I will have to do some computing."

"If Bogin holds his acceleration, plus and minus, we can take most of the day, and still beat him in." Jon had been doing some rapid preliminary figuring. "But it'll take a couple of hours—maybe more—to compute the last hop. It's tricky. Especially, I'll have to look in the ephemeris to find the position of Luna when we get near her orbit."

"Right, we don't want to hit her. Well, we can keep at one gravity for at least twelve hours, then," his father said, and Mrs. Carver breathed a sigh of relief. She was still a bit worried about their undergoing such untried experiments, even though she trusted the abilities of her menfolks, and knew they had all come safely through the first time.

"I'll make notes of all this, and ask each of you for your full reactions," Jak said animatedly. "Then when we get home I'll write up a complete report and send it to Dr. Svendholm. I'm sure he'll be tickled pink to get this added confirmation of his studies and experiments."

"That's thoughtful of you, Son." His father smiled. "You're developing into a true research scientist."

"He sure is!" His younger brother paid deserved tribute. Jak reddened a bit and hastily left the control room to help his mother with her work.

They all took warm baths and changed their clothing. As Jak was helping his father, he asked anxiously, "Now that we're alone, Father, did you really

come through all right? You look a bit more tired and worn than before we started this."

"Sure, I'm OK," Mr. Carver said quickly, but he could not meet his son's eyes.

"You're not, sir, and you know it and I know it," Jak smiled a bit strainedly. "I don't like it, but I know how you feel about this, so I'll keep quiet. How's your leg?"

"Thanks, Son. Our getting back first is very important to me, and I can rest and get well after we reach Earth and get the Board's confirmation on our claim. And don't forget that we might not get back at all if Bogin catches up with us. He's ruthless about anyone who gets in his way.... As for the leg, it aches some, but not like it did before. I really think it's healing in fine shape."

"Let's have a look." Jak threw back the covers, and peered closely at the leg, lifting it so he could better see all around it.

"Yes," he said finally, as he tucked in the blankets again. "It's almost healed, and there isn't a sign of inflammation. Not even a bump where the break was. I ... I sure hope I set it right, so that it won't bother you later on."

His father patted his hand. "You did a grand job, I know, Son, and I'm very grateful to you—as well as proud of having such a fine boy."

"Two fine boys, then, for Jon is certainly every bit as deserving of your praise as I am, sir."

"That I'll certainly buy!" Mr. Carver's eyes shone.

They all sat about during the day, eating as much as they could hold of Mrs. Carver's fine cooking, and relaxing gratefully in the comfortable one-gravity Earth-weight.

Jon and his father worked tirelessly until they had computed precisely where they were and how soon and how much more deceleration they would have to use to finish their trip. Then they, too, relaxed for the balance of the day.

Late that afternoon Jon suddenly swiveled his chair about to face his father's recline seat.

"I think I've figured out something on that new fuel and how to use it, Pop. Ships'll have to be changed, though. The bins will have to be heavily lead-lined, of course, and so would the injector tubes have to be shielded. The nozzles would have to be made smaller, so the pellets will fit better. I figure the people who used to handle the stuff made the nuggets that exact size on purpose—that we'd not want to try making them the same size as our copper ones."

"That sounds reasonable. What about shielding for the generators?"

"There'd have to be a lot more of that, too. Probably thick shields of neocarbolly and paraffin. But can't they surround the generators, bins and everything with force fields, as an added precaution?"

"*Mmmm*, maybe they could at that. We'd better put it up to the scientists and technies back on Terra. Neither of us knows enough to handle it ourselves, when it comes down to the actual work."

The boy's face fell, then he forced a smile. "I hate to give in to anyone else on this, but you're right as rain, Pop. It is too big a fish for us to handle alone. But I'm sure going to learn before I finish, and some day when I run up against anything like this again, you can bet I'll know what to do."

"What's the use of going to all that trouble when you only have that small amount of fuel you found?" their mother asked curiously.

"Ouch! You would have to think of that," Jon grumbled, but Mr. Carver smiled up at his wife.

"There's plenty on Planet Five, remember? And probably in other places about the universe. You can bet that prospectors will be hunting—and finding it—once we announce our discovery ... IF we and the scientists can figure out what it was before it started losing its half life, and IF we can learn how to use it," he said firmly. "Once metallurgists have had the chance to analyze it, they won't take long to figure out exactly what it is, and where it can probably be found—the type of sun and planet that would have it, I mean," he added.

"And under the Prime Discoverer's code, we'll get a percentage of the process, won't we, Pop?"

"I think so. That'll be up to the Board, but they're usually pretty square about such things."

When it was time, Jak again gave the family the dosages of nutrients, and then the shots. Jon had filled four thermos bottles with strength-regaining soup his mother had made, and these were placed at each pressure-pack, ready for their awakening. Again the four lapsed into the complete unconsciousness of suspended animation—knowing neither discomfort nor the long passage of time—while their little ship bored through the immensities of space at a constant negative acceleration of five gravities.

As before, when they awakened they felt as though they had just gone to sleep. As soon as they had taken their initial feedings of the thermos-hot broth, Jon and his father set to work taking observations and making long and intricate calculations of their present speed and placement. Where were they? How much of their utterly incomprehensible top speed did they have left?

"Practically perfect!" Jon exclaimed happily after nearly an hour of careful computations, as he read the last tapes from the calculator. "It works out at one point eight four G's to atmosphere."

"O positively K," Mr. Carver agreed. "A master computer couldn't have done any better. And Jak has certainly proved himself to be a grand doctor."

"It's not my credit. Dr. Svendholm's the one who...."

"But it was you who made up the fluid and induced us to take it." His mother came over and ran her hand gently through his hair. "I'm proud of you all."

Jon had been tuning his receptors carefully, but was unable to get any trace of Bogin's ship, and all were happy at his report.

Warm baths and changes of clothing, and the fine meals prepared by Mrs. Carver, plus the fact they were rapidly nearing Sol, which could be seen on their telescopic plates, made them all very gay and full of chatter.

"I've decided I want to go back to the hospital-school and really prepare myself to be a doctor," Jak said in no uncertain terms. "Later I want to go into medical research."

"And I still want to enroll at the Centropolitan Institute of Atomics." Jon's eyes were shining.

"Aren't you boys forgetting one little detail?" their father asked drily. While the long sleeps had relaxed his body, and had practically completed the healing of his broken leg, the pressure had not been good for him, and his condition as a whole was worse. But his spirits were high, and he was careful not to let any of his family know just how weak he felt.

At his question they all looked up, astonished, and he continued, "There's the small matter of getting the Colonial Board's approval of our claim against the counter-claim we feel so sure Bogin is coming in to make."

"Pooh, he hasn't got a chance," Jon said airily.

"You hope," Jak scoffed, suddenly serious and worried.

"How about it, Mr. C.?" Mrs. Carver asked.

"Our pictures and data are so detailed I don't see how Bogin can possibly match them," her husband answered slowly and thoughtfully. "I think we can prove our claim. Besides, their receivers there on Terra should have picked up the broadcasts of our signals, and then the change—and that should have made them wonder why, so our explanation ought to satisfy them."

"That reminds me." Jon swung back to his panel. "Let's see if we can pick up our signals from here ... or Bogin's, rather!" His lips tightened.

In a few moments his tubes had warmed up, but nothing came in over his ultra-range receivers. He stepped up the power, and swung his directional loops forth and back, although mostly he aimed them directly toward the Carveria system's known coordinates. For long, anxious minutes he worked, but still no sounds, save the noise of cosmic rays and the other forces of the void that made long-distance communication such a problem.

With a weary gesture Jon finally turned off the set, and swung about with a stricken face. "What do you suppose is wrong, Pop?"

The elder shook his head slowly. "Only thing I can think of is that we're so far away the senders can't reach this far."

"Won't that be in our favor?" Jak asked. "If they can't hear any signals at all, our records ought to be enough."

"Maybe yes, maybe no," Mr. Carver answered with a tired smile.

"And after all our hard work, too." Jon's tone was dispirited.

"And the dangers you were up against." Mrs. Carver's eyes were tear-dimmed.

Their father caught himself and looked at each with a disarming grin. "Hey, we're all crossing bridges where maybe there isn't even a creek to be spanned." He made his voice mockingly cheerful. "What's happened to the good old Carver spirit?"

"You're right, Pop." Jon shook away his dismay and began to smile. "We're not licked yet."

But while they were eating, a short time later, Jon turned his seat to face his father.

"Don't like to start worrying again, Pop," he said in a low voice, "but our receptor is picking up atomic activity behind us again. Of course," he added quickly, "this close to Terra it could be some other ship, not Bogin's."

"Could be, and probably is." His father stroked his chin reflectively.

"I don't see how he could've caught up with us, but we don't know what his ship can do."

"The guy's tricky and dirty, but he does have a brain and he has some darned good technies in his crew. He'd know, from his own receptors, when we started speeding up so fast, and he'd do something to counteract that, if he could."

"I've heard things like that about him, but I don't know him."

"I do," grimly. "We've had brushes before, when I was in other ships. He's a skunk and ought to be behind bars—but so far no one has been able to produce any real evidence of what all spacemen know must be true."

"If the Board accepts our claim and data against his, won't that be proof against him?"

"It should be. You can bet your tackle I'll work on that angle. Space will be cleaner if that hellhound isn't in it."

"You bet, Pop. I hope you sink your hook in him this time."

His father laughed grimly. "It won't be for lack of trying, that's for danged sure."

15

Jon Carver spoke into the microphone of his ship radio. "Exploration ship *Star Rover*, Tad Carver owner, Jon Carver pilot, asking permission to land. We are circling at ten miles up."

A moment's crackling noise from the speaker, then a cheery, feminine voice, "Centropolitan spaceport. Landing permission granted. What size is your ship?"

"A seventy-two foot space-yacht."

"Do you need servicing?"

"We will in a day or so, but not at the moment."

"Use cradle forty-three in section D. Land in four minutes."

"Instructions received with thanks. *Star Rover* off."

Carefully Jon sighted through his visiplat until he located the cradle marked with a large "43" in the section of the tremendous spaceport also clearly marked "D." He lowered the ship slowly and gently, keeping his eyes closely on the chronom and its big sweep-secondhand.

So expert had he become at handling the ship, and so well did his new automatic technique work, that the ship settled gently into the cradle dead center ... and only one point three zero seconds off the four minutes specified.

"Nice handling, Chubby," Jak cheered as they felt the mighty engines and generators shut off.

"Aw, it was rotten. I was almost a second and a half off in my timing."

"Who cares?" There was a lilt of joy and pure thankfulness in their mother's voice. "We are back on Earth—home—and all of us are whole. That's the best part of all."

Her husband looked up from the recline seat where he was still lying, and winked at his sons. Then he faced his wife. "The eternal mother." He smiled

gently at her, and his voice was soft with emotion. "Happiest when her brood is safe. And," he added hastily at the look coming into her eyes, "how thankful mankind is, or should be, that womenfolk have always had that feeling. Man would never have gone as far as he has if she hadn't."

Jak soon came in from the other part of the ship. "All our data books, pictures and specimens are packed and stashed by the inner lockdoor," he reported.

Jon jumped from his pilot's seat and started toward the living quarters. "Let's get our street clothes on, and get going to the Colonial Board headquarters."

"Yes," Mrs. Carver said eagerly. "After all we've gone through to make sure we beat that Bogin and his ship back home, let's not waste any time."

"Well," Mr. Carver's eyes twinkled, "go put on your prettiest frock and all your war paint, so you can make a good impression on the Board members."

She crinkled her nose at him, but went in to the bunkroom. Mr. Carver raised his chair to upright, and began struggling to get up. The two boys, watching closely, saw how weak he was, and ran to help him. With his arms across their shoulders, he finally managed to half-walk, half be carried, into the other room. The boys lowered him into a seat.

"I'll get your clean clothes, and your razor and some hot water," Jak said.

Jon went back into the control room, and turned on his radio-sender. "Service, please," he said when the operator came on, and in a moment, "*Star Rover*, cradle 43, section D. Please have a taxi-hopper here in thirty minutes, and a wheel chair with it. Thanks."

When the four got outside on the landing platform and Tad Carver saw the wheel chair he was indignant. "I'm not going to ride in any lousy perambulator," he grumbled, but the boys were insistent.

Finally his wife came over and put her hand on his arm. "You might as well give in, Mr. C. Besides, your leg is not strong enough to do without one—yet."

Still grumbling, he let the boys help him into the wheel chair ... but they noticed his sigh of relief when he was settled and the weight was taken off

his feet. His body trembled with weakness, in spite of his efforts to control himself.

The chair, their books and cases were soon loaded into the copter, then Jon directed, "Colonial Board building, please."

The little ship rose swiftly on her whirling vanes, then streaked through the clear air toward the center of the great city of Centropolis, while the four watched the familiar sights of "home" with eager, happy eyes.

"Look at the trees and flowers," Jak called excitedly, pointing at the riot of color below. "They're getting green and in full bloom. It's late spring here, yet it was fall back on Three."

"Different suns, different seasons on the various planets." There was amusement in his father's voice.

"Sure, you ought to know that," Jon said condescendingly.

"I do know it, you fathead. I was just...."

"Now, Boys," their mother interposed—and the two grinned covertly at each other. Poor mother never seemed to realize there was no real animosity behind their bickerings.

It took only a few minutes for the swift taxi-hopper to ferry them from the spaceport to the roof of the huge Colonial building. Tad Carver paid the fare, the boys again filled their arms with their books and cases, and Mrs. Carver pushed the wheel chair to the elevator. They descended to the Board headquarters' floor.

In the anteroom their father propelled his chair to the receptionist's desk.

"I'm Tad Carver, owner of the *Star Rover*, just back from a trip. We wish to present a claim as Prime Discoverers of a new planetary system."

"Oh, splendid!" The stately brunette's eyes lighted. "Is it a good one?" she asked as she reached into one of the drawers of her desk for a sheet of forms.

Mr. Carver smiled. "Five planets and seven moons. Two of the planets are very Earthlike, and there are lots of metal, wood and many other worthwhile things."

A distant look came into the girl's eyes. "I've never been out in space. It must be wonderful...." She straightened with determination. "Please answer these preliminary questions. Then I'll get your appointment with the Board." Rapidly she put the questions as listed on her forms, and filled in the vacant places as he answered her.

Finished, she rose, said, "Just a moment, please," and went in through a side door with the papers in her hand.

Mr. Carver wheeled himself back to his family, who were sitting stiffly in chairs against the further wall. "Are they going to allow our claim?" Jon asked nervously. The others leaned forward to hear the answer.

"Take it easy." Mr. Carver's eyes showed amusement. "The girl has merely gone in to make an appointment for us. This takes time, you know. We probably won't have the answer for several days."

"Oh!" It was a trio of disappointment, and they sat back to wait, glumly, impatiently.

But only a few minutes later they straightened expectantly as they saw the receptionist coming back. She crossed over to them.

"The Board is at liberty to hear your preliminary claim now," she told them. "Please follow me."

She led them through the same side door and into a large room beyond. The four looked eagerly about them, seeing a well-lighted, wood-paneled office. Across the room was a large, heavy table-desk, behind which were seated five men.

"Mr. and Mrs. Tad Carver, and their two sons," the girl introduced them before leaving.

"Please take those chairs." From his seat at the center of the table the chairman indicated comfortable chairs on the side of the table opposite him. Jon pushed one aside while Jak propelled the wheel chair into the vacant space. Then the other three Carvers seated themselves in adjoining seats.

"I am Robert Wilson, Chairman of the Board. The other members are Phil Silverman, James Dougherty, Will Irwin, and Sam Reardon." He indicated in turn the other men at the table. "I see you claim to be the Prime Discoverers of a new Solar System. That's wonderful! We're expanding so

rapidly, what with the increasing birth rate on Terra and the other colonized planets, that we already and always need more room. Tell us more about your find."

"It's a five-planet system with a sun much like Sol, only about a quarter larger. The coordinates are Right Ascension 17.45, Declination Minus 11.4, distance about sixty-two light years."

Swiftly Mr. Carver gave the pertinent facts about the habitability of planets Two and Three, and presented their books of data, and their cases of photographs.

"How come we haven't received your signals—or didn't you place any?" Irwin asked.

"We did place them, sir, but we noticed several days ago, coming in, that we could not hear them with our own receivers. It is my opinion that the distance is too great for the strength of the senders."

"That's possible," Silverman spoke up. "Your claim is farther away than any yet presented to us. I happen to know that the signal-senders furnished by our Board technicians ordinarily have a theoretical range of not quite fifty light-years."

Mr. Carver half-rose, then settled back and spoke with a level voice, while his eyes swept from one to the other of the five men.

"I want to report honestly on this case, sirs. Just before we left, we started back along a course that would take us fairly close to all our planets and the sun, to make sure our senders were functioning correctly. We started from Two, where we had just completed marking-out our city site, went past One and around the sun, planning to make a big swing to the other planets and so back home. The senders of One, Two and the sun were working all right, but as we neared Three we heard, instead of our own, signals stating that the system had been charted and claimed by Michael Bogin and his...."

"Slik Bogin!" Several of the Board members exclaimed in concert, and Chairman Wilson added, grimly, "So he's at work again."

Mr. Carver waited until they were silent, then continued, "We think he either destroyed our senders or substituted his own tapes in ours. However,

we put our sun-signal into an orbit so close to the sun's surface we doubt if he'll be able to do anything about it. It's only about ten million miles...."

"Ten million!" Reardon almost yelled the question, and the others sat upright in excited astonishment, doubt showing in their faces. "How could you do that?"

"I figured a van Sicklenberg, sir, to give our sender a circular orbit apexing at ten million miles," Jon Carver explained simply. "We used the servo-mechs in our lock to throw the sender out when at minimum distance."

"You?" There was a concerted expression of disbelief and Mr. Reardon said, witheringly, "Why, you're not a listed astrogator. How could you compute a ... a what was it you called it?"

"A van Sicklenberg throw-out orbit, sir. I...."

"Never heard of a van Sicklenberg. What is it ... what sort of nonsense are you talking?"

Jon opened his mouth to reply, but his mother forestalled him. She rose determinedly. "My Jon is 'only a boy,' gentlemen, but he has also become an expert pilot and an excellent astrogator, if I do say so myself. He is also an inventor, and will shortly apply for patents on a new automatic piloting system—which I don't pretend to understand anything about, but which I do know from watching its use is far in advance of anything you now have. You can be sure he knows how to do such a simple thing as plot an orbit." She sat down, eyes defiant, her mouth in a straight line.

The men's faces showed astonishment at her words as much as at her outburst.

"I had been knocked unconscious and my leg was broken," Mr. Carver took up the explanation, "so I was out of action for a long time. I'm not fully recovered yet, which is the reason for the discourtesy of this wheel chair. The two boys had to take over all the work of mapping the new system. But I have examined their books and pictures, and feel sure you will find everything in order and complete, and that it will prove our rights as Prime Discoverers, no matter what Bogin may have to say when he gets here. He is following us, but we managed to beat him in."

"Hmmm." The chairman frowned in thought, then whispered for some moments to the other men on either side of him. The four Carvers sat nervously, awaiting the decision of the final arbiters.

Finally Chairman Wilson addressed them directly. "You can well understand that we will have to make a rather more thorough examination than usual in this case, Mr. Carver, and that we will have to wait a few days to see whether or not Michael Bogin is going to make a counter-claim. Knowing you got here first, he may decide not to do so. Where are you located, so we can get in touch with you later?"

"We came directly from our ship, sir, so do not have an address as yet. However, as soon as we have found a place, I'll call your secretary and leave our address and visiphone number."

"You do that, please." Then, as the Carvers rose to depart, Chairman Wilson halted them, his voice kindly, yet grim. "This is a peculiar case, Carver, as you can well understand. We know the reputation of Bogin, but we also know he has never been found guilty of any of the things rumor claims he has done. We Board members try always to be fair and honest in these matters, and you can feel certain and confident you and your claims will be given careful consideration. We will get in touch with you in a few days."

"Thank you, Gentlemen. I'm content to rest my claim in your hands."

The four bowed, then left the office and the building.

"What do you think, Father?" Jak asked anxiously as they were riding a ground-cab whose driver had been instructed to find them a good apartment hotel.

"I don't know," Mr. Carver added a weary smile. "It's all in the future, and I'm not a seer. However, I'm sure we'll get an honest and unbiased hearing and decision, and I'm equally positive we have the better claim. So let's forget it until we're notified to appear, and enjoy our return to Terra."

"Mr. C.'s right, Boys," their mother agreed. "We've done our best, and thanks to you boys, it's a very fine best. Now we must wait, but not worry."

Their cabby found them a nice place where there was a vacancy, and soon the four were unpacking their gear and getting settled in their new home.

Mr. Carver visiphoned at once to give the new address and phone number to the Board's receptionist.

Then the Carvers settled down to wait, with as much patience as they could muster, for the call.

Jak insisted on having a good doctor called at once. The latter made a thorough examination. He had Mr. Carver taken to an X-Ray laboratory, where it was determined that the broken leg had been perfectly set, and was now practically healed, although it would be some time before the strength returned to it. He also prescribed a course of medications to bring back the invalid's full health and vigor.

The call came from the Board three days later, in the middle of a morning when, fortunately, the four Carvers were all in the apartment. They hurried down to the street, where they flagged a ground-cab and were driven swiftly to the Colonial building. The same brunette girl ushered them at once into the Board room.

Inside, they found the complete Board in session, and in chairs opposite them sat Slik Bogin and his chief lieutenant, who glared at the Carvers sullenly as they entered. Hardly were the four seated when Bogin sprang to his feet.

"What's the big idea, Carver," he almost yelled, "trying to claim our discovery? You've got a crust, trying something...."

"Sit down, Mr. Bogin, and keep quiet," Chairman Wilson spoke in a low but commanding voice. "We're here to judge the facts as presented, not to indulge in charges, countercharges and vituperation. Now, the Board has examined minutely both sets of claims. Both parties have presented all the data required by us, and these have been studied by each of us individually. Dougherty," he turned to one of the Board members, "please review the data sheets for us."

A tall, serious-faced man rose, and arranged the two sets of sheets before him.

"According to the Carver claim, as presented here," he gestured toward one set of books, "they arrived at the system and made their first landing on Planet Two on January 14th of this year, 2136. The Bogin claim is that they first discovered the system and landed on Planet Three on January tenth,

also of the year 2136. Both parties claim they set out the required signal-senders, although neither have been heard by our listening posts here. However, we know that signals from these senders cannot, ordinarily, be read at distances in excess of fifty light-years, and the system under consideration is said to be over sixty. We have asked the communications department to check with ships and planets nearer the system in question, to learn whether or not any signals from it have been received."

He paused a moment and looked at his fellow members first, then at the expectant six across the table. "I said that both parties have presented complete data. However, it seems to us, after careful scrutiny, that one set of data was obtained from the air and from the surface of three of the five planets, and from a height of less than twenty miles above the seven moons, and less than five from the other two planets, as our regulations specify. The other books clearly show that the observations were all taken from above the surface."

There was relief on the faces of the four Carvers, nervous side glances between Bogin and his henchman, none of which escaped the sharp eyes of the five Board members, watching closely the reactions of the two opposing parties.

"We have here the two sets of photographs, taken from a height of five miles as specified by our rulings, of the townsite that we require to be laid out." He held one set out to each party across the table. "Please examine them and let us know if you see any differences."

There was silence for several long, anxious moments, then after the two groups had studied the photographs handed them, Jon Carver suddenly let out a gasp, and looked up eagerly.

"May I speak, sirs?"

"You may."

"This is not the picture we took," indicating one. "If you will compare the two, you will find that this one was taken before the work was fully completed. See, there is a gap here along the east side where not all the stakes are in."

"Oh," his mother looked up quickly, and took up the story. "Then I was right, that time. I did hear a ship. You see, sirs," she addressed the Board

members more directly, "the boys ran out of stakes and had to go to the forest there some miles to the northwest of the ship to cut more. Look, you can see just the edge of our ship right here on the margin. While they were gone, I thought I heard a ship passing over ours, but when I got to the control room and could look through a plate, either it had gone out of sight or I was mistaken. The boys said, when asked after they got back, that they had neither heard nor seen it—they were in the woods at the time. But I believe now that I did hear a ship, that it was Mr. Bogin's, and that he took this picture at that time. It took the boys nearly a day and a half longer to complete their work, and not until after that was our picture taken."

The Board chairman smiled at her, then turned a severe face toward Bogin and his lieutenant. "That is exactly what the pictures show—that one is complete and the other is not. What have you to say about that?"

The man's face was black with fury. "I say they're liars," he shouted. "This one here," shaking the photograph he held, "is our picture. That one is theirs."

Mr. Carver started to rise, but Jak was before him, and it was the latter whose voice cut through the din. "Oh, no, and I can prove which is our picture, if you will examine all the rest. I did all the developing and printing, and you'll find a small 'C' down in the lower right-hand corner of all our pictures. I marked all our negatives that way, as you can determine if you'll send someone to our ship to get the negatives from the darkroom."

The Board members huddled over the sheaves of pictures for a moment, then turned severely toward Bogin. "The young man is correct. All the Carver pictures are so marked, and so is this one of the completed townsite."

Jon Carver broke in. "Ask them to describe the animal life they found on Planet Three," he suggested.

"Well?" the Chairman looked levelly at Bogin.

"Why ... why...." The latter was quite taken aback by this sudden challenge. "Why, there are several species ... and ... and they were more or less like ours here, although not exactly like them, of course."

"Ha, that proves he was never on the ground there," Jon cried out in triumph. "We found, and so will anyone else you care to send there, that

while Planet Three has a lot of vegetation and is perfectly habitable—though cold—there is absolutely no protoplasmic life to be found there. At least," he added honestly, "on any of the surface we covered, and our ship's log will show, as you can see there, that we flew at less than five miles up for eight complete but spiral revolutions about the planet, and were on the ground in several places, which we explored and photographed thoroughly."

"The young man is right," Mr. Silverman spoke up. "I noticed that fact mentioned in their records, and intended asking more about it, because this is the first planet of which I've heard, that is otherwise completely habitable by mankind, where such a condition has obtained—where there is voluminous floral life but no protoplasmic life of any kind. Being something of a botanist, that fact struck me at once."

Bogin rose, sneering, but also feeling safe in this part of his claim. "Bah! They just didn't happen to sit down in the same places we did ... if they were there at all. We saw lots of animal life there."

"And you took pictures of such life?" Mr. Carver asked pointedly.

"Why, no, it isn't required."

There was a discreet rap on the door, and when the chairman gave permission, the receptionist entered and handed him a sheet of paper. He examined it quickly, then passed it to his fellow members. The five conferred together in quick whispers for several minutes—while Bogin and his man glared in sullen anger at the Carvers.

Four of the Board members finally resumed their seats, while Chairman Wilson stood at his place. He pressed a buzzer, then took up his gavel. He struck three loud, solemn notes with it.

"It is the considered opinion of the Terran Colonial Board, here assembled in official meeting," he intoned, "that Tad Carver has proven his claim as Prime Discoverer of the Solar System henceforth to be known as 'Carveria,' and this decision shall be so entered in our records as of this date. Congratulations," he added, smiling as he turned to the happy four, who were attempting the almost impossible task of each hugging all the others at the same time.

Bogin and his lieutenant rose wrathfully and started to leave the room.

"Just a moment, Bogin," Chairman Wilson said authoritatively. "You are under arrest for an illegal attempt to defraud by false testimony."

Bogin, eyes blazing, suddenly seemed to go berserk. He drew a blaster from an underarm holster, and waved it about as he and his man backed toward the door. "You ain't gonna arrest nobody. We're leaving here—and we cinder the first one of you that moves."

But, unseen by them, the door behind had opened and three space marines, guns in hand, had entered in response to the chairman's buzzed call.

"Drop that gun, Mister," their leader said sharply, his own muzzle pushed against Bogin's back. The latter, face livid, did so. In moments the two pirates were handcuffed. The sergeant saluted the Board members. "Guards will be sent to the spaceport at once to arrest the other members of the Bogin crew, sir, pending examination and trial."

"Thank you, Sergeant. We will prefer charges at once."

As the marines started leading the two away, Bogin pulled back suddenly, and faced Mr. Carver.

"One thing I'd like to know. We were catching up with you, fast, and all of a sudden you pulled away from us as though we was standing still, yet we were all in slings, and doing three and a half G's. How'd you do it?"

Mr. Carver smiled lazily. "We're submitting a full report to *The Space Pilot's Gazette*. You can read it there—if they let you read where you're going."

He turned back to the Board members and again expressed the thanks of himself and his family.

Chairman Wilson held out a sheet of paper. "You may be interested in this report. It came from the Communications Center on Petrarch Three, and was the deciding factor in your case."

The four clustered close to read:

"Carver signals heard first, then ceased one by one and the Bogin signals began, although one Carver signal, the solar one, is still heard."

Mr. Carver turned to the Board members and said, "Like most crooks, Bogin was yellow. He didn't have nerve enough to run in as close to the sun

as these youngsters of mine did, and so couldn't change their signal there. The boys are great planet mappers—both of them."

EDWARD EVERETT EVANS

was born in Coldwater, Michigan, the youngest of the four children of John and Nellie Evans.

Enlisting in the U. S. Navy after leaving high school, he served as a musician before and during the First World War. He played in concert and dance bands for several years after leaving the service and still finds enjoyment in listening to good music, although he no longer performs on any instrument.

He learned to read even before he entered kindergarten and has always had a fondness for the strange and off-trail in stories. When science fiction and fantasy first gained their own magazines, he became a regular reader of them—and still is.

Evans began attending conventions of the "fans" of science fiction with the first "Chicon," or Chicago Convention, and he has not missed one since. He finds both enjoyment and profit from meeting the people who are also interested in this kind of literature.

From reading to writing was a logical step—although not an easy one, but Mr. Evans has made the step successfully. In addition to his novels, he has over forty short stories to his credit. All of these reflect his optimism about the future of the human race, and his firm belief that the great majority of people are "swell guys." He confidently expects to see man's first spaceship make a successful flight within his lifetime—and thinks it will not be long after that before many of the astonishing happenings and forward-looking inventions of his stories will be actualities.

*** END OF THE PROJECT GUTENBERG EBOOK THE PLANET
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